SAFETY DATA SHEET

Data source map: where every line of a generated SDS comes from

An SDS is now assembled when it is generated, from your own data. The formula and each raw material's constituents (CAS numbers and percentages), flash points, boiling points, physical states and other properties go through the hazard engine and the transport classifier. These are joined by the CAS determinations (hazard data, exposure limits, element, acrylate and target-organ flags), the product family (UV/LED flag, use text and SDS profile), the regulatory lists you maintain, and the company or private-label manufacturer settings. The engine then chooses standard, translated GHS and OSHA wording to match those results. Sections 2 to 15 agree with one another because they share one classification. A per-product, per-language text edit still replaces any editable line. Every row in this map ends with one of two labels. "Generated from real data" means the line comes from your data, including fixed wording that the data chooses. "Uses fixed text" means the line is the same whatever the data says.

Before the fixes, only about a third of a sheet came from data. Most paragraphs were canned: one fixed first-aid, fire, spill, handling, stability, toxicology, ecology and disposal text, used on nearly every product. Many lines were English only, exact percentages were printed, trade-secret identities leaked in several sections, flammability came from the lowest raw-material flash point, and transport copied one ingredient's DOT row. Now nearly every paragraph is built from hazard codes, physical state, family settings and list matches. Concentrations print as prescribed bands. Trade secrets are masked everywhere. All four languages are fully translated. Fixed text is left only where a regulation or a deliberate owner decision requires it.

Key points

- Publishing is blocked on every path and in every language by any of these: a blank emergency phone (the company's, or the manufacturer's on private label), a blank private-label manufacturer name, missing federal hazard data above the threshold, a trade-secret ingredient on the Prop 65 list, an inactive finished good, or a Section 14 result of 'Not determined' or an incomplete transport override. Warnings never block. That covers flash point, STOT organs, combustible dust, physical state, TSCA and composition warnings.
- Enter flash points on raw materials. The product flash point is the weight-averaged value of the raws that have one, so water and other non-flammable raws need a high flash point entered, or the average will understate the product's flash point. If no raw has a flash point, the product is treated as not flammable. Boiling points set Category 1 vs 2 and Packing Group I.
- Keep every raw material's constituent CAS numbers and percentages (min/max) complete. They drive the Section 3 bands, the classification, trade-secret masking, TSCA, SARA 313, HAP, Prop 65 and RCRA. Gaps raise warnings and force TSCA to 'not verified'.
- Maintain the CAS determinations: hazard data, exposure limits (with 'Skin' in the notes where it applies), the nitrogen, sulfur and halogen flags (they decide which combustion gases print), the RCRA metal flags, STOT target organs and route for H370-H373, and the Acrylate flag (it decides which acrylates are named on UV products).
- Mark the Combustible Dust status (Yes or No) on powder and solid raw materials, and set physical states on the product, the family default or the raw materials. A Powder or Solid product that contains a raw marked Yes gets the OSHA Combustible Dust class. A raw left as Not assessed only raises a warning.
- Maintain the regulatory lists (TSCA, SARA 313, HAP, SNUR, Prop 65, carcinogen lists, RCRA waste codes) and the product families. For families, that means the UV/LED flag, the SDS profile (storage range, water-based setting, default physical state, typical pH) and the Recommended Use and Restrictions text in each language. Also keep the disclaimers per language, in Admin Settings and on each manufacturer record.
- Some text stays fixed because a regulation or an owner decision requires it: GHS hazard and precautionary wording (including the official 'Call a POISON CENTER/doctor.'), the prescribed concentration bands and 0.1% cut-off, flammability and transport thresholds, Appendix D headings, standard regulatory sentences, and the UV glove sentence, which is always nitrile.

How to read this document

The pages that follow are laid out like a generated Safety Data Sheet: the same sixteen sections in the same order, under the same banners, with the same field labels. Where the sheet would print a value, this document shows where that value comes from.

Each entry has a coloured tag naming the kind of source, a plain description (when several sources compete, they are listed as A > B > C, and the first one that has a value wins), an example of what prints, **You maintain** (the information a person keeps current) and **If missing** (exactly what prints, or what happens, when that information is blank). The small grey line is the technical reference for the system team.

FORMULA / RAW MATERIALS	Calculated from the product's formula and the data entered on its raw materials.	HAZARD ENGINE	GHS classification computed from the formula and the competent person determinations.
REGULATORY LIST	Looked up in a maintained regulatory list (Prop 65, HAP, SARA 313, TSCA, RCRA, carcinogens).	PRODUCT FAMILY	Comes from the product's family, assigned by rules or inherited from its raw materials.
TRANSPORT RULES	Derived by the DOT rules from flash point, boiling point and hazard classes.	COMPANY / MANUFACTURER	Company settings, or the private label manufacturer on private label sheets.
ADMIN SETTING	An administrator setting that applies to every sheet.	PRODUCT OVERRIDE	Text typed for one product and language in the SDS editor; replaces the automatic value.
PUBLISH RECORD	Stamped when a version is published (version number, effective date).	TEXT CHOSEN BY HAZARD	Standard wording, selected and combined according to the product's hazards.
FIXED TEXT (TRANSLATED)	Standard wording, the same on every sheet, in the sheet's language.	FIXED TEXT	The same on every sheet (labels, regulatory citations).

At a glance

Fields mapped in Sections 1-16	312
Printed fields: generated from real data	254 of 294 (86%)
Printed fields: uses fixed text	40 of 294 (14%)
From your product data	155 (50%)
Standard wording chosen by the product's hazards	80 (26%)
Admin settings and per-product edits	20 (6%)
Fixed text (layout, labels, regulatory phrases)	57 (18%)
Conditions that block publishing	8
Languages	English, Spanish, French, German (same sources; labels and standard wording translated)

SECTION 1: IDENTIFICATION

Section 1 combines the product's code and description from the CMS import (alias sheets: the alias's base code), use text from a per-product override, the product's own text or its active product family (resale sheets: a standard resale sentence), and the supplier block and logo from Admin > Settings or the private-label manufacturer record. All of it is set in fixed, translated labels and page furniture. Batch F (0c9207b, 072cd54) did not change any printed Section 1 line or page furniture. Only the operator-only preview Warnings box gained new messages.

Product Identifier:

FORMULA / RAW MATERIALS The product code and description from the CMS import. Alias sheets use the alias's base code (before the first hyphen) and its description; resale sheets use the RM base code and supplier product name; private label uses custom > shared alias > product.

On a sheet: Product Identifier: BK1080 — Process Black Offset Ink

You maintain: Product code and description (CMS import / Finished goods screen); supplier product name on resale raw materials; alias code and description (CMS import; the Aliases screen is read-only); private-label custom code/description when used

If missing: A blank description, or one equal to the code, prints the code alone ("BK1080"). A resale RM with no supplier name prints its base code alone. The line is never omitted.

`SDSGenerator::productIdentifier :898-906; section1 :1157; resale :494-500; createAliasVariant :867 + AliasResolver::stripPack; PrivateLabelPublisher::resolveIdentity :51`

Recommended Use:

PRODUCT FAMILY Per-product override > the product's own text > its active family's text in the sheet language > the translated standard sentence. Resale sheets skip the family and use the resale sentence; their override is stored per raw material.

On a sheet: Recommended Use: UV-curable offset ink for sheetfed presses.

You maintain: Recommended Use text in every language on each active Product Family, plus family rules (Recompute now after an import or a rule change). Leave the product's own Recommended Use blank unless it is product-specific.

If missing: "Printing ink for commercial and industrial applications." (resale: "Raw material for industrial formulation."), translated. A family with no text in the sheet language also gives this sentence. Never blank.

`SDSGenerator::resolveUseText :1183: override(1,recommended_use) > fg.recommended_use > active family recommended_use_json[lang] (not resale) > section1.recommended_use[_resale]`

Restrictions on Use:

PRODUCT FAMILY The same order as Recommended Use: per-product override > product's own text > active family text in the sheet language > standard sentence. Resale sheets skip the family.

On a sheet: Restrictions on Use: For professional/industrial use only. Not for household consumer use.

You maintain: Restrictions text in every language on each active Product Family. Leave the product's own Restrictions blank unless it is product-specific.

If missing: "For professional/industrial use only. Not for household consumer use." (the same for resale), translated. A family with no text in the sheet language also gives this sentence. Never blank.

`resolveUseText(..,'restrictions') :1183: override(1,restrictions) > fg.restrictions_on_use > active family restrictions_json[lang] (not resale) > section1.restrictions[_resale]`

Manufacturer / Supplier Information:

FIXED TEXT (TRANSLATED) A fixed translated sub-heading above the supplier block.

You maintain: Nothing

If missing: Always printed, even when every supplier line below it is blank

`labels.manufacturer_info via getLabels() :5347 -> meta.labels; PDFService::renderSection1 :385; preview.php:67`

Company:

COMPANY / MANUFACTURER The company name from Admin > Settings. On a private-label sheet, the manufacturer's name.

On a sheet: Company: Your Ink Company, Inc.

You maintain: Company Name in Admin > Settings > Manufacturer Information; Name on each private-label manufacturer (required)

If missing: A saved blank omits the line, and the FG and resale previews and SDS Review warn; publishing is not blocked. A never-saved key prints the config.php placeholder with a warning. A blank private-label name blocks publishing.

`settings company.name (config.php only if never saved) -> buildManufacturerBlock :946; PL manufacturers.name via Manufacturer::toCompanyInfo; PDFService:389`

Address:

COMPANY / MANUFACTURER Street, city, state and ZIP, and country from Admin > Settings (or the private-label manufacturer), joined with commas and skipping blanks.

On a sheet: Address: 123 Industrial Blvd, Suite 100, Anytown, OH 44000, US

You maintain: Address, City, State, ZIP, Country in Admin > Settings; the same fields on each manufacturer

If missing: Blank parts are skipped; a part never saved in Admin > Settings prints the config.php placeholder. If every part is blank, the line is omitted. No warning.

`formatManufacturerAddress (SDSGenerator:918-930): company.address/city/state/zip/country; PL manufacturers.address/city/state/zip/country; PDFService:390`

Phone:

COMPANY / MANUFACTURER The company phone from Admin > Settings, or the private-label manufacturer's phone.

On a sheet: Phone: (555) 123-4567

You maintain: Phone in Admin > Settings; phone on each manufacturer

If missing: A saved blank omits the line; a never-saved key prints the config.php placeholder. Either way the FG and resale previews and SDS Review warn (private label: its preview). Publishing is not blocked.

`company.phone (config if never saved) / manufacturers.phone -> buildManufacturerBlock :948; PDFService:391; SDSReadinessService::companySupplierWarnings :528`

Email:

COMPANY / MANUFACTURER The company email from Admin > Settings, or the private-label manufacturer's email.

On a sheet: Email: sds@yourinkcompany.com

You maintain: Email in Admin > Settings; email on each manufacturer

If missing: A saved blank omits the line; a never-saved key prints the config.php placeholder. No warning.

`company.email (config if never saved) / manufacturers.email -> buildManufacturerBlock :949; PDFService:393; preview.php:59`

Website:

COMPANY / MANUFACTURER The company website from Admin > Settings, or the private-label manufacturer's website.

On a sheet: Website: https://www.yourinkcompany.com

You maintain: Website in Admin > Settings; website on each manufacturer

If missing: A saved blank omits the line; a never-saved key prints the config.php placeholder. No warning.

`company.website (config if never saved) / manufacturers.website -> buildManufacturerBlock :950; PDFService:394; preview.php:60`

Emergency:

COMPANY / MANUFACTURER The Emergency Phone saved in Admin > Settings, with no config fallback; it prints last in the block. A private-label sheet prints only the manufacturer's own emergency number.

On a sheet: Emergency: CHEMTREC: (800) 424-9300

You maintain: Emergency Phone in Admin > Settings (required); emergency phone on every private-label manufacturer (required)

If missing: The line is omitted, every publish path for that sheet refuses, and the FG, resale and private-label previews show "Publishing blocked: ...".

`settings company.emergency_phone only (getCompanySettings :5113-5160) -> section1 :1165; PL trim(manufacturers.emergency_phone) :980; PDFService:395`

Preview Warnings box (operator only):

FIXED TEXT English operator messages above the preview, chosen by the data: Section 1 blocks and warnings (emergency phone, supplier name and phone, inactive product) plus generator warnings from other sections, now including partial STOT target organs, combustible dust not assessed and the family default physical state. Never on the customer PDF.

On a sheet: Publishing blocked: the company emergency phone number is blank or has never been saved.

You maintain: Nothing (clear the underlying data)

If missing: No box when there are no warnings

`preview-pdf.php:32-41; SDSController::preview :78-91, previewResale :142-149; createManufacturerVariant :981-995; SDSGenerator :273-366, familyPhysicalStateWarning :1243`

SECTION 2: HAZARD(S) IDENTIFICATION

Section 2 is built from the formula's hazard classification. The inputs are: ingredient hazard records (PubChem, attributed per class), competent-person determinations, trade-secret entries, mixture calculations, a Flammable Liquids class derived from the product flash point and boiling point, the optional per-product hazard override, the carcinogen list and, since batch F, the Combustible Dust flag on raw materials. The wording comes from fixed GHS phrase and label tables in the sheet language. Two things change that wording: target organs and route entered per CAS fill the H370-H373 statements, and on a UV/LED product family the Other Hazards line prints a polymerization sentence and the glove and eyewear lines print the UV sentences (nitrile gloves always; goggles plus face shield stay for H314/H318). The product family's default physical state now counts when a product has no state of its own. Acrylate flags do not change Section 2. Operator warnings (flash point, STOT organs, combustible dust, physical state) appear only in the preview Warnings box or on the readiness page. The missing-data publish block applies on every publish path.

Not a hazardous substance or mixture.:

TEXT CHOSEN BY HAZARD A fixed sentence that prints only when, after every hazard step, the product has no signal word, pictogram, hazard class (the Combustible Dust class counts) or H-statement. P-statements alone do not count.

On a sheet: Not a hazardous substance or mixture.

You maintain: Nothing for the wording. Whether it prints depends on the hazard data listed under Signal word.

If missing: Snapshots stored without is_classified are judged on their own stored signal word, pictograms, classes and H-statements, so the sentence never appears above printed hazards.

`section2.not_classified (en.php:17); SDSGenerator::isClassified :1474-1480, section2IsClassified :1492-1498; PDFService.php:413-416; preview.php:73-75`

Signal word (DANGER / WARNING):

HAZARD ENGINE The most severe signal word of the classes that trigger: the GHS default for ingredient, mixture-rule and flash-point classes, and the word entered on determinations and trade-secret entries. Precedence: replace-mode hazard override > most severe of (computed, additive override, carcinogen-list rule) > powder carbon-black/TiO2 rule, then the combustible-dust rule (each sets Warning only when no word is set yet).

On a sheet: DANGER (red) or WARNING (orange)

You maintain: Hazard data per ingredient CAS (Raw materials, hazard refresh) or a determination (CAS determinations). On Raw materials: trade-secret hazards, flash point, boiling point, physical state and Combustible Dust. Hazard override and physical state (Finished goods); family default state (Product families). Carcinogen list (Regulatory lists).

If missing: No word: the line is omitted. Carbon black/TiO2 adds no WARNING when any raw material is not Solid/Powder (a blank state counts). A Warning left only by acute toxicity Category 5 is removed. Combustible dust adds nothing in replace mode.

`HazardEngine::classify :669-1234, applyFinishedGoodOverride :2064-2072; SDSGenerator :5806-5811, :6224-6227, applyCombustibleDust :5657-5683; PDF :419-426`

Pictograms:

FIXED TEXT (TRANSLATED) A fixed label in the sheet language, printed only when the product has pictograms.

On a sheet: Pictograms:

You maintain: Nothing

If missing: Fallback order: label stored with the sheet > sheet-language translation > the key. Not printed when there are no pictograms.

`labels.pictograms (getLabels SDSGenerator.php:5347; en.php:673); PDFService::renderPictogramRow :630-697; preview.php:84`

GHS pictogram images (GHS01-GHS09):

HAZARD ENGINE The pictograms of each triggered class, determination, trade-secret entry, mixture rule, the flash-point Flammable Liquids class (GHS02) and the hazard override. GHS07 is dropped when GHS06 or GHS05 is present, the rest are ordered GHS01, 05, 06, 02, 04, 03, 08, 07, 09, and GHS08 added by the carcinogen-list or powder carbon-black rule prints last (Combustible Dust adds none).

On a sheet: GHS02 flame + GHS07 exclamation mark diamonds

You maintain: The same hazard data as the signal word. Optional replacement artwork on Admin > Pictograms.

If missing: No pictograms: the block is omitted. In the PDF a code with no artwork is skipped; if no code has artwork, text such as 'GHS02 (Flame), ...' prints instead. The preview shows the code as a badge.

`contributionForRow :367; applyFlammableLiquidsFromFlashPoint :2868; applyPictogramPrecedence :2990-3009; SDSGenerator :5806, :6219; PictogramHelper::getPngPath :61-79`

Pictogram caption under each image:

FIXED TEXT (TRANSLATED) The fixed GHS pictogram name for each printed code, in the sheet language.

On a sheet: Flame / Exclamation Mark

You maintain: Nothing

If missing: Falls back to the English name, then to the code itself, e.g. 'GHS02'.

`GHSStatements::pictogramName :345-358 (ghs_{lang}.pictogram_names > PICTOGRAM_NAMES > code); PDFService::renderPictogramRow :630-697; preview.php:92, 97`

GHS Classification:

FIXED TEXT (TRANSLATED) A fixed heading, printed only when at least one hazard class exists.

On a sheet: GHS Classification:

You maintain: Nothing

If missing: Falls back to the sheet-language translation, then the key. Not printed when there is no class.

`labels.ghs_classification (en.php:674); PDFService.php:479; preview.php:104`

Physical Hazards: / Health Hazards: / Environmental Hazards:

FIXED TEXT (TRANSLATED) Three fixed headings, all printed whenever any class exists; each class goes under one by its English class name (standard GHS names, plus 'Combustible Dust' under Physical Hazards). Legacy free-text determination classes keep their typed name, and unknown names go under Health Hazards.

On a sheet: Physical Hazards:

You maintain: Nothing for the headings. A class typed into the hazard override must use the standard GHS name, spelled exactly, to land in the right group.

If missing: Falls back to the sheet-language translation. An unrecognised class name (e.g. 'Unclassified', a free-text determination class or a non-standard override class) is listed under Health Hazards.

`labels.physical_hazards/health_hazards/environmental_hazards (en.php:675-677); HazardEngine::groupByHazardType :3603-3616, HAZARD_GROUP_ORDER :3356; PDF :481-505`

None (under an empty hazard group):

FIXED TEXT (TRANSLATED) A fixed word in the sheet language, printed under any group that has no classes.

On a sheet: None

You maintain: Nothing

If missing: Falls back to the sheet-language translation, then the key.

`labels.none (en.php:839); PDFService.php:504-505; preview.php:150-151`

Hazard class lines (H-code — Class (Category)):

HAZARD ENGINE One line per standard hazard class (most severe category kept) from triggered ingredient records, determinations, trade-secret entries, mixture calculations, the flash-point class and the hazard override, plus carcinogen-list and powder carbon-black lines added afterwards. Each line carries the GHS default H-code for its class and category and lines sort by that code within each group.

On a sheet: H319 — Serious Eye Damage/Eye Irritation (Category 2A): Causes serious eye irritation

You maintain: Ingredient hazard data and determinations (Raw materials, CAS determinations); constituent CAS and %; carcinogen list (Regulatory lists); hazard override classes (Finished goods).

If missing: No classes: the block is omitted. Unrecognised class or category: no H-code prefix (its codes print under Hazard Statements). Override class without its H-code: code prefix, no phrase. Acute toxicity Category 5 never prints.

`classify :669-1234; HazardRowNormalizer::expandRows :301, entryFor; h_codes stamped :1148-1160; consolidateHazardClasses :3478; dropUnadoptedAcuteCategories :2515; PDF :440-540`

Flammable Liquids class line:

HAZARD ENGINE Derived from the product flash point (wt%-weighted average over raws that have one; a '>' value makes the product '> n') and the lowest raw boiling point: below 23 °C Cat 1 (boiling point ≤35 °C) or Cat 2, 23-60 °C Cat 3, above 60 to 93 °C Cat 4. Ingredient H224-H227 never classify the mixture, a Solid, Powder or Paste product (own state > family default > largest raw) gets no line, and override > flash point.

On a sheet: H226 — Flammable Liquids (Category 3): Flammable liquid and vapour

You maintain: On every raw material (Raw materials): flash point in °C (with the '>' flag) and initial boiling point; a high flash point for water and similar raws. Physical state on the finished good, else the family default (Product families > SDS profile), else the raw materials.

If missing: No raw with a flash point, a product flash point above 93 °C, or a Solid/Powder/Paste product: no line. Flash point below 23 °C with no boiling point: Category 2 (H225), with a preview warning.

`FormulaCalcService::weightedFlashPoint :693, lowest BP :551-557; HazardEngine::flammableLiquidCategory :2781, applyFlammableLiquidsFromFlashPoint :2868; SDSGenerator::flammabilityInputs :4546`

Combustible Dust: May form combustible dust concentrations in air.:

FORMULA / RAW MATERIALS Printed last under Physical Hazards when the product's resolved physical state is Powder or Solid and any raw material in the formula (any depth, above 0%) is marked Combustible Dust = Yes, unless the hazard override is in replace mode. The class name and statement are fixed sheet-language text with no H-code, pictogram or P-codes, and the signal word becomes Warning if none is set.

On a sheet: Combustible Dust: May form combustible dust concentrations in air.

You maintain: Combustible Dust (Not assessed / Yes / No) on each powder or solid raw material, from the vendor SDS Section 2 (Raw materials > Edit). Physical state on the finished good or its product family.

If missing: No line when no raw is marked Yes, the product is not Powder/Solid, or the override is in replace mode. Unassessed powder/solid raws only raise a preview and readiness warning.

`SDSGenerator::combustibleDustStatus :5600, applyCombustibleDust :5657-5683, localizeCombustibleDustClass :5709; section2.combustible_dust_* (en.php:19-20); PDF :522-524`

Unclassified (Health Hazards line):

HAZARD ENGINE A stored hazard row named 'Unclassified' whose H-codes match no GHS class. At 1% or more it prints under Health Hazards with no H-code; its codes go to the Hazard Statements list, and its stored P-codes, pictograms and signal word are used unfiltered.

On a sheet: Unclassified

You maintain: Fix the ingredient's hazard data with a determination (CAS determinations), or refresh PubChem.

If missing: Not printed when its H-codes map to GHS classes (those classes print instead) or when the CAS is below 1%.

`HazardRowNormalizer::rowsForStorage :488, expandRows :301-352, contributionForRow :373-376 (no class); HazardEngine::getCutoff 1% default :1390-1397; groupByHazardType`

H-statement phrase (text after the colon):	REGULATORY LIST The standard GHS wording for the line's H-code in the sheet language; the wording on the source record is always replaced. Precedence: H370-H373 text with target organs (next row) > sheet-language GHS phrase > English phrase. *On a sheet: Causes serious eye irritation* **You maintain:** Nothing (fixed phrase table). **If missing:** Code missing from the sheet-language table: English wording. Missing from both, or the line's code not among the product's H-statements: the line ends at '(Category n)'. `GHSStatements::hText :309-322 (ghs_{lang}.h_statements > H_STATEMENTS), resolveHStatements :410; translateHazardResult :447; SDSGenerator::applyStotTargetOrgans :1446`
H370-H373 statement with target organs and route:	HAZARD ENGINE For H370-H373, the organs entered for every CAS behind the code are listed in brackets and, when all those CAS give the same single route, 'by inhalation / ingestion / skin contact' is added, in the sheet language. It is filled only when every CAS behind the code has organs and no hazard override or trade-secret entry also carries it, and it also prints in the Hazard Statements list. *On a sheet: Causes damage to organs (lungs) through prolonged or repeated exposure by inhalation* **You maintain:** Target organs and route for single exposure (H370/H371) and repeated exposure (H372/H373) per CAS, on CAS determinations > STOT Target Organs, from the vendor SDS or EU CLP Annex VI. **If missing:** The standard phrase, e.g. 'Causes damage to organs through prolonged or repeated exposure'. Route 'All routes' or differing routes: no route words. Organs on only some sources: standard phrase plus a preview warning. `StotTargetOrgans::resolve :163-224, statementText :227-235; HazardEngine stot_details :1125-1137; SDSGenerator::applyStotTargetOrgans :1446-1463; stot.* (en.php:27-61)`
Hazard Statements: (H-codes without a class line):	HAZARD ENGINE Lists each H-statement that no class line above carries (e.g. override H-codes entered without a class, codes of an unrecognised class), worded from the fixed GHS table or the organ-filled H370-H373 text. Codes OSHA does not adopt (H303, H313, H333, H316, H305) are never listed. *On a sheet: Hazard Statements: H319: Causes serious eye irritation* **You maintain:** Nothing beyond the hazard data and the hazard override H-codes (Finished goods). **If missing:** When every H-code is on a class line, the heading and list are omitted. A code not in the phrase table prints alone. `labels.hazard_statements (en.php:679); GHSStatements::uncoveredHStatements :551-583, NOT_ADOPTED_H_CODES :496; PDFService.php:546-562; preview.php:176-183`
Precautionary Statements:	FIXED TEXT (TRANSLATED) A fixed heading, printed only when P-statements exist. *On a sheet: Precautionary Statements:* **You maintain:** Nothing **If missing:** Falls back to the sheet-language translation. Not printed when there are no P-statements. `labels.precautionary_statements (en.php:678); PDFService.php:569; preview.php:189`
Precautionary statement lines:	HAZARD ENGINE Each triggered ingredient class gives its GHS default P-codes plus the record's own codes for that class; determinations and trade-secret entries give the P-codes entered on them, and mixture rules, the flash-point class and the override add theirs, sorted by code. Withdrawn P281 becomes P280, and the carcinogen-list and powder carbon-black rules append their class defaults at the end (Combustible Dust adds none). *On a sheet: P210: Keep away from heat, hot surfaces, sparks, open flames and other ignition sources.* **You maintain:** The same hazard data as the class lines; P-codes on determinations and trade-secret entries; P-codes in the hazard override if used (Finished goods). **If missing:** No P-codes: the block is omitted. A code not in the phrase table prints alone. `contributionForRow :367; parseDeterminationStructure :3063-3066, :3269-3271; classify :1165 (replaceWithdrawnPCodes); appendClassDefaultPStatements SDSGenerator :6069; PDF :567-587`
Recommended Personal Protective Equipment (PPE):	FIXED TEXT (TRANSLATED) A fixed heading, printed only when at least one PPE field is driven by the product's H-codes. *On a sheet: Recommended Personal Protective Equipment (PPE):* **You maintain:** Nothing **If missing:** Falls back to the sheet-language translation. Not printed when no field is hazard-driven. `labels.ppe_recommendations (en.php:680); gate resolvePPE hazard_driven SDSGenerator.php:1507-1511; PDFService.php:591-597; preview.php:197-201`

PPE pictogram images (eye, hand, respiratory, skin):	TEXT CHOSEN BY HAZARD One icon for each PPE field whose level is set by the H-codes: for example H319 gives eye protection, H315 gloves and H335 a respirator. P-codes play no part, and H-codes Section 2 does not list (e.g. the non-adopted H316) still count. *On a sheet: Blue eye-protection and glove icons* **You maintain:** Ingredient hazard data that yields H-codes. Optional artwork on Admin > Pictograms. **If missing:** No hazard-driven field: no icons. Missing artwork: the PDF drops that icon and its caption; the preview keeps the caption without an image. `HazardEngine::derivePPE :1298-1377 via SDSGenerator::resolvePPE :2936-2966; PictogramHelper PPE-*.png; PDFService::renderPPEPictogramRow :698`
PPE icon captions:	FIXED TEXT (TRANSLATED) A fixed caption for each active PPE icon, in the sheet language. *On a sheet: Wear Eye Protection / Wear Gloves* **You maintain:** Nothing **If missing:** Falls back to the sheet-language translation. `labels.ppe_wear_eye/_gloves/_respiratory/_skin (en.php:682-685); PDFService::renderPPEPictogramRow :698; preview.php:203-218`
Respiratory: / Hand Protection: / Eye Protection: / Skin/Body Protection:	FIXED TEXT (TRANSLATED) Fixed sub-labels, printed before each hazard-driven PPE sentence. *On a sheet: Hand Protection:* **You maintain:** Nothing **If missing:** The line is skipped when its sentence is empty. `labels.respiratory/hand_protection/eye_protection/skin_body (en.php:724-728); PDFService.php:603-611; preview.php:226-235`
PPE recommendation sentences:	TEXT CHOSEN BY HAZARD Section 8 per-product override (if not blank) > on UV rule-pack products where the field is hazard-driven or H317 is present: the UV nitrile glove sentence alone (even on corrosive inks), the UV eyewear sentence alone (H314/H318 keep the goggles-and-face-shield sentence), and the respiratory and clothing level sentences with the UV sentence appended > the translated sentence for the H-code level. Printed in Section 2 only for hazard-driven fields, always the same text as Section 8. *On a sheet: For uncured acrylate product, wear chemical-resistant gloves (nitrile, minimum 0.4 mm …* **You maintain:** Ingredient hazard data (H-codes). Optional Section 8 PPE text per product and language (SDS editor). UV/LED flag on the product family (Product families) and the uv_acrylate_rule_pack setting. **If missing:** Baseline level (general/none): not printed in Section 2 (Section 8 prints it). A Section 8 override replaces the whole sentence, UV text included. UV pack off or non-UV family: the level sentence only. `resolvePPE SDSGenerator.php:2936-2966, UV_PPE_REPLACE_FIELDS/UV_PPE_TIER_FLOOR :2865-2878; section8.ppe.<field>.<tier> (en.php:246-273); section8.uv_* (en.php:239-242)`
Other Hazards:	FIXED TEXT (TRANSLATED) A fixed label, printed on every sheet with the Other Hazards text. *On a sheet: Other Hazards:* **You maintain:** Nothing **If missing:** Omitted only when a stored snapshot carries blank Other Hazards text; a newly generated sheet always has text (blank override rows are ignored and a default always applies). `labels.other_hazards (en.php:681); PDFService.php:619-623; preview.php:242-245`
Other Hazards text:	PRODUCT OVERRIDE Precedence: per-product override typed in the SDS editor (per language; per raw material on resale sheets) > on a UV/LED product family, the fixed sentence 'Uncured product may polymerize when exposed to UV light, elevated temperatures or contaminants.' > 'None known.'. The UV sentence follows the family flag alone, even with the UV rule pack switched off. *On a sheet: Uncured product may polymerize when exposed to UV light, elevated temperatures…* **You maintain:** Optional override in the SDS editor, per product (or resale raw material) and language, for hazards not otherwise classified. UV/LED flag on the product family (Product families). **If missing:** 'None known.' in the sheet language, or the polymerization sentence on a UV/LED family. A stored blank row is ignored and the default prints. `TextOverrideService::effective :144 (blank rows dropped) > section2.other_hazards_uv / other_hazards (en.php:15-16); SDSGenerator.php:1513, :1529; getOverrides :5208`
Flash point warnings (preview Warnings box only; not printed):	HAZARD ENGINE Shown when any of these applies: flash point below 23 °C with no boiling point; an override Flammable Liquids category that disagrees with the formula flash point; a Section 9 flash or boiling point edit dropped for contradicting the class; or a raw material with no flash point carrying a disclosed ingredient (1% or more) classified as a flammable liquid. *On a sheet: Flash point warning: raw material(s) R1234 contain an ingredient classified as a flammable…* **You maintain:** Flash points and boiling points on raw materials (Raw materials). Section 9 Flash Point edit (SDS editor) when an override reflects a tested flash point. **If missing:** Nothing to warn about: no message. Never printed, not shown in publish messages, and never blocks publishing. `SDSGenerator::flammabilityWarnings :4692, vetFlashPointEdit :4616, vetBoilingPointEdit :4658; sds['warnings'] shown in preview.php:670, preview-pdf.php:32`

STOT target organs warning (preview Warnings box only; not printed):

HAZARD ENGINE Shown for each of H370-H373 when some CAS behind the code have target organs but another CAS, a hazard override or a trade-secret entry carries the same code without them. The sheet then prints the standard phrase in Sections 2, 4 and 11.

On a sheet: Section 2 warning: H372 comes from CAS 14808-60-7 (target organs entered) and also from…

You maintain: Target organs for every listed CAS (CAS determinations > STOT Target Organs).

If missing: No organs entered for any source, or organs for all of them: no message. Never printed and never blocks publishing.

`StotTargetOrgans::warnings :249-271 (stot_incomplete from resolve :203-206); SDSGenerator.php:357-359, :779-781; preview.php:670`

Combustible dust warning (preview Warnings box and readiness page only; not printed):

FORMULA / RAW MATERIALS Shown when the product is Powder or Solid, no raw material is marked Combustible Dust = Yes, and at least one raw whose own state is Powder or Solid is still 'Not assessed'. It lists those raw material codes.

On a sheet: Combustible dust: this product's physical state is Powder and its formula contains a powder…

You maintain: Combustible Dust on each listed raw material (Raw materials > Edit).

If missing: Nothing unassessed, or a raw already marked Yes: no message. Never printed and never blocks publishing.

`SDSReadinessService::combustibleDustNotAssessedWarning :684, ...ForFinishedGood / ...ForRawMaterial; SDSGenerator.php:363-366, :785-788; sds-review/index.php:98-101`

SECTION 3: COMPOSITION / INFORMATION ON INGREDIENTS

Section 3 is built from the formula. Each raw material's ingredients (CAS number, name and percentage) are combined across the formula and any nested finished goods. The hazard engine, the hazard-data records and the competent-person determinations then decide which ingredients are listed and which H-codes they carry, and the rows print in a fixed table whose headings and notes come from the translation files. Every concentration prints as a prescribed band. A trade-secret identity is replaced by its description. A Substance sheet always names its main constituent. Batch F did not change any Section 3 wording; it changed two inputs: the product family's default physical state now decides whether a product without its own Physical State is a liquid (and so whether flammable solvents are listed), and a Powder/Solid product classified only as OSHA Combustible Dust counts as classified, so when no ingredient is listed it prints the classified-but-no-ingredients note instead of the no-hazardous-ingredients note. Operator-only warnings appear in the preview Warnings box and are never printed. A published sheet keeps a frozen copy of this section until a new version is published.

Type:

FIXED TEXT (TRANSLATED) A fixed label from the language file. The layout adds the colon.

On a sheet: Type:

You maintain: Nothing

If missing: The order is: snapshot label > sheet-language translation > English > the literal key "type".

`labels.type (en.php:688) via SDSGenerator::getLabels:5347; PDFService::renderSection3:763 -> label:1349 + labelValue:1369; preview.php:248`

Type (value: Mixture / Substance):

FORMULA / RAW MATERIALS The Substance/Mixture setting on the finished good. On Auto it is Substance only when the formula is one raw-material line marked Substance, otherwise Mixture; resale sheets use the raw material's own setting.

On a sheet: Type: Mixture

You maintain: Substance / Mixture on the finished-good form, and on the raw-material form for single-substance raws.

If missing: "Mixture" prints.

`finished_goods.substance_mixture > SubstanceMixtureResolver::resolveForFinishedGood:84-100 (1 RM line) > 'mixture' (SDSGenerator:146); resale resolveForRawMaterial:70 (:496); labels.substance/mixture…`

Hazardous-only disclosure note:

FIXED TEXT (TRANSLATED) A fixed sentence from the language file. It prints above the table only on Mixture sheets that list at least one row, never on Substance sheets.

On a sheet: Hazardous ingredients and ingredients with an occupational exposure limit are listed. ...

You maintain: Nothing

If missing: It does not print when no row is listed or when the sheet is a Substance. Older snapshots without the mixture_notes key are treated as Mixture.

`labels.hazardous_only_note (en.php:692); sections[3].mixture_notes (SDSGenerator:1708); PDFService.php:765-771; preview.php:250-251`

CAS Number | Chemical Name | Concentration | H-Codes (table header):

FIXED TEXT (TRANSLATED) Fixed column headings from the language file, in bold on a grey band, printed whenever the table prints.

On a sheet: CAS Number | Chemical Name | Concentration | H-Codes

You maintain: Nothing

If missing: The order is: snapshot label > sheet-language translation > English > the literal key. The table prints only when rows exist.

`labels.cas_number/chemical_name/concentration/h_codes (en.php:689-691,696); PDFService.php:773-781; preview.php:254-258`

CAS Number (ingredient row):

FORMULA / RAW MATERIALS The CAS number on each raw-material ingredient, combined across every formula line and every nested finished-good component.

On a sheet: 108-88-3

You maintain: A CAS number on every ingredient of every raw material (Raw materials > constituents), and a current formula on every finished good used as a component.

If missing: An ingredient with a blank CAS gets no row of its own and the preview shows a Composition warning (a CAS-less trade secret with H-codes feeds the synthetic TRADE SECRET row instead). A finished-good component with no current formula adds nothing; only the readiness page warns. A CAS ticked trade secret on any raw material prints "TRADE SECRET".

`raw_material_constituents.cas_number via Formula::getExpandedComposition:280 (resale: FormulaCalcService::buildResaleComposition:199); SDSGenerator::section3:1611-1679`

Chemical Name (ingredient row):

FORMULA / RAW MATERIALS Prop 65 list name > CAS master preferred name > name typed on the raw-material ingredient. Resale sheets use the name stored on the ingredient, which is replaced by the canonical CAS name when the raw material is saved.

On a sheet: Toluene

You maintain: The ingredient name on the raw material, and the Prop 65 list and CAS master names (Regulatory lists).

If missing: The name as typed on the ingredient prints, or the cell is blank if every source is blank. It is never translated.

`COALESCE(NULLIF(p65.chemical_name,''), NULLIF(cm.preferred_name,''), rmc.chemical_name) Formula.php:293; resale rmc.chemical_name; RawMaterial::canonicalCasName:507`

Concentration (ingredient row):

FORMULA / RAW MATERIALS Each contribution's lower and upper bound (line % x ingredient %, an exact % counting as both) is summed per CAS, and the widest prescribed band containing [min, max] prints. If none contains it, the band containing the max with the lowest lower end prints.

On a sheet: 10 - 30%

You maintain: Formula line percentages (Finished goods > formula), and an exact % or a min + max % on each raw-material ingredient.

If missing: An ingredient with no % counts as 0 and is dropped. Max only counts as [0, max], min only as [min, min]. A max above 100 % prints "80 - 100%". "<0.1%" prints only on a Substance identity row whose max is below 0.1 %.

`SDSGenerator::formatConcentration:5292 over PRESCRIBED_RANGES:5278; Formula::constituentBounds:821 + mergeSubEntryExtras:853; concentration_min/max (Formula.php:376-379)`

H-Codes (ingredient row):

HAZARD ENGINE The H-codes credited to that CAS before the engine reduces to one entry per class: its own classes (federal data > competent-person determination), mixture-rule contributions, carcinogen-list and powder H351 entries, and its own H224-H227.

On a sheet: H225, H304, H315, H336

You maintain: Federal hazard data for each CAS, or an active competent-person determination (CAS determinations). Also the carcinogen list and the physical state on each raw material.

If missing: The cell is blank. A replace-mode finished-good hazard override blanks every cell except a flammable solvent's own H224-H227, and the rows stay listed. Carbon black and TiO2 keep only the codes left after the wet-mixture rule.

`TransportClassifier::casHCodeMap:452 over hazardResult.cas_h_codes (HazardEngine::buildCasHCodeMap:3408) + hazard_classes; SDSGenerator::section3:1560-1575,1678`

Which ingredients are listed (rows):

HAZARD ENGINE An ingredient is listed when it has a CAS, is at least 0.1 % and has an exposure limit or a hazard class that meets its cut-off or feeds a mixture rule; a flammable solvent counts only when it is at least 1 % and the product flash point makes the product a flammable liquid (state: product Physical State > family default > largest raw material; Paste, Solid and Powder never qualify). An inhalation-only CAS with no remaining H-code and no exposure limit is dropped.

On a sheet: Toluene listed; water and talc in a liquid not listed

You maintain: Hazard data or a CPD for every CAS, exposure limits, the physical state on raw materials, the product Physical State or the family default physical state (Product families), and the inhalation-only CAS list (Settings).

If missing: A CAS with no hazard data, no CPD and no exposure limit is left out; publishing is blocked only at or above the missing-data threshold (default 1.0 %). An ingredient whose only hazard is Acute Tox. Cat 5 is not listed unless it has an exposure limit.

`SDSGenerator::section3:1624-1641 (hazardous_cas, exposure_limits, getInhalationOnlyCas:5489); flammabilityInputs:4546; HazardEngine::flammableLiquidCategory:2781 (NOT_LIQUID_STATES:472), recordFlamma…`

Substance identity row:

FORMULA / RAW MATERIALS On a Substance sheet, the CAS with the largest percentage is always listed, even when it is unclassified, has no exposure limit or is below 0.1 %. Other constituents follow the normal rules.

On a sheet: 64742-47-8 | Distillates (petroleum), hydrotreated light | 80 - 100%

You maintain: Substance / Mixture setting, and the CAS constituent on the raw material.

If missing: With no CAS constituent, the table and notes do not print, and the preview shows the Section 3 Substance warning.

`SDSGenerator::section3:1597-1609 ($identityCas), :1614-1641; formatConcentration:5292`

TRADE SECRET (trade-secret row, CAS cell):

FIXED TEXT (TRANSLATED) A fixed word from the language file, printed in place of the CAS number only on rows whose ingredient is a trade secret.

On a sheet: TRADE SECRET

You maintain: Nothing. The row appears when a listed ingredient (hazardous or with an exposure limit, at least 0.1 %) is ticked trade secret, or when a raw material marked hazardous with no CAS declares hazards.

If missing: The order is: sheet-language translation > English > the literal key.

`labels.trade_secret_cas (en.php:836); SDSGenerator::section3:1648; PDFService.php:787-809 (CAS cell height measured); preview.php:261+`

Chemical Name (trade-secret row):

FORMULA / RAW MATERIALS The trade-secret description on the raw-material ingredient > the translated words "Trade Secret"; the real chemical name is never used. There is one row per distinct description, so a blank description and the synthetic row can print as two rows both named "Trade Secret".

On a sheet: Proprietary Resin Blend

You maintain: The trade-secret description on each trade-secret ingredient (Raw materials > constituents; pick-list in Settings).

If missing: "Trade Secret" (translated).

`SDSGenerator::tradeSecretName:3072-3079 (rmc.trade_secret_description; 'Trade Secret' sentinel -> labels.trade_secret en.php:835); bucket key section3:1645-1655; Formula.php:424,487-489`

Concentration (trade-secret row):

FORMULA / RAW MATERIALS The summed lower and upper bounds of trade-secret rows sharing one description, banded like normal rows; each trade-secret CAS must reach 0.1 % on its own. CAS-less trade-secret ingredients and hazardous-with-no-CAS raw materials (whole line %, exact) are first summed into one synthetic row, which must reach 0.1 % as a whole.

On a sheet: 1 - 5%

You maintain: Formula line percentages and ingredient percentages on the raw material.

If missing: Same as normal rows. Several trade-secret CAS that are each below 0.1 % are dropped, even when together they reach 0.1 %.

`SDSGenerator::section3:1656-1662,1683-1690 then formatConcentration:5292; TRADE_SECRET bucket bounds Formula.php:415-437, :465-487`

H-Codes (trade-secret row):

HAZARD ENGINE A masked row with a real CAS shows that CAS's own H-codes, combined across every CAS sharing the description. The synthetic trade-secret row shows the codes of the declared trade-secret classes that survive the one-entry-per-class reduction.

On a sheet: H317, H319

You maintain: Manual hazard classification on raw materials marked hazardous with no CAS, and trade-secret H-codes on CAS-less trade-secret ingredients (Raw materials).

If missing: The cell is blank, including when another ingredient is equally or more severe in every declared class or a replace-mode override applies. Declared H224-H227 never print on the synthetic row; a masked real CAS shows its own H224-H227.

`SDSGenerator::section3:1577-1587,1663-1669; final hazard_classes cas='TRADE_SECRET' (HazardEngine::consolidateHazardClasses:3478); raw_materials.manual_hazard_json; rmc.trade_secret_h_codes`

Trade-secret identity masking:

FORMULA / RAW MATERIALS A CAS ticked trade secret on any raw material in the formula, including inside a nested finished good, is withheld on the whole sheet. Sections 8, 11, 12 and 15 print the same description and TRADE SECRET label, and Section 13 names only the description.

On a sheet: 25068-38-6 on one RM prints TRADE SECRET / Proprietary Resin Blend

You maintain: The trade-secret checkbox on each raw-material constituent, kept consistent for the same CAS on every raw material.

If missing: Unticked on every raw material: the real CAS and name print.

`Formula.php:381-385 + sub-FG merge :593-598; SDSGenerator::maskTradeSecretRow:3092; section3:1644; section13 :4380-4392`

Row order:

FORMULA / RAW MATERIALS Rows are sorted by calculated percentage, highest first. A ranged ingredient sorts by its range maximum, and trade-secret rows by their summed percentage.

You maintain: Formula and ingredient percentages.

If missing: Not applicable.

`usort on components[].concentration_pct desc (SDSGenerator:1694); Formula::resolveConstituentPct:791 (exact > max > min > max-only), resale same (#38)`

No-hazardous-ingredients note:	FIXED TEXT (TRANSLATED) A fixed sentence from the language file. It prints instead of the table only on an unclassified Mixture sheet that lists no ingredient. *On a sheet: No hazardous ingredients above disclosure thresholds.* **You maintain:** Nothing **If missing:** It shows only when no row is listed and Section 2 is unclassified (a Powder/Solid product with a raw material marked combustible dust Yes counts as classified, unless a replace-mode hazard override applies). It never prints on Substance sheets. `sections[3].empty_note = labels.no_hazardous_note (en.php:693) when !isClassified (SDSGenerator:1474,1712-1716); PDFService.php:813-818; preview.php:271-272`
Classified-but-no-ingredients note:	FIXED TEXT (TRANSLATED) A fixed sentence from the language file. It prints instead of the table on a Mixture sheet when Section 2 is classified (including by OSHA Combustible Dust alone) but no single ingredient qualifies for listing. *On a sheet: No ingredient is listed: no single ingredient meets the disclosure thresholds. ...* **You maintain:** Nothing; the combustible-dust case follows the raw-material combustible-dust setting and the product physical state. **If missing:** Older snapshots without the key print the no-hazardous-ingredients note. It never prints on Substance sheets. `section3.classified_no_ingredients_note (en.php:67) when isClassified(hazardResult) (SDSGenerator:1474,1712-1716); applyCombustibleDust:5657 (cas MIXTURE); PDFService.php:813-818`
Trade-secret / withheld-percentage statement:	FIXED TEXT (TRANSLATED) A fixed sentence from the language file, citing 29 CFR 1910.1200(i). It prints below the table whenever any row is listed, because every row shows a band. *On a sheet: Specific chemical identity and/or exact percentage ... withheld as a trade secret ...* **You maintain:** Nothing **If missing:** It does not print when no row is listed. `sections[3].trade_secret_note = section3.trade_secret_note (en.php:65) when components non-empty (SDSGenerator:1724-1726); PDFService:820-825; preview.php:274-275`
Section 3 warning: Substance with no CAS constituent (preview only):	FORMULA / RAW MATERIALS An operator warning in the preview Warnings box and the publish flash, shown when a Substance sheet has no CAS constituent to name. It is never printed. *On a sheet: Section 3 warning: this sheet is typed Substance but no constituent with a CAS ...* **You maintain:** The CAS constituent on the raw material, or set the product to Mixture. **If missing:** No warning when the substance has a CAS. Publishing is not blocked. `sections[3].identity_missing (SDSGenerator:1719) -> warnings[] :319-322 (base/resale path :741-744); key unset before snapshot`
Trade-secret warning: CAS secret on one raw material, disclosed on another (preview only):	FORMULA / RAW MATERIALS An operator warning naming the raw materials that disagree about one CAS. The sheet withholds the identity anyway. *On a sheet: Trade-secret warning: CAS 25068-38-6 is marked as a trade secret on raw material(s) ...* **You maintain:** The trade-secret checkbox, made consistent on every raw material that supplies the CAS. **If missing:** No warning when the raw materials agree. Publishing is not blocked. `SDSGenerator::tradeSecretDisclosureWarnings:6262 over contributing_materials[].is_trade_secret; warnings[] :336-338, :758-760`
Composition warning: incomplete constituent data (preview only):	FORMULA / RAW MATERIALS An operator warning listing formula raw materials (including inside nested finished goods) with no constituents, a constituent with no CAS, a CAS-less trade secret with no H-codes, or a constituent with no percentage. Raw materials marked hazardous with no CAS are not checked. *On a sheet: Composition warning: 1 raw material has incomplete constituent data — RM-123 (...)* **You maintain:** Every constituent with its CAS number and percentage (Raw materials > constituents). **If missing:** No warning when every constituent is complete. Publishing is not blocked. `SDSReadinessService::incompleteCompositionWarning:597 over TSCAService::incompleteRawMaterials:275; SDSGenerator:303-306, :725-728`
Trade-secret Prop 65 block message (preview and publish):	REGULATORY LIST Shown when a trade-secret constituent's CAS is on the Prop 65 list. The message appears in the preview Warnings box and every publish path refuses. *On a sheet: Publishing blocked: a trade-secret constituent is on the California Prop 65 list: ...* **You maintain:** The trade-secret checkbox and CAS on the raw material, or have the vendor disclose the chemical. **If missing:** No message when no trade-secret CAS is on the Prop 65 list. `Prop65Service trade_secret_conflicts -> SDSReadinessService::tradeSecretProp65Error:386; SDSGenerator:339-342, :761-764`

SECTION 4: FIRST-AID MEASURES

Section 4 is built almost entirely from the product's final GHS health hazard codes. Each route paragraph, the symptoms line and the notes to physician are put together from translated sentences chosen by those codes. Notes to Physician now also adds the official GHS "If exposed or concerned" first-aid phrases (P308+P313 / P308+P311) for CMR and STOT codes. Where target organs are entered for every CAS behind an H370-H373 code, the symptoms line names those organs and the route. Three other things can change Section 4. An optional per-product, per-language text override replaces a whole field, and resale raw materials have their own overrides. On UV-family products an uncured-product sentence is added to Skin Contact, naming the acrylates picked out by the CAS acrylate flag. The symptoms line is also reused in Section 11. Only the banner and the field labels are the same on every sheet.

Inhalation:

TEXT CHOSEN BY HAZARD Per-product override > sentences picked by hazard codes. H330 or H331 replaces the base paragraph, otherwise H332 adds a sentence; H334, H335 and H336 each add a sentence in every case.

On a sheet: Move to fresh air. If breathing difficulty persists, seek medical… Harmful if inhaled.

You maintain: Formula lines and raw-material constituents (CAS, %), hazard data per CAS (classification sources, competent-person determinations), trade-secret H-codes and manual hazard JSON on raws, raw physical states with the inhalation-only CAS setting, and any FG hazard override. Optional override on the SDS edit screen.

If missing: With no matching codes: "Move to fresh air. If breathing difficulty persists, seek medical attention." A non-blank override prints as typed and replaces every hazard sentence. Blank override rows are ignored.

`TextOverrideService::effective(4.inhalation) > section4.inhalation[_fatal|_toxic] + _harmful(H332) + _resp_sensitizer/_irritant/_narcotic; SDSGenerator.php:1750-1772`

Skin Contact:

TEXT CHOSEN BY HAZARD Per-product override > hazard sentences: H314 corrosive paragraph (plus poison-center sentence for H310/H311, else H312 sentence) > H310/H311 toxic paragraph > base paragraph + H312/H315 sentences. H317 always adds the sensitiser sentence; UV-family products then add the uncured-product sentence (next row).

On a sheet: Remove contaminated clothing. Wash skin thoroughly… May cause an allergic skin reaction…

You maintain: Same hazard inputs as Inhalation. Optional override on the SDS edit screen.

If missing: "Remove contaminated clothing. Wash skin thoroughly with soap and water. If irritation persists, seek medical attention." A non-blank override replaces the whole field, including the UV sentence.

`effective(4.skin) > section4.skin[_corrosive|_toxic] + poison_center_immediate/skin_harmful/_irritant + skin_sensitizer(H317) + UV fragment; SDSGenerator.php:1774-1801`

Skin Contact: UV/EB-curable sentence (end of the field):

PRODUCT FAMILY Added when the UV rule pack is on and the resolved family is flagged UV/LED, with or without H317. Acrylates at >= 0.1 % are named in brackets: the CAS acrylate flag decides > if the flag is not set, the old CAS list or an 'acrylate' name; without any, the generic sentence prints.

On a sheet: …containing acrylate monomers/oligomers (Tripropylene glycol diacrylate): remove uncured…

You maintain: UV/LED flag, active status, rules and manual picks on Product families. UV rule-pack switch in Settings. Acrylate box per CAS on CAS determinations (CAS Descriptions tab, or the seed preview/apply). Constituent CAS, names and trade-secret descriptions on Raw materials.

If missing: Not added when the pack is off or the family is not UV. Names: Prop 65 name > CAS registry name > constituent name (resale: constituent name); a nameless row prints its CAS. A trade secret prints its description or "Trade Secret". Unticked flag = never named.

`UVAcrylateRulePack::isApplicable :105, detectAcrylates :198 (effectiveFlag, cm.is_acrylate), section4SkinFragment :241; section4.uv_skin / uv_skin_names; SDSGenerator.php:1798-1800`

Eye Contact:

TEXT CHOSEN BY HAZARD Per-product override > hazard sentences: H314 (30-minute rinse) or H318 (20-minute rinse) replaces the base paragraph. H314, H318 or H319 adds the contact-lens sentence.

On a sheet: Flush eyes with large amounts of water for at least 15 minutes… Remove contact lenses…

You maintain: Same hazard inputs as Inhalation. Optional override on the SDS edit screen.

If missing: The base 15-minute flush sentence prints, with no contact-lens sentence. A non-blank override replaces the field.

`effective(4.eyes) > section4.eyes[_corrosive|_serious_damage] + eyes_contact_lenses(H314/H318/H319); SDSGenerator.php:1803-1816`

Ingestion:

TEXT CHOSEN BY HAZARD Per-product override > hazard sentences: H304 aspiration paragraph (plus poison-center sentence for H300/H301, else the H302 sentence) > H300/H301 toxic paragraph > base paragraph + H302 sentence. H305 does not select the aspiration paragraph.

On a sheet: Do not induce vomiting. Rinse mouth with water… Harmful if swallowed. Call a poison…

You maintain: Same hazard inputs as Inhalation. Optional override on the SDS edit screen.

If missing: "Do not induce vomiting. Rinse mouth with water. Seek medical attention if symptoms develop." A non-blank override replaces the field.

`effective(4.ingestion) > section4.ingestion_aspiration(H304) | _toxic(H300/H301) | ingestion; + poison_center_immediate / ingestion_harmful(H302); SDSGenerator.php:1818-1838`

Most Important Symptoms/Effects, Acute and Delayed:	HAZARD ENGINE Per-product override > the product's health hazard statements (H3xx), restated in the sheet language. Sensitisation, CMR, lactation and repeated-exposure codes (H317, H334, H340-H362, H372, H373) are grouped as Delayed, all others as Acute; the same text prints in Section 11. *On a sheet: Acute: Causes serious eye irritation. Delayed: May cause an allergic skin reaction.* **You maintain:** Same hazard inputs as Inhalation, plus the carcinogen lists, which add H350/H351 (skipped in 'replace' hazard-override mode), and STOT target organs per CAS (next row). H-statement translations live in the ghs_es/fr/de files. **If missing:** With no health codes: "No significant symptoms or effects are known or expected under normal conditions of use." A missing translation prints the English text. A sub-coded statement (e.g. H360D) prints the base code's text. H303/H313/H333 never print. `effective(4.symptoms) > deriveSymptoms(hazard.h_statements) SDSGenerator.php:1840-1846, :1916-1956; DELAYED_EFFECT_H_PREFIXES :107; GHSStatements::hText; Section 11 reuse :3976`
Symptoms line: target organs and route in H370-H373 statements:	HAZARD ENGINE Organs (and route) entered on every CAS behind the code are filled into the H370-H373 statement, which then prints in this line. Route prints only when all those CAS give the same single route other than 'All routes'; H370/H371 sit under Acute, H372/H373 under Delayed. *On a sheet: …Causes damage to organs (lungs) through prolonged or repeated exposure by inhalation.* **You maintain:** Target organs and route per CAS on CAS determinations > STOT Target Organs tab (from the vendor SDS or EU CLP Annex VI). **If missing:** The standard statement without organs prints (e.g. "Causes damage to organs through prolonged or repeated exposure."). Same when only some CAS behind the code have organs, or an FG hazard override / trade-secret entry also carries it; the preview then shows a warning. `StotTargetOrgans::resolve (HazardEngine.php:1129, stot_details) + statementText :227; SDSGenerator::applyStotTargetOrgans :1446; stot.h37x[_route], stot.organ.*, stot.route.*`
Acute: / Delayed: (prefixes inside the symptoms line):	FIXED TEXT (TRANSLATED) Fixed words from the translation file. Each prints only when that group has at least one health statement. *On a sheet: Acute: … Delayed: …* **You maintain:** Nothing (driven by the same hazard inputs as the symptoms line) **If missing:** An empty group is left out. If both groups are empty, the 'No significant symptoms…' sentence prints instead. `section4.symptoms_acute_prefix / symptoms_delayed_prefix; SDSGenerator.php:1948-1955; en.php:99-100`
Notes to Physician:	TEXT CHOSEN BY HAZARD Per-product override > hazard sentences in this order: aspiration (H304), corrosive burns (H314) or, without H314, serious eye damage (H318), delayed lung effects (H330/H331), then the P308 first-aid phrases (next row). The default closing sentence always follows. *On a sheet: Do not induce vomiting; risk of aspiration into the lungs. Treat symptomatically. …* **You maintain:** Same hazard inputs as Inhalation. Optional override on the SDS edit screen. **If missing:** "Treat symptomatically. Show this SDS to medical personnel." A non-blank override replaces the field, including the P308 phrases. `effective(4.notes) > notes_aspiration + notes_corrosive | notes_eye_damage + notes_inhalation_delayed + P308 phrases + section4.notes; SDSGenerator.php:1848-1895`
Notes to Physician: If exposed or concerned (P308+P313 / P308+P311):	TEXT CHOSEN BY HAZARD Official GHS P-phrase wording chosen by H-codes. Any H340-H373 code other than H370/H371 (sub-codes such as H360FD included) prints P308+P313; H370/H371 alone print P308+P311; both groups print P308+P313 then "Call a POISON CENTER/doctor." *On a sheet: If exposed or concerned: Get medical advice/attention.* **You maintain:** Same hazard inputs as Inhalation (including carcinogen lists and STOT data). Optional Notes override on the SDS edit screen. **If missing:** Not printed when the product has no H340-H373 code. A non-blank Notes override replaces it. `section4.exposed_medical_advice / exposed_poison_center / call_poison_center (en.php:108-110); SDSGenerator.php:1869-1892`

SECTION 5: FIRE-FIGHTING MEASURES

Section 5 is standard translated wording. Each paragraph is chosen by the product's final hazard data, the same data Section 2 prints. Flammability has one source. The Flammable Liquids category (Cat 1-4) comes from the product flash point, which is the wt%-weighted average over the raw materials that have one, together with the initial boiling point and the physical state. The physical state is the product's own, else the family default, else the largest raw material's. A finished-good hazard override can replace that category, so Sections 2, 5, 7, 9, 10, 13 and 14 agree. The flash point sentence prints the same string as Section 9: either the formula value rounded to a whole degree and marked '(calculated)', or a kept measured edit exactly as typed. Batch F made the combustion sentence name nitrogen oxides, sulfur oxides or hydrogen halides when a constituent CAS carries that element flag, matching Section 10. It also added two combustible-dust fragments for Powder/Solid products that contain a raw material marked as combustible dust. UV/LED-family products get one extra UV sentence inside Specific Hazards. A per-product override can replace any of the four paragraphs, and the flash point sentence inside a Specific Hazards override is kept current.

Suitable Extinguishing Media:

TEXT CHOSEN BY HAZARD One of four fixed sentences, chosen by the final hazard codes. Order: product override > water-reactive (H260/H261) > oxidizer (H271/H272) > flammable liquid (H224-H227 from the product flash point or the FG override) > default.

On a sheet: Alcohol-resistant foam, dry chemical, carbon dioxide (CO2), or water spray (fog).

You maintain: Flash point, '>' tick and boiling point on raw materials. Physical state on the finished good or the family default. Constituent CAS and % on raw materials, with H-codes on CAS determinations. Optional FG hazard override.

If missing: With no water-reactive, oxidizer or flammable code (including a product with no flash point data): "Water spray, dry chemical, carbon dioxide (CO2), foam."

`text_overrides(5,'suitable_media') > section5.suitable_water_reactive > suitable_oxidizer > suitable_flammable > suitable_media; SDSGenerator.php:2189-2202; extractHCodes :1106`

Unsuitable Extinguishing Media:

TEXT CHOSEN BY HAZARD Fixed sentence chosen by the final hazard codes. Order: product override > water-reactive (H260/H261) > flammable liquid (flash-point-derived Cat 1-4 or FG override) > default. Oxidizer codes have no tier here, so an oxidizer gets the flammable or default sentence.

On a sheet: Do not use a solid water stream; it may scatter the burning liquid and spread the fire.

You maintain: Same data as Suitable Extinguishing Media.

If missing: "Do not use direct water stream as it may spread fire."

`text_overrides(5,'unsuitable_media') > section5.unsuitable_water_reactive > unsuitable_flammable > unsuitable_media; SDSGenerator.php:2204-2213`

Unsuitable Extinguishing Media - combustible dust addition:

FORMULA / RAW MATERIALS Appended to the automatic Unsuitable sentence when the product is classified OSHA Combustible Dust. That means a Powder or Solid product whose formula (any depth) contains a raw material marked Combustible Dust = Yes, with no replace-mode FG override.

On a sheet: Avoid high-pressure water jets and other media that could raise dust and form a dust cloud.

You maintain: Combustible Dust (Not assessed / Yes / No) on raw materials (Raw materials); any raw marked Yes counts, whatever its own state. Physical state on the finished good, else the family default (Product families), else the largest raw material's state.

If missing: Not printed when the product is not Powder/Solid, no raw material is marked Yes, a replace-mode FG override is set, or an Unsuitable override exists.

`section5.unsuitable_combustible_dust :2214-2216 when hasCombustibleDust :5637; class added by applyCombustibleDust :5657 from combustibleDustStatus :5600 (resolved FG state + enriched_lines.combustib…`

Specific Hazards - flammable liquid sentence:

HAZARD ENGINE Printed for Flammable Liquids Cat 1-4. Category: FG hazard override with a Flammable Liquids class or H224-H227 > product flash point. FP <23 °C: Cat 1 if IBP <=35 °C, else Cat 2; 23-60 °C Cat 3; 60-93 °C Cat 4.

On a sheet: Flammable liquid and vapor (Flammable Liquids, Category 3). Vapors can form ignitable…

You maintain: Flash point (°C) and '>' tick on each raw material, boiling point on solvent raws (Raw materials). Physical state (Finished goods) or the family default physical state (Product families). Optional FG hazard override.

If missing: Omitted, with the flash point line, when no raw has a flash point, the weighted value is above 93 °C, the resolved state is Solid/Powder/Paste, or a replace-mode FG override has no Flammable Liquids class. A Specific Hazards override replaces it.

`section5.specific_hazards_flammable_cat{1-4}; SDSGenerator.php:2167-2180, :2232-2233; HazardEngine::flammableLiquidCategory :2781, applyFlammableLiquidsFromFlashPoint :2868; flammabilityInputs :4546`

Specific Hazards - Flash point line:	FORMULA / RAW MATERIALS Same string as Section 9. A Section 9 Flash Point edit that is a temperature and agrees with the classification prints exactly as typed > the wt%-weighted raw average, rounded to a whole degree with '(calculated)' and '> ' if any raw is ticked. *On a sheet: Flash point: 39 °C (102 °F) (calculated).* **You maintain:** Flash point (°C) and the '>' tick on raw materials (Raw materials screen). Optionally a measured Section 9 Flash Point edit with a unit (e.g. '24 °C') in the SDS editor, per language. **If missing:** Line omitted when the product is not flammable, or when it is flammable only through the FG override and no raw material or kept edit has a flash point. A contradicting edit is not printed: the formula value prints and the preview warns. `section5.flash_point_line ':fp' :2224-2226; resolveFlashPointDisplay :2034 > flashPointDisplay :2010 + section9.flash_point_calculated; roundFlashPointC :1977; vetFlashPointEdit :4616`
Specific Hazards - water-reactive sentence:	TEXT CHOSEN BY HAZARD Fixed sentence added when the final hazard codes include H261. The H260 version adds 'which may ignite spontaneously'. *On a sheet: Reacts with water, moisture, and water-based extinguishing agents to release flammable gases.* **You maintain:** H260/H261 in the constituent CAS hazard data (CAS determinations), a trade-secret manual hazard entry on a no-CAS raw material, or the FG hazard override. **If missing:** Omitted when neither H260 nor H261 is present. `section5.specific_hazards_water_reactive / _water_reactive_h260; SDSGenerator.php:2238-2242`
Specific Hazards - oxidizer / organic peroxide sentence:	TEXT CHOSEN BY HAZARD One lead sentence placed before the combustion sentence. Order: oxidizer (H271/H272) > organic peroxide (H240-H242, 'May catch fire or explode upon heating.'). *On a sheet: Oxidizer — may cause or intensify fire.* **You maintain:** H271/H272 or H240-H242 in the CAS hazard data (CAS determinations), a manual trade-secret entry, or the FG hazard override. **If missing:** Omitted when no oxidizer or organic peroxide code is present. `section5.specific_hazards_oxidizer > specific_hazards_organic_peroxide; SDSGenerator.php:2247-2251`
Specific Hazards - combustion products sentence:	FORMULA / RAW MATERIALS Always printed in the automatic text. It always names carbon monoxide and carbon dioxide, then adds nitrogen oxides, sulfur oxides or hydrogen halides for each element flag carried by any constituent CAS, at any %. The products listed are the same as in the Section 10 decomposition products. *On a sheet: Combustion may produce carbon monoxide, carbon dioxide, nitrogen oxides, and other toxic gases.* **You maintain:** Nitrogen / sulfur / halogen flags per CAS (CAS Determinations > CAS Descriptions). Set from the chemical name the first time a new CAS is saved on a raw material; the /determinations/element-flags seed sets them from the molecular formula, else the names. **If missing:** No flagged constituent (trade-secret buckets carry no flags): "Combustion may produce carbon monoxide, carbon dioxide, and other toxic gases." A Specific Hazards override replaces it. `combustionProductsSentence :2299 > section5.combustion_products ':products' (section10.decomp_nitrogen|sulfur|halogen) else section5.specific_hazards; decompositionElements :2476`
Specific Hazards - UV/EB sentence:	PRODUCT FAMILY A fixed sentence added after the combustion sentence when the UV rule pack is on and the product's resolved family is active and flagged UV/LED. No acrylate detection is needed. *On a sheet: When heated, uncured UV/EB-curable product may polymerize uncontrollably, releasing heat…* **You maintain:** UV rule pack setting (Settings). UV/LED flag and active status on the family (Product families). Family on the finished good (automatic from the formula or a manual pick), or on the raw material for a resale sheet. **If missing:** Not printed when the pack is off, the family is unresolved, inactive or not UV, or a Specific Hazards override exists. `section5.uv_specific_hazards; SDSGenerator.php:2253-2255; $this->uvPack = UVAcrylateRulePack::isApplicable (isEnabled + familyIsUv) :223-228, setUvState :1295`
Specific Hazards - combustible dust sentence:	FORMULA / RAW MATERIALS Added last in the automatic Specific Hazards text when the product is classified OSHA Combustible Dust (a Powder/Solid product with a raw material marked Yes, and no replace-mode FG override). *On a sheet: Fine dust dispersed in air in sufficient concentration can form an explosive mixture…* **You maintain:** Combustible Dust on raw materials (Raw materials); any raw marked Yes counts, whatever its own state. Physical state on the finished good, else the family default (Product families), else the largest raw material's state. **If missing:** Not printed when the product is not combustible dust or a Specific Hazards override exists. `section5.specific_hazards_combustible_dust :2256-2258 when hasCombustibleDust :5637; applyCombustibleDust :5657, combustibleDustStatus :5600`

Advice for Firefighters:

TEXT CHOSEN BY HAZARD Fixed sentence chosen by the final hazard codes. Order: product override > explosive (H200-H205, plus a no-water sentence if also H260/H261) > water-reactive (H260/H261) > flammable liquid (flash-point-derived Cat 1-4 or FG override) > default.

On a sheet: Wear self-contained breathing apparatus (SCBA)… Eliminate all ignition sources…

You maintain: The hazard codes and flash point data above. Optional per-product override.

If missing: "Wear self-contained breathing apparatus (SCBA) and full protective gear. Cool containers exposed to fire with water spray."

`text_overrides(5,'firefighter_advice') > section5.firefighter_advice_explosive (+ firefighter_no_water) > _water_reactive > _flammable > firefighter_advice; SDSGenerator.php:2261-2278`

Preview Warnings box - flash point / flammability warnings (not printed):

FORMULA / RAW MATERIALS Operator-only messages that never print. They cover a dropped or ignored Section 9 flash or boiling point edit, a flash point below 23 °C with no raw boiling point (Cat 2 assumed), an FG override category the formula flash point does not support, and raw materials with a disclosed flammable-liquid CAS (>=1%) but no flash point.

On a sheet: Flash point warning: raw material(s) R123 contain an ingredient classified as a flammable…

You maintain: Clear them by entering flash points and boiling points on raw materials, or by correcting or resetting the Section 9 edit in the SDS editor.

If missing: No box when there is nothing to warn about. Publishing is never blocked by these warnings.

`vetFlashPointEdit :4616, temperatureOverrideWarnings :2109, flammabilityWarnings :4692 into $sds['warnings'] (:345-353); preview.php Warnings box`

Preview Warnings box - combustible dust not assessed (not printed):

FORMULA / RAW MATERIALS An operator-only message that never prints. It appears for a Powder/Solid product with no raw material marked Yes when at least one Powder/Solid raw material is still 'Not assessed'. The same message appears on SDS Creation Readiness.

On a sheet: Combustible dust: this product's physical state is Powder and its formula contains a powder…

You maintain: Combustible Dust on each listed raw material (Raw Materials > Edit), taken from the vendor SDS Section 2.

If missing: No warning once every Powder/Solid raw material is Yes or No, or the product is not Powder/Solid. Publishing is never blocked.

`SDSReadinessService::combustibleDustNotAssessedWarning :684 (combustibleDustStatus :5600) into $sds['warnings'] :363-366`

SECTION 6: ACCIDENTAL RELEASE MEASURES

Section 6 is built from standard, translated sentences. The hazard codes, the product flash point, the physical state and the UV family decide which ones print. The aquatic sentences now use the Section 2 hazard statement wording. Operators can replace any of the three fields for one product (or one resale raw material) in one language.

Personal Precautions:

FIXED TEXT (TRANSLATED) Field label from the language file. It is stored with the sheet when the sheet is generated.

On a sheet: Personal Precautions:

You maintain: Nothing

If missing: Stored label > sheet-language translation > English > 'Personal Precautions'. No label prints when the value is empty.

`labels.personal_precautions -> meta.labels (getLabels :5347); PDFService FIELD_LABEL_MAP + label(); preview $genericLabelMap`

Personal Precautions (standard paragraph):

TEXT CHOSEN BY HAZARD Standard sentence used when the product has no acute toxicity category 1-3 code (H300/H301/H310/H311/H330/H331) and no H314. Operator text > this sentence.

On a sheet: Use appropriate PPE (see Section 8). Avoid contact with skin and eyes. Ensure adequate…

You maintain: Hazard data on the CAS determinations and Raw materials screens (classifications, ATE values, competent-person determinations, trade-secret H-codes, constituent %), plus any finished-good hazard override.

If missing: This is the default. If hazard data is missing, it also prints for a toxic or corrosive product. Publishing is then blocked when a no-data CAS is at or above the Missing Data Threshold.

`text_overrides(6,'personal_precautions') > section6.personal_precautions; SDSGenerator::section6 :2376-2389; codes via extractHCodes(HazardEngine::classify) :1106`

Personal Precautions (acute toxicity paragraph):

TEXT CHOSEN BY HAZARD Replaces the standard paragraph when any of H300, H301, H310, H311, H330 or H331 applies. The acute category is the more severe of the ingredient cut-off result and the ATEmix result.

On a sheet: Use appropriate PPE including self-contained breathing apparatus (SCBA) (see Section 8)…

You maintain: Acute toxicity classifications, ATE values and percentages of constituents (CAS determinations, Raw materials); finished-good hazard override.

If missing: If no acute toxicity code results, the corrosive or standard paragraph prints.

`section6.precautions_acute_toxic; SDSGenerator :2379-2380 (checked ahead of the H314-only branch); acute categories from HazardEngine::classify`

Personal Precautions (corrosive paragraph):

TEXT CHOSEN BY HAZARD Replaces the standard paragraph when H314 applies and no acute toxicity 1-3 code applies. H318 alone does not trigger it.

On a sheet: Use appropriate PPE including chemical-resistant suit, gloves, and face shield…

You maintain: Skin corrosion classifications of constituents; finished-good hazard override.

If missing: Standard paragraph prints.

`section6.precautions_corrosive; SDSGenerator :2385-2386`

Personal Precautions (corrosive add-on):

TEXT CHOSEN BY HAZARD Added after the acute toxicity paragraph when H314 also applies, because that paragraph has no skin or eye wording.

On a sheet: … Wear a chemical-resistant suit, gloves and face shield. Avoid all contact with skin…

You maintain: Acute toxicity and skin corrosion classifications of constituents; finished-good hazard override.

If missing: Not added.

`section6.precautions_corrosive_addon; SDSGenerator :2381-2384 (#32)`

Personal Precautions (flammable add-on):

FORMULA / RAW MATERIALS Added after the chosen paragraph for H220-H226 or H228. H224-H226 come from the product flash point (wt%-weighted over raw materials that have one: below 23 °C = Cat 1-2, 23-60 °C = Cat 3), not from ingredient codes.

On a sheet: … Remove all sources of ignition. Use only non-sparking tools and explosion-proof equipment…

You maintain: Flash Point (and the '>' flag) on every raw material on the Raw materials screen; give water and similar raws a high flash point. Also the physical state (product, family default or largest raw) and the finished-good hazard override, where a Flammable Liquids entry replaces the derived category.

If missing: Not added when no raw material has a flash point (treated as not flammable), when the average is above 60 °C, or when the resolved state is Solid, Powder or Paste (including a family default of Paste, Solid or Powder).

`section6.precautions_flammable; SDSGenerator :2390-2391; flammabilityInputs :4546 -> HazardEngine::flammableLiquidCategory :2781; FormulaCalcService::weightedFlashPoint :693`

Personal Precautions (ignition add-on):

FORMULA / RAW MATERIALS Added instead of the flammable add-on for H227 or H250/H260/H261, when no H220-H226/H228 applies. H227 means a product flash point above 60 °C and up to 93 °C (Flammable Liquids Cat 4); '> 60' also counts.

On a sheet: … Eliminate all ignition sources (no smoking, sparks, open flames or hot surfaces)…

You maintain: Raw material flash points (Raw materials screen); pyrophoric and water-reactive classifications (CAS determinations); finished-good hazard override.

If missing: Not added.

`section6.precautions_ignition; SDSGenerator :2392-2396 (#64); H227 from HazardEngine::flammableLiquidCategory Cat 4`

Environmental Precautions:

FIXED TEXT (TRANSLATED) Field label from the language file. It is stored with the sheet when the sheet is generated.

On a sheet: Environmental Precautions:

You maintain: Nothing

If missing: Stored label > sheet-language translation > English > 'Environmental'.

`labels.environmental_precautions (data key 'environmental'); PDFService FIELD_LABEL_MAP + label(); preview $genericLabelMap`

Environmental Precautions (drains sentence):

FIXED TEXT Always the first sentence on every sheet. Operator text for this product > this sentence.

On a sheet: Prevent entry into drains, sewers, and waterways.

You maintain: Nothing

If missing: Always prints unless the field has operator text.

`text_overrides(6,'environmental') > section6.environmental; SDSGenerator :2400-2402`

Environmental Precautions (aquatic acute sentence):

TEXT CHOSEN BY HAZARD One sentence for the acute route, chosen H400 > H401 > H402 and worded exactly as Section 2 prints that hazard statement, with a full stop. Section 2 text > GHS phrase list for the sheet language > English GHS phrase list > old Section 6 sentence.

On a sheet: Very toxic to aquatic life.

You maintain: Aquatic acute classifications and M-factors of constituents (CAS determinations); finished-good hazard override.

If missing: No acute sentence.

`S6_AQUATIC_ACUTE :2314; section2StatementSentence :2333 (h_statements text > GHSStatements::hText :309, ghs_<lang>.php then H_STATEMENTS > section6.environmental_aquatic_*); section6 :2410-2416`

Environmental Precautions (aquatic chronic sentence):	TEXT CHOSEN BY HAZARD Chosen separately from the acute sentence: H410 > H411 > H412 > H413, worded exactly as the Section 2 hazard statement (Section 2 text > sheet-language GHS phrase > English GHS phrase > old Section 6 sentence), with a full stop. Prints after the acute sentence when both apply. *On a sheet: Harmful to aquatic life with long lasting effects.* **You maintain:** Aquatic chronic classifications and M-factors of constituents; finished-good hazard override. **If missing:** No chronic sentence. `S6_AQUATIC_CHRONIC :2319; section2StatementSentence :2333 (h_statements > GHSStatements::hText :309 > section6.environmental_aquatic_chronic_*); section6 :2410-2416`
Environmental Precautions (notify sentence):	TEXT CHOSEN BY HAZARD Added last whenever any aquatic sentence printed. *On a sheet: Notify authorities if the product enters drains or waterways.* **You maintain:** Aquatic classifications. **If missing:** Not printed for products with no aquatic code. `section6.environmental_notify; SDSGenerator :2418-2421`
Containment and Cleanup:	FIXED TEXT (TRANSLATED) Field label from the language file. It is stored with the sheet when the sheet is generated. *On a sheet: Containment and Cleanup:* **You maintain:** Nothing **If missing:** Stored label > sheet-language translation > English > 'Containment'. `labels.containment_cleanup (data key 'containment'); PDFService FIELD_LABEL_MAP + label(); preview $genericLabelMap`
Containment and Cleanup (solid / powder):	FORMULA / RAW MATERIALS Used when the physical state is Solid or Powder. Operator text > finished good's Physical State > active product family's default physical state > largest raw material that has a state > 'Liquid'. *On a sheet: Avoid generating dust. Sweep or vacuum up material. Collect in suitable containers…* **You maintain:** Physical State on the finished good, or Default physical state on its Product families record, or Physical State on its raw materials (the largest one that has a state is used). A resale sheet uses the raw material's own state, then its family default. **If missing:** With no state anywhere, 'Liquid' is assumed, the liquid text prints and the preview shows a physical-state warning. `section6.containment_solid; SDSGenerator :2427-2429; resolvePhysicalState :2980 (fg > family_profile.default_physical_state > formula_props.physical_state > 'Liquid')`
Containment and Cleanup (gel / paste):	FORMULA / RAW MATERIALS Used when the resolved physical state is Gel or Paste (same order as above). *On a sheet: Stop leak if safe to do so. Scrape up or shovel material and collect in suitable…* **You maintain:** Physical State = Gel or Paste on the finished good, Default physical state = Gel or Paste on the family (e.g. offset = Paste), or Paste on the raw material that sets the state (raw materials have no Gel option). **If missing:** Liquid text prints. `section6.containment_paste; SDSGenerator :2430-2431`
Containment and Cleanup (gas):	FORMULA / RAW MATERIALS Used when the resolved physical state is Gas. Families cannot default to Gas. *On a sheet: Stop the leak if this can be done without risk. Ventilate the area and allow the gas…* **You maintain:** Physical State = Gas on the finished good, or on the raw material that sets the state. **If missing:** Liquid text prints. `section6.containment_gas; SDSGenerator :2432-2433 (#64); ProductFamily::PHYSICAL_STATES has no Gas`
Containment and Cleanup (liquid / everything else):	FORMULA / RAW MATERIALS Used for Liquid (including a family default of Liquid), for custom states such as Aerosol, and when no state is recorded (assumed 'Liquid'). *On a sheet: Stop leak if safe to do so. Dam or contain spill. Absorb with inert absorbent material…* **You maintain:** Physical state data as above. When no state is recorded anywhere, this text prints and the preview shows a warning. **If missing:** This is the fallback text. `section6.containment_liquid; SDSGenerator :2434-2437`
Containment and Cleanup (UV add-on):	PRODUCT FAMILY Added to whichever containment text printed when the UV acrylate setting is on and the product's resolved family is active and flagged UV/LED. No acrylate has to be found in the formula, and the CAS acrylate flags do not affect it. *On a sheet: … Keep collected uncured product in closed containers away from UV light and direct…* **You maintain:** UV/LED flag and Active status on the Product families screen. Family assignment: automatic (UV families pooled by total share) or a manual pick on the finished good or resale raw material. UV acrylate toggle in Settings. **If missing:** Not added when the family is missing, inactive or not UV, or when the setting is off. Operator text for Containment also removes it. `section6.uv_containment; SDSGenerator :2438-2440; uvPack = UVAcrylateRulePack::isApplicable :105 (isEnabled && familyIsUv, from familyFields :1265)`

Physical-state default warning (preview only):

FORMULA / RAW MATERIALS Shown in the preview Warnings box and never printed. It appears when the finished good, its product family default and every raw material in its formula all have no physical state, so Sections 6, 8, 9 and 14 assume 'Liquid'.

On a sheet: Physical state: no physical state is recorded on this product or on any raw material…

You maintain: Physical State on the finished good or its raw materials (on the raw material for a resale sheet), or Default physical state on the product family.

If missing: No warning when any state is recorded.

`SDSGenerator::PHYSICAL_STATE_DEFAULT_WARNING :2996, physicalStateIsDefault :2999; added to calcResult warnings in generate / computeBase / resale base`

Family physical-state note (preview only):

PRODUCT FAMILY Shown in the preview Warnings box and never printed. It appears when the product has no state of its own and its family's default physical state differs from the state of its largest raw material, which changes the Section 6 containment text.

On a sheet: Physical state: this product has no Physical State of its own, so the sheet uses its product…

You maintain: Default physical state on the Product families screen; Physical State on the finished good to override it.

If missing: No note when the product has its own state, the family has no default, or both states agree.

`SDSGenerator::familyPhysicalStateWarning :1243, added by applyFamilyProfile :1229 before resolvePhysicalState`

SECTION 7: HANDLING AND STORAGE

Section 7 is built automatically from the product's final GHS hazard codes, the P-statements Section 2 prints, the OSHA Combustible Dust class and the product family. Fixed base sentences come first, then a sentence for each matching hazard class, then UV sentences for products in an active UV/LED family. Storage also adds the family's storage-temperature range and 'Protect from freezing.' for water-based products, and ends by repeating the Section 10 incompatible-materials list. Flammable-liquid wording follows the product flash point (the wt%-weighted average over raws that have one) and the resolved physical state (product > family default > largest raw), so a product with no flash point data gets no flammable-liquid wording. A per-language override in the SDS editor (finished goods and resale raw materials) replaces either paragraph entirely, and a Storage override puts a warning on the preview.

Handling:

FIXED TEXT (TRANSLATED) Fixed label from the language files.

On a sheet: Handling:

You maintain: Nothing

If missing: English 'Handling', then the literal 'labels.handling'. Old snapshots without meta.labels fall back to the sheet-language label, then 'Handling'.

`meta.labels.handling = labels.handling (en.php:718; getLabels SDSGenerator.php:5347); PDFService.php:47; preview.php:623`

Handling (paragraph as a whole):

TEXT CHOSEN BY HAZARD Published sheet snapshot > the Section 7 Handling override for that language (finished good: SDS > Edit; resale raw material: Readiness Check > Edit SDS text), used word for word > base sentence + every matching hazard, combustible-dust and UV sentence. Alias and private-label sheets copy the base product's text, and a blank override counts as automatic.

On a sheet: Use only in well-ventilated areas. Avoid contact with skin, eyes, and clothing. ...

You maintain: GHS classifications on each CAS, flash points, physical state and Combustible Dust on raw materials, trade-secret hazard codes, the finished good's physical state and hazard override, the family UV flag and default physical state, an optional per-language override, and a republish after any change

If missing: Never blank. With no matching hazard the base sentence prints alone. An old stock paragraph saved by the pre-update editor keeps printing until the second cleanup pass (cleanup-legacy-overrides.php --apply) deletes it.

`snapshot_json > text_overrides(fg.id | raw_material_id,7,'handling',lang,NULL) > handling_base + fragments; SDSGenerator.php:2599-2700; getOverrides :5208`

Handling: base sentence (always first):

FIXED TEXT Fixed text that every sheet prints unless Handling is overridden. It ends 'Wear the personal protective equipment specified in Section 8.'

On a sheet: Use only in well-ventilated areas. Avoid contact with skin, eyes, and clothing.

You maintain: Nothing

If missing: English, then the literal key (all four languages define it)

`section7.handling_base; SDSGenerator.php:2640; en.php:180; es/fr/de.php:169`

Handling: heat and ignition sources:	TEXT CHOSEN BY HAZARD Added for flammable (H220-H226, H228), combustible (H227), pyrophoric (H250), self-reactive (H240-H242), oxidizer (H270-H272), explosive (H200-H205) codes or the OSHA Combustible Dust class. H224-H227 come only from the product flash point plus boiling point and resolved physical state, or from the finished good's hazard override. *On a sheet: Keep away from heat, hot surfaces, sparks, open flames, and other ignition sources. ...* **You maintain:** Flash point (with '>' flag) on every raw material, including a high value for water; physical state on the product, family default or raws; Combustible Dust on Powder/Solid raws; other physical-hazard classifications on CAS records; the FG hazard override **If missing:** Not printed. No flash point on any raw, or a Solid/Powder/Paste product state, gives no H224-H227 and so no fire wording unless another listed class applies (a Gel is still treated as a liquid). `section7.handling_ignition when $ignition (SDSGenerator.php:2612-2633, 2641-2643); en.php:181; HazardEngine::flammableLiquidCategory :2781; resolvePhysicalState :2980`
Handling: grounding and static:	TEXT CHOSEN BY HAZARD Added for flammable codes (H220-H226, H228), not for H227 combustible. A flammable-liquid code (H224-H226) means a product flash point at or below 60 °C (a '> n' value counts as just above n) on a product whose resolved state is not Solid/Powder/Paste, or the FG override. *On a sheet: Keep container tightly closed. Ground and bond container and receiving equipment. ...* **You maintain:** Flash points on raw materials; physical state (product > family default > largest raw); flammable gas/aerosol/solid classifications on CAS records; the FG hazard override **If missing:** Not printed `section7.handling_flammable_static; SDSGenerator.php:2612, 2644-2646; en.php:182; HazardEngine::flammableLiquidCategory :2781`
Handling: combustible dust:	TEXT CHOSEN BY HAZARD Added when the product carries the OSHA Combustible Dust class: resolved state Powder or Solid, a formula raw marked Combustible Dust = Yes, and no 'replace' FG hazard override. A flammable product prints the version without the closing grounding sentence, which the static sentence already gave. *On a sheet: Prevent dust accumulation and avoid dispersing dust in the air. Clean up with methods ...* **You maintain:** Combustible Dust (Not assessed / Yes / No) on each Powder/Solid raw material (Raw Materials > Edit); physical state on the product or family default **If missing:** Not printed. 'Not assessed' raws give only a preview/readiness warning, never the sentence. `section7.handling_combustible_dust | _no_ground; SDSGenerator.php:2631, 2647-2654; combustibleDustStatus :5600; applyCombustibleDust :5657; en.php:222-223`
Handling: aerosol:	TEXT CHOSEN BY HAZARD Added for aerosols (H222, H223, H229). *On a sheet: Pressurized container: do not pierce or burn, even after use.* **You maintain:** Aerosol classification, usually through the finished good's hazard override **If missing:** Not printed `section7.handling_aerosol; SDSGenerator.php:2614, 2655-2657; en.php:183`
Handling: explosive:	TEXT CHOSEN BY HAZARD Added for explosives (H200-H205). *On a sheet: Do not subject to grinding, shock or friction. Keep only in the original packaging.* **You maintain:** Explosive classification on the CAS records or the FG override **If missing:** Not printed `section7.handling_explosive; SDSGenerator.php:2623, 2658-2660; en.php:212`
Handling: gases under pressure:	TEXT CHOSEN BY HAZARD Added for gases under pressure (H280, H281). *On a sheet: Handle pressurized containers with care; do not drop or damage them. Close the valve ...* **You maintain:** Gas-under-pressure classification on the CAS records or the FG override **If missing:** Not printed `section7.handling_gas_pressure; SDSGenerator.php:2624, 2661-2663; en.php:213`
Handling: oxidizer:	TEXT CHOSEN BY HAZARD Added for oxidizers (H270, H271, H272). *On a sheet: Keep away from clothing and other combustible materials. Do not mix with flammable ...* **You maintain:** Oxidizer classification on the CAS records or the FG override **If missing:** Not printed `section7.handling_oxidizer_combustibles; SDSGenerator.php:2615, 2664-2666; en.php:184`
Handling: self-reactive:	TEXT CHOSEN BY HAZARD Added for self-reactive substances and organic peroxides (H240-H242). *On a sheet: Keep cool. Handle and open container with care. Keep only in original container ...* **You maintain:** Self-reactive / organic peroxide classification **If missing:** Not printed `section7.handling_self_reactive; SDSGenerator.php:2616, 2667-2669; en.php:178`
Handling: self-heating:	TEXT CHOSEN BY HAZARD Added for self-heating substances (H251, H252). *On a sheet: Keep cool. Protect from sunlight. Maintain air gap between stacks or pallets. ...* **You maintain:** Self-heating classification **If missing:** Not printed `section7.handling_self_heating; SDSGenerator.php:2617, 2670-2672; en.php:185`

Handling: pyrophoric and water-reactive:	TEXT CHOSEN BY HAZARD Printed instead of the separate air and water sentences when the product is both pyrophoric (H250) and water-reactive (H260/H261), so 'inert gas' appears once. *On a sheet: Do not allow contact with air or water. Handle under inert gas.* **You maintain:** Pyrophoric and water-reactive classifications **If missing:** Not printed `section7.handling_pyrophoric_water_reactive; SDSGenerator.php:2673-2674; en.php:211`
Handling: pyrophoric (air):	TEXT CHOSEN BY HAZARD Added for pyrophoric substances (H250) that are not also water-reactive. *On a sheet: Do not allow contact with air. Handle under inert gas.* **You maintain:** Pyrophoric classification **If missing:** Not printed `section7.handling_pyrophoric_air; SDSGenerator.php:2618, 2675-2676; en.php:186`
Handling: water-reactive:	TEXT CHOSEN BY HAZARD Added for water-reactive substances (H260, H261) that are not also pyrophoric. *On a sheet: Do not allow contact with water. Handle under inert gas.* **You maintain:** Water-reactive classification **If missing:** Not printed `section7.handling_water_reactive_water; SDSGenerator.php:2619, 2677-2678; en.php:187`
Handling: moisture:	TEXT CHOSEN BY HAZARD Printed once when the product is pyrophoric (H250) or water-reactive (H260/H261). *On a sheet: Protect from moisture.* **You maintain:** Pyrophoric / water-reactive classification **If missing:** Not printed `section7.handling_moisture; SDSGenerator.php:2680-2682; en.php:188`
Handling: corrosive:	TEXT CHOSEN BY HAZARD Added for skin-corrosive products (H314). *On a sheet: Do not breathe mist, vapors, or spray. Do not get in eyes, on skin, or on clothing.* **You maintain:** Skin Corr. 1 classification (including the engine's summation) **If missing:** Not printed `section7.handling_corrosive; SDSGenerator.php:2620, 2683-2685; en.php:189`
Handling: corrosive to metals:	TEXT CHOSEN BY HAZARD Added for products corrosive to metals (H290). *On a sheet: Keep only in original container.* **You maintain:** Met. Corr. 1 classification **If missing:** Not printed `section7.handling_corrosive_metals; SDSGenerator.php:2621, 2686-2688; en.php:190`
Handling: repeated-exposure organ toxicity:	TEXT CHOSEN BY HAZARD Added for STOT RE (H372, H373) unless the product is also H314, whose sentence already says 'Do not breathe'. It names no target organ. *On a sheet: Do not breathe dust, mist, vapors, or spray.* **You maintain:** STOT RE classification on the CAS records **If missing:** Not printed `section7.handling_stot_re; SDSGenerator.php:2625, 2689-2691; en.php:214`
Handling: sensitizer:	TEXT CHOSEN BY HAZARD Added for skin or respiratory sensitizers (H317, H334). When H314 or H372/H373 already printed a 'Do not breathe' sentence, only the work-clothing half prints. *On a sheet: Avoid breathing vapors, mist, or spray. Contaminated work clothing must not be ...* **You maintain:** Skin / respiratory sensitizer classification **If missing:** Not printed `section7.handling_sensitizer | handling_sensitizer_clothing; SDSGenerator.php:2622, 2635 ($breatheSaid), 2692-2695; en.php:191, 215`
Handling: UV uncured product:	PRODUCT FAMILY Added when the UV acrylate rule pack is on in admin settings and the product's resolved family is active and flagged UV/LED. No acrylate needs to be detected or flagged. Alias, private-label and resale sheets follow the same rule. *On a sheet: Avoid prolonged or repeated skin contact with uncured product.* **You maintain:** Product family UV/LED flag, active flag and membership rules or manual picks (Product Families); the admin 'UV acrylate rule pack' toggle **If missing:** Not printed. It is also dropped when Handling is overridden. `section7.uv_handling when $this->uvPack (UVAcrylateRulePack::isApplicable; familyFields SDSGenerator.php:1265); SDSGenerator.php:2696-2698; en.php:219`

Storage:	FIXED TEXT (TRANSLATED) Fixed label from the language files. *On a sheet: Storage:* **You maintain:** Nothing **If missing:** English 'Storage', then the literal 'labels.storage'. Old snapshots fall back to the sheet-language label, then 'Storage'. `meta.labels.storage = labels.storage (en.php:719; getLabels SDSGenerator.php:5347); PDFService.php:48; preview.php:624`
Storage (paragraph as a whole):	TEXT CHOSEN BY HAZARD Published sheet snapshot > the Section 7 Storage override for that language (finished good or resale raw material), used word for word, which also replaces the incompatibles list > base sentence + family temperature/freezing sentences + hazard sentences + UV sentence + 'Store locked up.' + Section 10 incompatibles. Alias and private-label sheets copy the base product, and a blank override counts as automatic. *On a sheet: Store in a cool, dry, well-ventilated area. ... (see Section 10): strong oxidizing ...* **You maintain:** CAS GHS classifications, raw-material flash points and Combustible Dust, the FG hazard override, the family UV flag and SDS profile (storage temperatures, water-based), optional Section 7 Storage and Section 10 Incompatible Materials overrides, and a republish after changes **If missing:** Never blank. The minimum is the base sentence plus 'Store away from incompatible materials (see Section 10): strong oxidizing agents, strong acids, strong bases.' An old stock override prints until the second cleanup pass deletes it. `snapshot_json > text_overrides(fg.id | raw_material_id,7,'storage',lang,NULL) > storage_base + fragments + storage_locked + storage_incompatible; SDSGenerator.php:2702-2762`
Storage: base sentence (always first):	FIXED TEXT Fixed text that every sheet prints unless Storage is overridden. *On a sheet: Store in a cool, dry, well-ventilated area. Keep container tightly closed when not ...* **You maintain:** Nothing **If missing:** English, then the literal key `section7.storage_base; SDSGenerator.php:2705; en.php:192; es/fr/de.php:181`
Storage: storage temperature:	PRODUCT FAMILY The resolved active family's SDS profile storage minimum and maximum (°C), with Fahrenheit rounded to a whole degree. Printed right after the base sentence. Alias and private-label sheets use the base product's family. *On a sheet: Store between 5 and 30 °C (41–86 °F).* **You maintain:** Product Families > Edit > SDS profile: Storage temperature minimum and maximum (°C), both or neither, minimum below maximum, -50 to 100 **If missing:** Not printed when the family has no range, the family is inactive or the product has no family. The min-only and max-only wordings (en.php:207-208) never print because the profile needs both values. `section7.storage_temp_range (en.php:206); storageTemperatureSentence SDSGenerator.php:1364-1374, 2708-2711; setProfileState :1311; ProductFamily::profile()`
Storage: protect from freezing:	PRODUCT FAMILY Added for water-based products: family Water-based setting Yes > No > Auto, where Auto (also used with no family or an inactive one) means water (CAS 7732-18-5, through intermediates) is at or above the Water-Based Threshold (default 20%) of the formula. *On a sheet: Protect from freezing.* **You maintain:** Product Families > SDS profile > Water-based (auto / yes / no); Settings > Water-Based Threshold (%); water as a CAS constituent on raw materials **If missing:** Not printed. Water held only in a raw's water content, with no 7732-18-5 constituent, does not count under Auto. `section7.storage_protect_freezing (en.php:209); SDSGenerator.php:1229-1236, 1322, 2712-2714; WaterBasedRule::forProduct / isWaterBased; ProductFamily::PROFILE_EMPTY water_based 'auto'`
Storage: heat and ignition sources:	TEXT CHOSEN BY HAZARD Same trigger as the Handling ignition sentence: flammable, combustible (H227), pyrophoric, self-reactive, oxidizer, explosive or OSHA Combustible Dust. H224-H227 come only from the product flash point and physical state, or the FG override. *On a sheet: Keep away from heat and ignition sources.* **You maintain:** Flash points and Combustible Dust on raw materials; physical state; physical-hazard classifications on the CAS records; the FG hazard override **If missing:** Not printed `section7.storage_ignition when $ignition; SDSGenerator.php:2633, 2715-2717; en.php:193`
Storage: flammables storage area:	TEXT CHOSEN BY HAZARD Added for flammable codes (H220-H226, H228), not for H227 or combustible dust. H224-H226 mean a product flash point at or below 60 °C on a product whose resolved state is not Solid/Powder/Paste, or the FG override. *On a sheet: Store in an approved flammable-materials storage area. Ground and bond containers ...* **You maintain:** Flash points on raw materials; physical state (product > family default > largest raw); flammable gas/aerosol/solid classifications; the FG hazard override **If missing:** Not printed `section7.storage_flammable_area; SDSGenerator.php:2612, 2718-2720; en.php:194; HazardEngine::flammableLiquidCategory :2781`

Storage: aerosol: TEXT CHOSEN BY HAZARD Added for aerosols (H222, H223, H229).
On a sheet: Protect from sunlight. Do not expose to temperatures exceeding 50 °C / 122 °F.
You maintain: Aerosol classification
If missing: Not printed
`section7.storage_aerosol; SDSGenerator.php:2721-2723; en.php:195`

Storage: explosive: TEXT CHOSEN BY HAZARD Added for explosives (H200-H205).
On a sheet: Store in accordance with the applicable explosives storage regulations, away from ...
You maintain: Explosive classification
If missing: Not printed
`section7.storage_explosive; SDSGenerator.php:2724-2726; en.php:216`

Storage: gases under pressure: TEXT CHOSEN BY HAZARD Added for gases under pressure (H280, H281).
On a sheet: Protect from sunlight. Store in a well-ventilated place. Secure cylinders against falling.
You maintain: Gas-under-pressure classification
If missing: Not printed
`section7.storage_gas_pressure; SDSGenerator.php:2727-2729; en.php:217`

Storage: oxidizer: TEXT CHOSEN BY HAZARD Added for oxidizers (H270-H272).
On a sheet: Store separately from combustible materials, reducing agents, and flammable substances.
You maintain: Oxidizer classification
If missing: Not printed
`section7.storage_oxidizer_separate; SDSGenerator.php:2730-2732; en.php:196`

Storage: self-reactive: TEXT CHOSEN BY HAZARD Added for self-reactive substances and organic peroxides (H240-H242). It does not use the family storage temperatures.
On a sheet: Store at the recommended temperature. Keep away from heat and sunlight.
You maintain: Self-reactive classification
If missing: Not printed
`section7.storage_self_reactive; SDSGenerator.php:2733-2735; en.php:197`

Storage: self-heating: TEXT CHOSEN BY HAZARD Added for self-heating substances (H251, H252).
On a sheet: Keep cool and protect from sunlight. Maintain air gap between stacks or pallets.
You maintain: Self-heating classification
If missing: Not printed
`section7.storage_self_heating_cool; SDSGenerator.php:2736-2738; en.php:198`

Storage: pyrophoric (air): TEXT CHOSEN BY HAZARD Added for pyrophoric substances (H250).
On a sheet: Protect from air.
You maintain: Pyrophoric classification
If missing: Not printed
`section7.storage_pyrophoric_air; SDSGenerator.php:2739-2741; en.php:199`

Storage: water-reactive: TEXT CHOSEN BY HAZARD Added for water-reactive substances (H260, H261).
On a sheet: Protect from water.
You maintain: Water-reactive classification
If missing: Not printed
`section7.storage_water_reactive_water; SDSGenerator.php:2742-2744; en.php:200`

Storage: inert gas and moisture: TEXT CHOSEN BY HAZARD Printed once for pyrophoric (H250) or water-reactive (H260/H261) products.
On a sheet: Store under inert gas. Protect from moisture.
You maintain: Pyrophoric / water-reactive classification
If missing: Not printed
`section7.storage_inert_moisture; SDSGenerator.php:2745-2747; en.php:201`

Storage: corrosive to metals: TEXT CHOSEN BY HAZARD Added for products corrosive to metals (H290).
On a sheet: Store in a corrosion-resistant container with a resistant inner liner.
You maintain: Met. Corr. 1 classification
If missing: Not printed
`section7.storage_corrosive_metals; SDSGenerator.php:2748-2750; en.php:202`

Storage: UV light: PRODUCT FAMILY Added when the product's resolved family is active and flagged UV/LED. The admin rule-pack toggle is ignored, as for the Section 10 UV items.
On a sheet: Store away from UV light sources and direct sunlight to prevent premature polymerization.
You maintain: Product family UV/LED flag, active flag and membership rules or manual picks (Product Families)
If missing: Not printed. It is also dropped when Storage is overridden.
`section7.uv_storage when $this->uvFamily (UVAcrylateRulePack::familyIsUv); SDSGenerator.php:2751-2754; en.php:220`

Storage: Store locked up:

TEXT CHOSEN BY HAZARD Added when the P-statements Section 2 prints include P405, alone or in a combined code. P405 comes from the class defaults for Acute Tox. 1-3, Skin Corr. 1, CMR, STOT SE 1-3 and Asp. Tox. 1-2, or from source data or the FG override.

On a sheet: Store locked up.

You maintain: Health-hazard classifications on the CAS records; P-codes in the FG hazard override

If missing: Not printed

`section7.storage_locked when $lockUp = P405 in resolvedPCodes() (SDSGenerator.php:1131, 2629, 2755-2757); en.php:203; GHSHazardData p_codes`

Storage: incompatible materials (see Section 10):

TEXT CHOSEN BY HAZARD The Section 10 Incompatible Materials override (trailing full stop removed; first letter lower-cased except on German sheets or acronyms) > a list built from the hazard classes and family. It always has acids and bases, has oxidizing agents unless the product is an oxidizer, adds items per class, and ends with 'peroxides and free-radical initiators' for an active UV/LED family.

On a sheet: Store away from incompatible materials (see Section 10): strong oxidizing agents, ...

You maintain: Physical-hazard classifications and raw-material flash points; the family UV/LED flag; optional Section 10 Incompatible Materials override in the SDS editor

If missing: Always printed: with no physical hazard and no UV family the list is 'strong oxidizing agents, strong acids, strong bases'

`section7.storage_incompatible :list = incompatibleMaterialsInline() SDSGenerator.php:2589-2597 + inlineCase() :1425; incompatibleMaterialItems() :2520-2567; en.php:204, 367`

Section 7 warning (preview only):

PRODUCT OVERRIDE Shown in the preview Warnings box, never printed on the SDS, whenever a non-blank Section 7 Storage override is stored for the sheet language. It says to list the incompatibles in the override or reset Storage to automatic.

On a sheet: Section 7 warning: the Storage text is a per-product override, so it no longer names ...

You maintain: The Section 7 Storage override (SDS editor)

If missing: No warning when Storage is automatic

`storageOverrideWarning() SDSGenerator.php:1410-1417, appended at :291 (generate) and :713 (generateFromBase); preview.php Warnings box`

SECTION 8: EXPOSURE CONTROLS / PERSONAL PROTECTION

Section 8 lists the federal and CAS-determination exposure limits for each ingredient, with the Section 3 concentration band and trade-secret identities masked. Under the table it names the components whose limit carries a Skin notation. When no limit applies, it prints a fixed "no exposure limits" sentence instead. The engineering-control and PPE sentences are chosen automatically from the product's final hazard codes (flammability comes from the product flash point), its physical state (product > family default > largest raw material) and whether any limit rows exist. On UV-family products, on lines whose tier is hazard-driven (or on all lines when the product has H317), the nitrile glove and UV eyewear sentences replace the tier sentences, and the UV respiratory and clothing sentences are added to theirs. Goggles and a face shield stay for H314/H318. Each line can be replaced by a per-product override.

Exposure limits table (which rows appear):

HAZARD ENGINE One row per exposure limit on file for each ingredient at 0.01% or more. Rows come from federal limits (NIOSH, ACGIH, OSHA), from the active CAS determination when that CAS has no current federal hazard classification, and from a vendor trade-secret hazard declaration that carries hazards. Inhalation-only CAS (carbon black, TiO2) and ingredients supplied only by solid or powder raw materials get no rows when any raw material is not solid or powder. A limit repeated with the same CAS, type, value and units prints once.

On a sheet: 108-88-3 | Toluene | PEL-TWA | 754 | mg/m3 | 10 - 30% | Skin; 29 CFR 1910.1000 Table Z-1

You maintain: Keep the federal data imports current. Add exposure limits on Admin > CAS determinations for chemicals with no federal data, and on the vendor hazard declaration of trade-secret raw materials. Set the physical state on every raw material, keep constituent CAS numbers and percentages complete, and keep the inhalation-only CAS list in Admin Settings current.

If missing: With no rows left, the table is replaced by the italic sentence "No occupational exposure limits apply to the components of this product in the form supplied."

`HazardEngine::classify :718-730 (hsr.is_current=1), trade-secret JSON :804, CPD :3322-3338; excludeInhalationOnlyCas :586; dedupeExposureLimits :1243; SDSGenerator :5562`

Exposure limits table headers: CAS \| Chemical \| Type \| Value \| Units \| Conc% \| Notes:

FIXED TEXT (TRANSLATED) Fixed column headings from the language file, printed above the limit rows only when at least one exposure-limit row exists.

On a sheet: CAS | Chemical | Type | Value | Units | Conc% | Notes

You maintain: Nothing directly (the exposure-limit rows decide whether the headings print)

If missing: English heading, then the key name. Not printed when there are no limit rows.

`labels.el_cas/el_chemical/el_type/el_value/el_units/el_conc_pct/el_notes; PDFService.php:833-840 widths [22,42,22,20,17,15,37] mm; preview.php:282`

CAS (exposure limits table):

FORMULA / RAW MATERIALS The ingredient's CAS number from the raw-material constituent. It prints "TRADE SECRET" instead when any raw material marks that CAS as a trade secret, or when the limit comes from a vendor trade-secret hazard declaration.

On a sheet: 108-88-3

You maintain: CAS number and the trade-secret box on each raw-material constituent (Raw materials screen)

If missing: A constituent with no CAS never gets a row. Vendor trade-secret declarations print "TRADE SECRET".

`rmc.cas_number via Formula::getExpandedComposition; SDSGenerator::maskTradeSecretRow :3092 -> labels.trade_secret_cas (is_trade_secret or CAS 'TRADE_SECRET')`

Chemical (exposure limits table):

FORMULA / RAW MATERIALS Ingredient name: Prop 65 list name > CAS master preferred name > constituent name typed on the raw material. It is never translated. For a trade-secret ingredient: trade-secret description > "Trade Secret".

On a sheet: Toluene

You maintain: Preferred name in the CAS master, or the constituent name on the raw material; the trade-secret description on trade-secret constituents

If missing: Blank cell. A trade-secret row with no description prints "Trade Secret" (translated).

`COALESCE(p65.chemical_name, cm.preferred_name, rmc.chemical_name) Formula.php:293; tradeSecretName() SDSGenerator.php:3072`

Type (exposure limits table):

REGULATORY LIST Limit type as stored in the federal data or chosen on the determination, printed exactly as stored.

On a sheet: TLV-TWA

You maintain: Federal data imports; the limit type on each determination row

If missing: Blank cell (determination rows only)

`exposure_limits.limit_type (NIOSH REL-/PEL-, ACGIH TLV-, OSHA PEL-…); CPD exp_limit_type[] PEL-TWA/TLV-TWA/REL-TWA/STEL/Ceiling/IDLH/Other`

Value (exposure limits table):

REGULATORY LIST Numeric limit from the federal data or the determination, printed as stored.

On a sheet: 20

You maintain: Federal data imports; the value on each determination limit

If missing: A determination row with no value is dropped

`exposure_limits.value (NOT NULL); CPD determination_json.exposure_limits[].value, rows with empty() value (incl. '0') skipped HazardEngine.php:3326`

Units (exposure limits table):

REGULATORY LIST Unit stored with the limit. Importers keep mg/m3 when the source gives it and ppm otherwise, so each limit prints one unit.

On a sheet: mg/m3

You maintain: Federal data imports; the units on each determination limit

If missing: "mg/m3"

`exposure_limits.units (DB default 'mg/m3'; NIOSHConnector and load-seed-data prefer *_mgm3); CPD units mg/m3/ppm/f/cc/mppcf, default 'mg/m3' HazardEngine.php:3334`

Conc% (exposure limits table):

FORMULA / RAW MATERIALS The ingredient's prescribed concentration band, the same band Section 3 prints, never the exact %. It is the widest band that contains the ingredient's summed lower and upper bounds. When no band contains both, it is the band containing the upper bound with the lowest lower end.

On a sheet: 10 - 30%

You maintain: Exact % or min/max % on raw-material constituents; formula line percentages

If missing: "<0.1%" when the upper bound is below 0.1%; "80 - 100%" when it is above 100%. Blank on snapshots published before #8.

`section8() SDSGenerator.php:2781-2799 -> formatConcentration :5292 (PRESCRIBED_RANGES :5278); bounds Formula::constituentBounds; concentration_pct unset`

Notes (exposure limits table):

REGULATORY LIST ACGIH rows print the notation. OSHA rows print the notation plus the CFR reference. Determination rows print free text, and NIOSH rows are blank. When the same limit is on file twice, the first row's notes print, or the first non-blank notes of a duplicate.

On a sheet: Skin; A4

You maintain: Federal seed files; notes on determination limits (a "Skin" note also feeds the skin-notation line)

If missing: Blank cell

`exposure_limits.notes from load-seed-data.php (notation [+ '; ' + cfr_ref]); CPD exposure_limits[].notes; merged in dedupeExposureLimits HazardEngine.php:1243`

Skin notation sentence (components with a skin notation):	REGULATORY LIST Printed in italics under the table when any limit row's Notes carry a Skin notation ("Skin", "[skin]", "Skin; A4"). It lists those rows' printed names (trade secrets by their description) in table order, each once. Notes such as "skin sensitizer", "skin irritant", "Eye & skin irr", "skin damage" and "no skin notation" do not count. *On a sheet: Components with a skin notation (absorption through the skin can …): Toluene, Xylene.* **You maintain:** Federal ACGIH/OSHA notations (seed files) and the notes on determination limits; trade-secret descriptions on constituents **If missing:** Not printed when no limit row carries a Skin notation, or when there are no limit rows. A row with a blank name is listed by its CAS. Snapshots published before batch F print nothing. `section8() SDSGenerator.php:2811-2819 -> skinNotationNames :2898 (SKIN_NOTATION_REGEX :2887 on masked rows) -> section8.skin_notation; PDFService.php:874-880; preview.php:305`
No-exposure-limits sentence:	FIXED TEXT (TRANSLATED) Fixed sentence from the language file, printed in italics in place of the table only when no exposure-limit row remains. *On a sheet: No occupational exposure limits apply to the components of this product in the form…* **You maintain:** Nothing directly (the exposure-limit rows decide whether it prints) **If missing:** Not printed when any limit row exists. Snapshots published before #64 print nothing in its place. `section8.no_exposure_limits -> exposure_limits_none SDSGenerator.php:2817; PDFService.php:882-888; preview.php:308-309`
Engineering Controls:	TEXT CHOSEN BY HAZARD A base ventilation sentence chosen by whether limit rows exist. Extra sentences follow: dust when the resolved physical state is Powder, explosion-proof and grounding for H224-H226 (Flammable Liquid Cat 1-3 from the flash point, or a finished-good hazard override), and eyewash for H314/H318. Precedence: per-product override > automatic sentences. *On a sheet: Use local exhaust ventilation or other engineering controls to maintain airborne…* **You maintain:** Physical State on the finished good, or the default physical state in the product family's SDS profile (else the largest raw material that has one). Flash points on raw materials, the finished-good hazard override, and the hazard data that produces H314/H318. Optional override in the SDS editor. **If missing:** Only the base sentence. With no state on the product, its family or any raw material, the state counts as Liquid (no dust sentence) and the preview warns. No flash point data means no flammable sentence. `non-blank text_overrides[8]['engineering'] > engineeringControls() SDSGenerator.php:2846-2862 (no limit rows -> engineering_no_limits, :2821); state = resolvePhysicalState :2980`
Engineering Controls: no-exposure-limits ventilation sentence:	TEXT CHOSEN BY HAZARD The base Engineering Controls sentence on a product with no exposure-limit rows (the same condition that prints the "No occupational exposure limits apply…" sentence). It replaces the sentence that refers to exposure limits, and the dust, flammable and eyewash sentences follow it unchanged. *On a sheet: Use adequate general ventilation. Use local exhaust ventilation where mist, vapor or dust…* **You maintain:** Nothing directly (the exposure-limit rows decide it); an Engineering Controls override in the SDS editor replaces the whole line **If missing:** With any limit row, "Use local exhaust ventilation or other engineering controls to maintain airborne concentrations below exposure limits." prints instead `section8.engineering_no_limits, chosen in engineeringControls() SDSGenerator.php:2851 when section8() passes $exposureLimits === [] (:2821)`
Respiratory Protection:	TEXT CHOSEN BY HAZARD One of five sentences chosen from the final H-codes (SCBA, sensitizer, cartridge respirator, general, none). On UV products the UV respiratory sentence is added after it when the tier is hazard-driven or the product has H317. Precedence: per-product override > automatic. *On a sheet: If exposure limits are exceeded or irritation is experienced, use a NIOSH-approved…* **You maintain:** Hazard data that drives the H-codes (federal data, determinations, vendor trade-secret declarations, finished-good hazard override). Optional override per language in the SDS editor (on the raw material for resale sheets). **If missing:** "No special respiratory protection required under normal conditions of use with adequate ventilation." `non-blank text_overrides[8]['respiratory'] > section8.ppe.respiratory.<tier> (HazardEngine::derivePPE :1298) in SDSGenerator::resolvePPE :2936 (+ section8.uv_respiratory)`

Hand Protection:

TEXT CHOSEN BY HAZARD Tier sentence chosen from the final H-codes (impervious, sensitizer, resistant, general, none). On UV products, when the tier is hazard-driven or the product has H317, the UV nitrile glove sentence replaces it, including the impervious (butyl/Viton) tier on corrosive UV inks. Precedence: per-product override > automatic.

On a sheet: Chemical-resistant gloves (nitrile recommended) meeting 29 CFR 1910.138…

You maintain: Hazard data that drives the H-codes; the product family's UV/LED flag; optional override in the SDS editor

If missing: "No special hand protection required under normal conditions of use. Gloves may be worn as good industrial hygiene practice."

`override > section8.ppe.hand_protection.<tier> (H310/311/314 impervious; H317 sensitizer; H312/313/315/316/systemic resistant); UV: uv_hand_protection replaces, resolvePPE :2958`

Eye Protection:

TEXT CHOSEN BY HAZARD Tier sentence chosen from the final H-codes (goggles and face shield, goggles, glasses, general, none). On UV products, when the tier is hazard-driven or the product has H317, the UV eyewear sentence replaces it. The goggles-and-face-shield tier (H314/H318) stands alone. Precedence: per-product override > automatic.

On a sheet: Chemical splash goggles or safety glasses with side shields meeting ANSI/ISEA Z87.1…

You maintain: Hazard data that drives the H-codes; the product family's UV/LED flag; optional override in the SDS editor

If missing: "No special eye protection required under normal conditions of use. Safety glasses may be worn as good industrial hygiene practice."

`override > section8.ppe.eye_protection.<tier> (H314/318 goggles_faceshield; H319 goggles; H320 glasses); UV: uv_eye_protection replaces, except goggles_faceshield (UV_PPE_TIER_FLOOR :2875)`

Skin Protection:

TEXT CHOSEN BY HAZARD Tier sentence chosen from the final H-codes (suit, clothing, general, none). On UV products the UV clothing sentence is added after it when the tier is hazard-driven or the product has H317. Precedence: per-product override > automatic.

On a sheet: Wear protective clothing (long sleeves and an impervious apron) to prevent skin contact…

You maintain: Hazard data that drives the H-codes; optional override in the SDS editor

If missing: "No special protective clothing required under normal conditions of use."

`non-blank text_overrides[8]['skin_protection'] > section8.ppe.skin_protection.<tier> (H310/H311/H314 suit; H312/H313/H315/H316/H317/systemic clothing) (+ section8.uv_skin_protection)`

UV acrylate PPE sentences (replace the glove and eyewear sentences; added to the respiratory and clothing sentences):

PRODUCT FAMILY Used when the UV rule pack is on and the product's resolved, active family is marked UV/LED, only on lines whose tier is hazard-driven, or on all four lines when the product has H317. The nitrile glove and eyewear sentences replace the tier sentence (goggles and face shield for H314/H318 stay instead), and the respiratory and clothing sentences are added after it. No acrylate ingredient is needed. Section 2 PPE prints the same text, and a line with an override gets none.

On a sheet: For uncured acrylate product, wear chemical-resistant gloves (nitrile, minimum 0.4 mm…

You maintain: The UV/LED flag and Active status on the product family (Product families screen) and the product's family. The UV rule pack toggle in Admin Settings.

If missing: No UV text; the tier sentences print. An unclassified UV product without H317 keeps its "No special … required" sentences.

`UVAcrylateRulePack::isApplicable :105 (rule pack on && family_is_uv, familyFields :1265) -> resolvePPE :2936; UV_PPE_REPLACE_FIELDS :2865, UV_PPE_TIER_FLOOR :2875`

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Section 9 prints every HazCom Appendix D property, then VOC and solids. Most lines come from the finished good, its product family and its raw materials: the product's own fields, the family SDS profile (default physical state, typical pH), the largest raw material, the lowest value, a weighted average or weighted sums. The automatic flash point is rounded to a whole degree and marked "(calculated)". Eleven Appendix D properties have no data field, so they print "Not determined" unless a value is typed for the product. Seventeen lines can be replaced per product and per language. Labels, the title and the fallback wording come from the translation files.

Physical State:

FORMULA / RAW MATERIALS The product's own physical state > its product family's default physical state > the state of the largest raw material (by summed wt%) that has one > 'Liquid'. The six standard states are translated.

On a sheet: Physical State: Paste

You maintain: Physical state on the finished good (Finished Goods > Edit), the default physical state in the family SDS profile (Product Families > Edit), or the raw materials' state (Raw Materials > Edit).

If missing: 'Liquid' (Líquido / Liquide / Flüssig), and the preview shows a warning. A raw with no state is skipped and the next-largest raw with a state is used. An inactive family's default is ignored.

`resolvePhysicalState (SDSGenerator.php:2980-2993): finished_goods.physical_state > family_profile.default_physical_state (A11) > formula_props.physical_state (FormulaCalcService.php:605-626) > 'Liqui…`

Color:

FORMULA / RAW MATERIALS The product's colour dropdown. The seven default colours are translated; a custom colour prints as entered.

On a sheet: Color: Black

You maintain: Colour on the finished good (Finished Goods > Edit).

If missing: The Color line is left off. It is always left off on resale raw-material sheets, because raw materials have no colour field.

`finished_goods.color (vocab setting sds.color_options) -> localizeSection9Enum('color') (SDSGenerator.php:3124-3125, 3044-3062); PDF labelValue skips ''`

Appearance:

FORMULA / RAW MATERIALS Per-product edit > product colour + resolved state, in each language's word order > the state alone. Resale sheets use the raw material's appearance text before falling back to the state.

On a sheet: Appearance: Black paste (ES: Pasta, negro)

You maintain: Colour and physical state on the finished good (or the family default state). For resale items, appearance on the raw material. Optional edit in SDS > Edit Text, per language.

If missing: The state alone, e.g. 'liquid' (EN, lower case) or 'Líquido' (ES). Never blank.

`text_overrides(9,'appearance',lang) > section9.appearance_color_state if finished_goods.color > (is_resale) formula_props.appearance > section9.appearance_state (SDSGenerator.php:3160-3184)`

Odor:

FORMULA / RAW MATERIALS Per-product edit > odour of the raw material with the highest summed wt% > 'Not determined'. Raw-material text is not translated.

On a sheet: Odor: Mild solvent

You maintain: Odor on the raw material that dominates each formula (Raw Materials > Edit). Optional edit in SDS > Edit Text.

If missing: 'Not determined' (No determinado / Non déterminé / Nicht bestimmt). The second-largest raw is not tried.

`text_overrides(9,'odor',lang) > formula_props.odor (dominant raw_materials.odor; FormulaCalcService.php:577-627) > labels.not_determined (SDSGenerator.php:3190-3196)`

Odor Threshold:

PRODUCT OVERRIDE 'Not determined' on every sheet, unless a value is typed for the product in the editor. No raw-material or formula data feeds it.

On a sheet: Odor Threshold: Not determined

You maintain: Only when a tested value exists: SDS > Edit Text, Section 9, per language (resale items: Edit Resale SDS, stored on the raw material).

If missing: 'Not determined'

`SECTION9_APPENDIX_D_FIELDS (SDSGenerator.php:3013-3017); text_overrides(9,'odor_threshold',lang) > labels.not_determined (:3207-3219)`

pH:

PRODUCT FAMILY Per-product value typed in the editor > the product family's typical pH range from its SDS profile, marked '(typical)' (one value when min = max) > 'Not determined'. Display only; never used for classification or RCRA D002.

On a sheet: pH: 8.0 – 9.5 (typical)

You maintain: Typical pH minimum and maximum in the family SDS profile (Product Families > Edit > SDS profile), entered from verified QC ranges. A tested value per product: SDS > Edit Text, Section 9.

If missing: 'Not determined'. A half pair, inverted range, value outside 0-14 or an inactive family also gives 'Not determined'.

`text_overrides(9,'ph',lang) > typicalPhText (SDSGenerator.php:1382-1403; section9.ph_typical_range / ph_typical_value; family_profile.ph_min/ph_max, 0-14, min<=max) > labels.not_determined (:3207-321…`

Melting Point/Freezing Point:

PRODUCT OVERRIDE 'Not determined' on every sheet, unless a value is typed for the product in the editor. No raw-material or formula data feeds it.

On a sheet: Melting Point/Freezing Point: Not determined

You maintain: Only when a tested value exists: SDS > Edit Text, Section 9, per language.

If missing: 'Not determined'

`SECTION9_APPENDIX_D_FIELDS (SDSGenerator.php:3013-3017); text_overrides(9,'melting_point',lang) > labels.not_determined (:3207-3219)`

Initial Boiling Point and Boiling Range:

FORMULA / RAW MATERIALS Per-product edit, if it reads as a temperature and does not move a Category 1/2 product across 35 °C > lowest boiling point of any raw in the full formula (wt% ignored) > 'Not determined'.

On a sheet: Initial Boiling Point and Boiling Range: 78.4 °C (173.1 °F)

You maintain: Boiling point (°C) on solvent and other volatile raw materials (Raw Materials > Edit). Pigments and resins can stay blank. Optional edit in SDS > Edit Text, with a °C/°F unit.

If missing: 'Not determined'. If the flash point is below 23 °C, the product is assumed Flammable Liquids Category 2 / PG II and the preview warns.

`resolveBoilingPointDisplay (SDSGenerator.php:2068-2076) after vetBoilingPointEdit (:4658); MIN(raw_materials.boiling_point_c) (FormulaCalcService.php:552, 623); TemperatureParser`

Flash Point:

FORMULA / RAW MATERIALS Per-product edit, kept only if it is a temperature giving the same flammability result, printed exactly as typed > wt%-weighted average over raws with a flash point, rounded to a whole degree, '>' if any is flagged, marked '(calculated)' > 'Not determined'. Section 5 prints the same text on a flammable product.

On a sheet: Flash Point: > 71 °C (159 °F) (calculated)

You maintain: Flash point and '>' flag on every raw material, including a high value (e.g. 100) for water and other non-flammable liquid raws (Raw Materials > Edit). A measured value only in SDS > Edit Text.

If missing: 'Not determined'. The product is then treated as not flammable (no Flammable Liquids class, Section 14 not Class 3) unless a finished-good hazard override sets one. Publishing is not blocked.

`resolveFlashPointDisplay (SDSGenerator.php:2034-2043) after vetFlashPointEdit (:4616); flashPointDisplay :2010 + roundFlashPointC :1977 (guards 23/60/93); weightedFlashPoint (FormulaCalcService.php:6…`

Evaporation Rate:

PRODUCT OVERRIDE 'Not determined' on every sheet, unless a value is typed for the product in the editor. No raw-material or formula data feeds it.

On a sheet: Evaporation Rate: Not determined

You maintain: Only when a tested value exists: SDS > Edit Text, Section 9, per language.

If missing: 'Not determined'

`SECTION9_APPENDIX_D_FIELDS (SDSGenerator.php:3013-3017); text_overrides(9,'evaporation_rate',lang) > labels.not_determined (:3207-3219)`

Flammability (Solid, Gas):

PRODUCT OVERRIDE Per-product value typed in the editor > 'Not applicable' when the resolved physical state is Liquid > 'Not determined' (Solid, Powder, Paste, Gel, Gas).

On a sheet: Flammability (Solid, Gas): Not applicable

You maintain: Physical state on the finished good, its family default or its raws. Optional value in SDS > Edit Text.

If missing: 'Not applicable' for liquids; otherwise 'Not determined'.

`SDSGenerator.php:3207-3219: text_overrides(9,'flammability_solid_gas') > strcasecmp(physicalState,'Liquid')==0 -> labels.not_applicable > labels.not_determined`

Upper/Lower Flammability or Explosive Limits:

PRODUCT OVERRIDE 'Not determined' on every sheet, unless a value is typed for the product in the editor. No raw-material or formula data feeds it.

On a sheet: Upper/Lower Flammability or Explosive Limits: Not determined

You maintain: Only when a tested value exists: SDS > Edit Text, Section 9, per language.

If missing: 'Not determined'. The label is too wide for the column, so the value prints on the next line.

`SECTION9_APPENDIX_D_FIELDS (SDSGenerator.php:3013-3017); text_overrides(9,'flammability_limits',lang) > labels.not_determined (:3207-3219)`

Vapor Pressure:

PRODUCT OVERRIDE 'Not determined' on every sheet, unless a value is typed for the product in the editor. No raw-material or formula data feeds it.

On a sheet: Vapor Pressure: Not determined

You maintain: Only when a tested value exists: SDS > Edit Text, Section 9, per language.

If missing: 'Not determined'

`SECTION9_APPENDIX_D_FIELDS (SDSGenerator.php:3013-3017); text_overrides(9,'vapor_pressure',lang) > labels.not_determined (:3207-3219)`

Vapor Density:

PRODUCT OVERRIDE 'Not determined' on every sheet, unless a value is typed for the product in the editor. No raw-material or formula data feeds it.

On a sheet: Vapor Density: Not determined

You maintain: Only when a tested value exists: SDS > Edit Text, Section 9, per language.

If missing: 'Not determined'

`SECTION9_APPENDIX_D_FIELDS (SDSGenerator.php:3013-3017); text_overrides(9,'vapor_density',lang) > labels.not_determined (:3207-3219)`

Specific Gravity:

FORMULA / RAW MATERIALS Weight-averaged mixture specific gravity, Σw/Σ(w/SG), from every raw material's specific gravity. A raw with no value counts as 1.0.

On a sheet: Specific Gravity: 1.052

You maintain: Specific gravity on every raw material (Raw Materials > Edit).

If missing: Each missing raw counts as 1.0, and the preview names it in a warning. 'Not determined' never prints in practice.

`VOCCalculator::getMixtureSG (VOCCalculator.php:248-279), applyDefaults SG NULL -> 1.0 (:613-622); round(,3) (SDSGenerator.php:3237); dataGapWarnings (FormulaCalcService.php:661)`

Solubility:

FORMULA / RAW MATERIALS Per-product edit > band from the soluble fraction (Soluble 1, Partially 0.5, Negligible 0.03, Insoluble 0; raws without a value skipped): ≥90 / ≥5 / ≥1 / below > 'Not determined'.

On a sheet: Solubility: Negligible solubility in water

You maintain: Solubility dropdown on raw materials (Raw Materials > Edit). Optional edit in SDS > Edit Text.

If missing: 'Not determined' when no raw material in the formula has a solubility value.

`text_overrides(9,'solubility') (SDSGenerator.php:3147-3153) > section9.solubility_<key> (solubilityText :3024); SOLUBILITY_WEIGHTS (FormulaCalcService.php:30-35), solubility_key :624`

Partition Coefficient: n-Octanol/Water:	PRODUCT OVERRIDE 'Not determined' on every sheet, unless a value is typed for the product in the editor. No raw-material or formula data feeds it. *On a sheet: Partition Coefficient: n-Octanol/Water: Not determined* **You maintain:** Only when a tested value exists: SDS > Edit Text, Section 9, per language. **If missing:** 'Not determined'. The label is too wide for the column, so the value prints on the next line. `SECTION9_APPENDIX_D_FIELDS (SDSGenerator.php:3013-3017); text_overrides(9,'partition_coefficient',lang) > labels.not_determined (:3207-3219)`
Auto-Ignition Temperature:	PRODUCT OVERRIDE 'Not determined' on every sheet, unless a value is typed for the product in the editor. No raw-material or formula data feeds it. *On a sheet: Auto-Ignition Temperature: Not determined* **You maintain:** Only when a tested value exists: SDS > Edit Text, Section 9, per language. **If missing:** 'Not determined' `SECTION9_APPENDIX_D_FIELDS (SDSGenerator.php:3013-3017); text_overrides(9,'auto_ignition_temp',lang) > labels.not_determined (:3207-3219)`
Decomposition Temperature:	PRODUCT OVERRIDE 'Not determined' on every sheet, unless a value is typed for the product in the editor. No raw-material or formula data feeds it. *On a sheet: Decomposition Temperature: Not determined* **You maintain:** Only when a tested value exists: SDS > Edit Text, Section 9, per language. **If missing:** 'Not determined' `SECTION9_APPENDIX_D_FIELDS (SDSGenerator.php:3013-3017); text_overrides(9,'decomposition_temp',lang) > labels.not_determined (:3207-3219)`
Viscosity:	PRODUCT OVERRIDE 'Not determined' on every sheet, unless a value is typed for the product in the editor. No raw-material or formula data feeds it. *On a sheet: Viscosity: Not determined* **You maintain:** Only when a tested value exists: SDS > Edit Text, Section 9, per language. **If missing:** 'Not determined' `SECTION9_APPENDIX_D_FIELDS (SDSGenerator.php:3013-3017); text_overrides(9,'viscosity',lang) > labels.not_determined (:3207-3219)`
VOC (lb/gal) (EPA Method 24):	FORMULA / RAW MATERIALS VOC weight fraction (normalised to the formula total) × mixture specific gravity × 8.345. A raw's VOC % is 0.99 if flagged '<1%', otherwise its VOC %, and 0 when blank. *On a sheet: VOC (lb/gal) (EPA Method 24): 2.15* **You maintain:** VOC % (or the <1% flag) and specific gravity on every raw material (Raw Materials > Edit). **If missing:** Missing VOC counts as 0 and missing specific gravity as 1.0, and the preview names each such raw. A number always prints (e.g. '0'). `getVOCLbPerGal (VOCCalculator.php:289-306) = getTotalVOCWeightPercent Σ(pct·effVOC)/Σpct (:181-200) /100 · getMixtureSG · 8.345; effVOC :626-671; round(,2) SDSGenerator.php:3243`
VOC (wt%):	FORMULA / RAW MATERIALS '<1' when every raw material in the formula is flagged '<1% VOC'. Otherwise the weighted average of raw VOC %, normalised to the formula total (0.99 for flagged raws, 0 when blank). *On a sheet: VOC (wt%): 12.35 or VOC (wt%): <1* **You maintain:** VOC % or the <1% flag on every raw material (Raw Materials > Edit). **If missing:** Missing VOC counts as 0, and the preview names each such raw. A formula with no valid lines prints '<1'. `formula_props.all_voc_less_than_one ? '<1' : round(total_voc_wt_pct,2) (SDSGenerator.php:3140-3144); getTotalVOCWeightPercent Σ(pct·effVOC)/Σpct (VOCCalculator.php:181-200)`
Solids (wt%):	FORMULA / RAW MATERIALS Weighted average of each raw's solids %, normalised to the formula total. When a raw has no solids %, it is calculated as 100 − VOC − exempt solvent − water. *On a sheet: Solids (wt%): 64.2* **You maintain:** Solids % on every raw material, or accurate VOC/exempt/water % and constituents (Raw Materials > Edit). Keep the exempt VOC CAS list current. **If missing:** Back-calculated with no preview warning; only the formula Calculate page lists it as an assumption. A number always prints. `getSolidsWeightPercent Σ(pct·effSolids)/Σpct (VOCCalculator.php:390-404); effSolids :709-724, water 7732-18-5 :684-707; round(,1) SDSGenerator.php:3245`
'Not determined' value:	FIXED TEXT (TRANSLATED) Fixed no-data wording from the language file. It prints for Odor, Boiling Point, Flash Point, Solubility, pH and the Appendix D properties that have no value. *On a sheet: Not determined* **You maintain:** Nothing **If missing:** English wording, then the key text. `labels.not_determined via TranslationService (SDSGenerator.php:3118); en.php:830, es/fr.php:793, de.php:789`

'Not applicable' value:	FIXED TEXT (TRANSLATED) Fixed wording from the language file, printed for Flammability (Solid, Gas) only when the resolved physical state is Liquid. *On a sheet: Not applicable (ES: No aplica; FR: Non applicable; DE: Nicht zutreffend)* **You maintain:** Physical state on the product, its family default or its raws. **If missing:** English wording, then the key text. `labels.not_applicable (SDSGenerator.php:3215); en.php:832, es/fr.php:795, de.php:791`
'(calculated)' marker on Flash Point:	FIXED TEXT (TRANSLATED) Added after the flash point only when it is the formula value. A kept per-product edit is treated as measured and prints without it; Section 5 uses the same text. *On a sheet: 74 °C (165 °F) (calculated) (ES: calculado; FR: calculé; DE: berechnet)* **You maintain:** Nothing (enter a tested flash point in SDS > Edit Text to drop the marker). **If missing:** Not printed when the flash point is an edit or 'Not determined'. `section9.flash_point_calculated ':fp (calculated)' (en.php:306, es/fr/de.php:287) passed to resolveFlashPointDisplay (SDSGenerator.php:3132, Section 5 :2163)`
'(typical)' marker on pH:	FIXED TEXT (TRANSLATED) Wraps the product family's typical pH range (or single value when min = max), one decimal, with the language's 'typical' word. *On a sheet: 8.0 – 9.5 (typical) (ES: típico; FR: typique; DE: typisch)* **You maintain:** Typical pH in the family SDS profile (Product Families > Edit). **If missing:** Not printed when the family sets no pH or a per-product pH edit exists. `section9.ph_typical_range ':ph_min – :ph_max (typical)' / ph_typical_value (en.php:303-304, es/fr/de.php:284-285); typicalPhText (SDSGenerator.php:1382)`
Section 9 preview warnings (not printed):	PUBLISH RECORD The preview Warnings box lists Section 9 data gaps and dropped edits: raws missing SG or VOC, the 'Liquid' default, a family default state that differs from the largest raw, temperature edits that are not temperatures or contradict the classification, and flash/boiling-point gaps. *On a sheet: Physical state: this product has no Physical State of its own, so the sheet uses its product family default "Paste" ...* **You maintain:** Act on each warning on the screen it names (Raw Materials > Edit, Finished Goods > Edit, Product Families > Edit, or SDS > Edit Text). **If missing:** Nothing prints on the sheet. None of these warnings blocks publishing. `FormulaCalcService::dataGapWarnings (:661); PHYSICAL_STATE_DEFAULT_WARNING (SDSGenerator.php:2996); familyPhysicalStateWarning :1243; temperatureOverrideWarnings :2109; vetFlashPointEdit :4616; flamm…`

SECTION 10: STABILITY AND REACTIVITY

Section 10 is built mostly from the product's final hazard classification (H-codes), the nitrogen, sulfur and halogen flags on its constituent CAS numbers, and its product family. Flammability has one source, the Flammable Liquids category worked out from the product flash point (wt%-weighted over the raws that have one), the initial boiling point and the physical state, so Section 10 agrees with Sections 2, 5, 7, 9, 13 and 14. A UV/LED family adds the UV-light condition, the hazardous-polymerization Reactivity sentence and the peroxides/initiators incompatible (plus an inhibitor sentence when acrylates are present), and the family SDS profile adds a named storage maximum and, for water-based products, freezing. The translation files supply the wording, and a per-product, per-language override can replace any of the five lines.

Reactivity / Chemical Stability / Conditions to Avoid / Incompatible Materials / Hazardous Decomposition Products (labels):	FIXED TEXT (TRANSLATED) Fixed label text in the sheet's language. The decomposition label is wider than the 50 mm PDF label column, so its value prints on the next line. *On a sheet: Hazardous Decomposition Products:* **You maintain:** Nothing **If missing:** A label key missing from every translation file prints the raw key, e.g. "labels.decomposition_products:"; an old snapshot without the label and no translation prints the field key in title case, e.g. "Decomposition". All five values are always generated and blank override rows are ignored, so no Section 10 line is left out. `meta.labels[key] = t->get('labels.<key>') (getLabels SDSGenerator:5347) > SDSDocumentStrings::translate (old snapshots) > ucwords(key); PDFService FIELD_LABEL_MAP :52-56; preview:626-630`

Reactivity:

TEXT CHOSEN BY HAZARD Comes from the product's hazard classification and family. Each reactive class present adds its own sentence (explosive, self-reactive or organic peroxide, pyrophoric, self-heating, water-reactive, unstable gas, oxidizer), then a UV/LED family adds the polymerization sentence. Precedence: per-product override > class and UV sentences > standard sentence.

On a sheet: Oxidizer: may cause or intensify fire in contact with combustible materials.

You maintain: Hazard data that produces H200-H205, H230-H232, H240-H242, H250-H252, H260/H261 or H270-H272: CAS hazard data and determinations (/determinations) count at 1% or more; manual and trade-secret H-codes (Raw materials screen) and the finished-good hazard override count at any level. The family UV/LED flag (Product families screen).

If missing: "No dangerous reaction known under conditions of normal use." (only when no reactive class applies and the product is not in an active UV/LED family)

`override(10,'reactivity') > section10.reactivity_<class> (H200-205/230-232/240-242/250-252/260-261/270-272) + reactivity_uv_polymerization joined ' ' > section10.reactivity; SDSGenerator:3306-3320`

Reactivity: UV hazardous polymerization sentence:

PRODUCT FAMILY Added after any class sentences when the product's assigned family is active and flagged UV/LED. Alone it replaces the 'No dangerous reaction known' sentence. It ignores the UV acrylate rule-pack setting and acrylate detection.

On a sheet: Hazardous polymerization may occur when uncured product is exposed to UV light, ...

You maintain: The UV/LED flag and Active status of each family, and family rules or manual picks so products land in the right family (Product families screen). Recompute families after imports or rule changes.

If missing: Sentence omitted when there is no family, the family is inactive or not flagged UV, or a Reactivity override exists.

`section10.reactivity_uv_polymerization when $isUv = UVAcrylateRulePack::familyIsUv($fg) (SDSGenerator:222/:446/:557); appended :3314-3316`

Chemical Stability:

TEXT CHOSEN BY HAZARD The same reactive classes as Reactivity except oxidizers, listed as the conditions under which the product is unstable; a UV/LED product with acrylates then adds the inhibitor sentence. Precedence: per-product override > 'Unstable under ...' sentence > standard sentence.

On a sheet: Unstable under the following conditions: contact with water or moisture (releases ...).

You maintain: Same hazard inputs as Reactivity (oxidizer codes H270-H272 do not change this line), plus the family UV/LED flag and the CAS acrylate flags for the inhibitor sentence.

If missing: "Stable under recommended storage conditions."

`override(10,'stability') > section10.stability_unstable(:conditions = stability_cond_<class> joined '; ', no oxidizer) > section10.stability, + stability_uv_inhibitor; SDSGenerator:3322-3338`

Chemical Stability: UV inhibitor / oxygen sentence:

PRODUCT FAMILY Appended to the stability sentence when the family is active and flagged UV/LED and at least one acrylate is found in the composition at 0.1% or more. The CAS acrylate flag decides (Yes or a manual No); when it is not set, the old CAS list or the name rule is used. Trade-secret acrylates count.

On a sheet: Contains acrylates stabilized with a polymerization inhibitor that requires oxygen ...

You maintain: Family UV/LED flag and Active status (Product families screen). The acrylate flag on each CAS: 'Acrylate' box on the CAS Descriptions tab (/determinations), seeded through /determinations/acrylate-flags or scripts/seed-cas-acrylate-flags.php.

If missing: Sentence omitted when the product is not in an active UV/LED family, no constituent counts as an acrylate at 0.1% or more (flag No, or no flag and neither the CAS list nor the name matches), or a Stability override exists.

`section10.stability_uv_inhibitor when $isUv && UVAcrylateRulePack::detectAcrylates($composition) !== [] (cas_master.is_acrylate via effectiveFlag, NAME_CUTOFF_PCT 0.1); SDSGenerator:3335-3337`

Conditions to Avoid:

TEXT CHOSEN BY HAZARD A list picked by the hazard classification and family. Heat is always first. A storage-temperature item, freezing, ignition sources, air, water and shock are added for the products that need them, oxidizers get combustibles instead of oxidizing agents, and UV families get UV light. Precedence: per-product override > generated list.

On a sheet: Excessive heat, sparks, open flames and other ignition sources, water and moisture, ...

You maintain: Same hazard inputs as Reactivity, the flash-point and physical-state inputs behind the ignition item, flammable gas/aerosol/solid codes (H220-H223, H228), and the family UV/LED flag and SDS profile (storage temperatures, Water-based) on the Product families screen.

If missing: "Excessive heat, contact with strong oxidizing agents." (never empty)

`override(10,'conditions_avoid') > section10.cond_<key> in COND_ORDER (SDSGenerator:2457-2459), items chosen at :3341-3389 from extractHCodes (:1106), family profile and $isUv`

Conditions to Avoid: storage temperature item:	PRODUCT FAMILY When the active family's SDS profile has a storage range, its maximum is named in °C and °F for every product in the family. Otherwise a self-reactive or self-heating product gets the generic 'temperatures above the recommended storage limit'. Precedence: per-product override > family maximum > generic item. *On a sheet: Excessive heat, temperatures above 30 °C (86 °F), contact with strong oxidizing agents.* **You maintain:** Storage temperature minimum and maximum (°C, both or neither, min below max) in each family's SDS profile (Product families > Edit). Saving the profile bumps the family's products for republish. **If missing:** With no range (no family, inactive family, or blank profile) the item prints only for H240-H242/H251/H252, as "temperatures above the recommended storage limit"; otherwise it is omitted. `cond_storage_max(:max_c/:max_f from profileTempPlaceholders :1349) when profileStorageMaxC set (setProfileState :1311), else cond_storage_temp on H240-242/H251-252; :3343-3351`
Conditions to Avoid: freezing item:	PRODUCT FAMILY Added for water-based products. Precedence: family Water-based Yes or No > Auto, where Auto (or no or inactive family) means water (CAS 7732-18-5, including inside intermediates) is at or above the Water-Based Threshold share of the formula. *On a sheet: Excessive heat, freezing, contact with strong oxidizing agents.* **You maintain:** Water-based (Auto / Yes / No) in each family's SDS profile (Product families screen); Water-Based Threshold (%) in Admin Settings (default 20); water as a CAS 7732-18-5 constituent on raw materials (Raw materials screen). **If missing:** Item omitted when the family says No, or on Auto when water is below the threshold. Water held only in a raw's water weight field, without a 7732-18-5 constituent, is not counted. `cond_freezing when profileWaterBased = WaterBasedRule::forProduct(family_profile.water_based, composition, threshold()) via applyFamilyProfile (:1229-1236); :3353-3355`
Conditions to Avoid: sparks, open flames and other ignition sources:	HAZARD ENGINE Added for a flammable liquid (Cat 1-4 from the wt%-weighted product flash point, initial boiling point and physical state), a flammable gas, aerosol or solid, or a pyrophoric, self-reactive, oxidizing, explosive or water-reactive product. Precedence for the liquid category: finished-good hazard override > flash-point category; physical state: product > family default > largest raw. *On a sheet: Excessive heat, sparks, open flames and other ignition sources, contact with strong ...* **You maintain:** Flash point (with its '>' flag) and physical state on every raw material (Raw materials screen). Physical state on the finished good (Finished goods screen) or the family's Default physical state (Product families screen). A Flammable Liquids class or H224-H227 in the finished-good hazard override replaces the derived category. **If missing:** Item omitted when no raw has a flash point (not flammable), the weighted flash point is above 93 °C, or the resolved state is Solid, Powder or Paste, unless another ignition class applies. A Section 9 flash point override does not change it. `cond_ignition on H220-228/H200-205/H240-242/H250/H260-261/H270-272 (:3358); H224-H227 only from HazardEngine::applyFlammableLiquidsFromFlashPoint (flammabilityInputs :4546) or FG override`
Conditions to Avoid: UV light item:	PRODUCT FAMILY Added at the end of Conditions to Avoid when the product's assigned family is active and flagged UV/LED. It does not depend on the UV acrylate setting or on acrylate detection. *On a sheet: ..., exposure to UV light and direct sunlight (risk of premature polymerization; ...).* **You maintain:** The UV/LED flag and Active status of each family, plus family rules or manual picks so that finished goods and resale raws land in the right family (Product families screen). Recompute families after imports or rule changes; use 'Reset legacy manual picks to Auto' for picks carried over from the old family name. **If missing:** Item omitted when there is no family, the family is inactive, or it is not flagged UV. A conditions_avoid override also removes it. `fg.family_is_uv from attachFamily/familyFields (SDSGenerator:1215-1284; inactive family -> null); UVAcrylateRulePack::familyIsUv :95-98; section10.cond_uv (:3378-3380)`
Incompatible Materials:	TEXT CHOSEN BY HAZARD A list picked by the hazard classification and family: acids and bases always, oxidizing agents unless the product is an oxidizer, halogens for flammables (including flash-point Flammable Liquids), extra items for self-reactive, pyrophoric, water-reactive, oxidizer and H290, and peroxides/initiators for UV families. Precedence: per-product override > generated list; Section 7 Storage repeats the value. *On a sheet: Strong oxidizing agents, strong acids, strong bases, halogens.* **You maintain:** Same hazard inputs as Reactivity, the flash-point inputs (a flammable product adds halogens), corrosive-to-metals (H290) in hazard data, and the family UV/LED flag. An override here also changes Section 7 Storage. **If missing:** "Strong oxidizing agents, strong acids, strong bases." (never empty) `resolveIncompatibleMaterials (:2574): override(10,'incompatible') > section10.incompat_<key> in INCOMPAT_ORDER (:2502-2506), items :2520-2567; Section 7 inline copy :2589`

Incompatible Materials: peroxides and free-radical initiators:

PRODUCT FAMILY Added at the end of the list when the product's family is active and flagged UV/LED, regardless of the rule-pack setting or acrylate detection. Section 7 Storage repeats it.

On a sheet: Strong oxidizing agents, strong acids, strong bases, peroxides and free-radical initiators.

You maintain: Family UV/LED flag and Active status, and correct family assignment (Product families screen).

If missing: Item omitted when the product is not in an active UV/LED family, or an Incompatible override exists.

`incompatibleMaterialItems($hazardResult, $uv): $uv ??= uvFamily (:2523); section10.incompat_peroxides_initiators (SDSGenerator:2556-2558); Section 10 passes $isUv (:3392)`

Hazardous Decomposition Products:

FORMULA / RAW MATERIALS Comes from the nitrogen, sulfur and halogen flags on each ingredient's CAS number, at any concentration. One element names that gas and two or more list them all; Section 5 names the same gases. Precedence: per-product override > element sentence > standard sentence.

On a sheet: Carbon monoxide, carbon dioxide, nitrogen oxides, and other toxic gases may be ...

You maintain: Element flags for every constituent CAS (/determinations, CAS Descriptions tab, 'Flags'), and the molecular formula in the CAS registry, which the seeder uses first. A CAS first saved on a raw material gets flags guessed from its name. Re-run the element-flag seeder (/determinations/element-flags or scripts/seed-cas-element-flags.php) after CAS rows arrive through CAS Descriptions or TSCA saves, seed data or direct inserts.

If missing: "Carbon monoxide, carbon dioxide, and other toxic gases may be released upon thermal decomposition." This prints when no constituent is flagged, the CAS is not in the registry or its flags are 0, the constituent is a no-CAS trade secret or sits in a hazardous no-CAS raw, or the CAS is blank (that constituent is dropped). A trade secret with a CAS keeps its flags.

`override(10,'decomposition') > decomposition_<el> / decomposition_multi / decomposition; decompositionElements (:2476) over cas_master has_*; SDSGenerator:3395-3410`

Decomposition base wording (carbon monoxide, carbon dioxide, other toxic gases):

FIXED TEXT Fixed wording inside every decomposition sentence. There is no carbon flag, so it prints even for wholly inorganic products unless the whole line is overridden.

On a sheet: Carbon monoxide, carbon dioxide, ...

You maintain: Nothing

If missing: Always printed unless overridden

`Fixed text inside section10.decomposition, decomposition_<el>, decomposition_multi (en.php:312-315, :349; ES/FR/DE word order differs)`

SECTION 11: TOXICOLOGICAL INFORMATION

Section 11 opens with the likely routes of exposure and the same symptoms line as Section 4. Three acute-toxicity route lines follow, then the chronic-effect fragments and the carcinogen and component block. All of this comes from the hazard classification (federal data, CPDs, trade-secret declarations, ATE summation, the Finished good hazard override), the carcinogen registry and the exposure-limit datasets. Since batch F, the STOT statements also name the target organs and route entered on CAS Determinations. Trade-secret identities are masked. Each product can override the Acute Toxicity, Chronic Effects, Carcinogenicity and UV Acrylate Information texts, and the symptoms line follows the Section 4 override.

Likely Routes of Exposure:

FIXED TEXT (TRANSLATED) Fixed label from the translation files.

On a sheet: Likely Routes of Exposure:

You maintain: Nothing

If missing: Sheet language > English > the literal key 'labels.routes_of_exposure'. Snapshots made before this line existed skip the whole line.

`labels.routes_of_exposure (en.php:775; getLabels :5347); PDFService::renderSection11 :935 labelValue :1369 (skips a blank value)`

Likely routes of exposure (text):

TEXT CHOSEN BY HAZARD Skin Contact and Eye Contact always print. Inhalation is added for H330-H332 or H334-H336, or when any ingredient keeps an exposure limit after the physical-state filters. Ingestion is added for H300-H302, H304 or H305 (not H303).

On a sheet: Inhalation, Skin Contact, Eye Contact, Ingestion.

You maintain: Nothing directly. It follows the hazard data on ingredient CAS and the exposure-limit data.

If missing: Never blank. At minimum it prints 'Skin Contact, Eye Contact.'

`SDSGenerator::deriveRoutesOfExposure :3722-3737 (extractHCodes, hazard.exposure_limits); labels.inhalation/skin_contact/eye_contact/ingestion (en.php:699-702)`

Most Important Symptoms/Effects, Acute and Delayed:

FIXED TEXT (TRANSLATED) Fixed label shared with Section 4. It is wider than the label column, so it prints on its own line above the text.

On a sheet: Most Important Symptoms/Effects, Acute and Delayed:

You maintain: Nothing

If missing: Sheet language > English > the literal key 'labels.symptoms_effects'. Older snapshots skip the line.

`labels.symptoms_effects (en.php:703); PDFService::renderSection11 :936; labelValue :1369-1389 (a long label goes on its own line)`

Symptoms/effects (text):	TEXT CHOSEN BY HAZARD The product's health H-statements (H3xx) print in 'Acute:' and 'Delayed:' groups as in Section 4, with H370-H373 naming the target organs (and the route when all sources share one) when every source CAS has organs entered and no hazard override or trade-secret declaration carries the code. Precedence: Section 4 Symptoms override > derived text > the no-symptoms sentence. *On a sheet: Delayed: Causes damage to organs (lungs) through prolonged or repeated exposure by…* **You maintain:** Health-hazard data on ingredient CAS (federal, CPD or trade-secret declaration). STOT target organs and route on CAS determinations (STOT Target Organs tab). Optionally, the Section 4 Symptoms override in the SDS editor. **If missing:** 'No significant symptoms or effects are known or expected under normal conditions of use.' When organs are not entered on every source CAS, the standard H370-H373 statement prints without organs. `overrides[4]['symptoms'] > deriveSymptoms :1916-1956 (DELAYED_EFFECT_H_PREFIXES :107); applyStotTargetOrgans :1446 (called :202/:640) via StotTargetOrgans::statementText :227; section4.symptoms_* (en…`
Acute Toxicity:	FIXED TEXT (TRANSLATED) Fixed label from the translation files. *On a sheet: Acute Toxicity:* **You maintain:** Nothing **If missing:** Sheet language > English > the literal key 'labels.acute_toxicity'. The value is never blank, because the three route lines always print. `labels.acute_toxicity (en.php:771; getLabels :5347); PDFService::renderSection11 :937 labelValue`
Acute toxicity (oral):	HAZARD ENGINE The most severe oral Category 1-4 in the classification: the ingredient cut-off or the ATEmix, whichever is more severe, or an H300-H302 that prints in Section 2. Precedence: per-product Acute Toxicity override > classification > 'Not classified' sentence. *On a sheet: Acute toxicity (oral): Category 4 — Harmful if swallowed (H302).* **You maintain:** Oral acute-toxicity data on each ingredient CAS (federal data, or a CPD on CAS determinations where there is none); constituent % on Raw materials; optional hazard override on the Finished good. **If missing:** 'Acute toxicity (oral): Not classified based on available data.' Category 5 (H303) prints the same sentence. H300 alone prints no category. `text_overrides(11,acute_toxicity) > buildAcuteToxicity :3538-3720 (acute_toxicity_oral; H-code fallback #40; GHSStatements::hText); section11.acute_route_* (en.php:374-385)`
Acute toxicity (dermal):	HAZARD ENGINE Same as the oral line, for the skin route (H310-H312). Precedence: per-product override > classification > 'Not classified' sentence. *On a sheet: Acute toxicity (dermal): Category 3 — Toxic in contact with skin (H311).* **You maintain:** Dermal acute-toxicity data on ingredient CAS (federal or CPD); constituent %. **If missing:** 'Acute toxicity (dermal): Not classified based on available data.' `buildAcuteToxicity acute_toxicity_dermal; defaults H310/H310/H311/H312; section11.acute_route_dermal / acute_route_line(_no_category) en.php:375-378`
Acute toxicity (inhalation):	HAZARD ENGINE Same as the oral line, for inhalation (H330-H332). Precedence: per-product override > classification > 'Not classified' sentence. *On a sheet: Acute toxicity (inhalation): Category 4 — Harmful if inhaled (H332).* **You maintain:** Inhalation acute-toxicity data on ingredient CAS (federal or CPD); constituent %. **If missing:** 'Acute toxicity (inhalation): Not classified based on available data.' `buildAcuteToxicity acute_toxicity_inhalation; defaults H330/H330/H331/H332; section11.acute_route_inhalation en.php:376`
ATEmix value (after a classified route):	HAZARD ENGINE The calculated mixture acute toxicity estimate. It is appended when the ATEmix falls in the printed category, including a tie with the ingredient cut-off, but not for a route known only from a bare H-code. Precedence: explicit LD50/LC50 on the ingredient > GHS category default estimate. *On a sheet: ATEmix = 1250 mg/kg bw. / ATEmix = 0.8 mg/L (4 h, dusts/mists).* **You maintain:** LD50/LC50 (ATE) values on ingredient CAS data or in the CPD; constituent %; trade-secret raw material line %. Physical State on the Finished good > family Default physical state (Product families) > largest raw material's state > Liquid; Solid/Powder switches inhalation to the dust unit. **If missing:** No number. A cut-off or trade-secret route ends with 'Classified based on ingredient concentration.' instead (next row); an override-only or bare H-code route ends after the statement. `hazard_classes ate_mix > ate_results at same cat; HazardEngine::applyATECalculation :2322; CATEGORY_DEFAULT_ATES :207; inhalationAteRoute :1546 / productIsDustForm :1566; formatAte :3744`
Classified based on ingredient concentration.:	TEXT CHOSEN BY HAZARD Added to a classified route when no ATEmix at its category exists: an ingredient cut-off with a lower, dominated or missing ATEmix, or a trade-secret declaration. Not added when the category came only from the Finished good hazard override or a bare H-code. *On a sheet: Classified based on ingredient concentration.* **You maintain:** Nothing. It follows the classification basis. **If missing:** Not printed on unclassified routes or when an ATEmix prints. `buildAcuteToxicity :3702-3706 (override_only / source h_code skip); section11.acute_basis_concentration (en.php:386)`

Chronic Effects:	FIXED TEXT (TRANSLATED) Fixed label from the translation files. *On a sheet: Chronic Effects:* **You maintain:** Nothing **If missing:** Sheet language > English > the key name. `labels.chronic_effects (en.php:772); PDFService::renderSection11 :938`
Chronic effects: sensitisation:	TEXT CHOSEN BY HAZARD A standard sentence prints when the product carries H334 or H317. A per-product Chronic Effects override replaces the whole paragraph. *On a sheet: Skin sensitizer: may cause an allergic skin reaction (H317). Repeated or prolonged…* **You maintain:** Sensitisation data on ingredient CAS (federal, CPD or trade-secret declaration), or a Finished good hazard override. **If missing:** Fragment omitted. `text_overrides(11,chronic_effects) > buildChronicEffects :3436-3529 (H334 chronic_resp_sens, H317 chronic_skin_sens; en.php:413-414)`
Chronic effects: germ cell mutagenicity:	TEXT CHOSEN BY HAZARD Standard sentence for H340, otherwise for H341. *On a sheet: Germ cell mutagenicity: suspected of causing genetic defects (H341).* **You maintain:** Mutagenicity data on ingredient CAS. **If missing:** Fragment omitted. `buildChronicEffects :3464-3468; section11.chronic_muta_1 / chronic_muta_2 (en.php:415-416)`
Chronic effects: carcinogenicity:	TEXT CHOSEN BY HAZARD H350 sentence > H351 sentence > 'listed' sentence when a reportable registry listing survives. 'See Carcinogenicity below' is added only when that paragraph lists components and is not overridden. *On a sheet: Carcinogenicity: suspected of causing cancer (H351). See Carcinogenicity below for…* **You maintain:** Carcinogen registry (seed CSV only, no admin screen); carcinogenicity data on CAS or CPD; physical state on Raw materials; Settings: Airborne/Unbound Particles Override (sds.inhalation_only_cas). **If missing:** Fragment omitted. With a 'replace' Finished good hazard override, the registry and the powder rule add no H350/H351. `buildChronicEffects :3475-3481 (chronic_carc_1/2/listed + _no_ref, en.php:417-423, #66); applyPostClassifyCarcinogenSteps :5731; applyCarbonBlackLogic :5754; applyCarcinogenFindings :6106`
Chronic effects: reproductive toxicity / lactation:	TEXT CHOSEN BY HAZARD Standard sentence for any H360 code, otherwise for any H361 code, plus a lactation sentence for H362. *On a sheet: Reproductive toxicity: suspected of damaging fertility or the unborn child (H361).* **You maintain:** Reproductive-toxicity data on ingredient CAS. **If missing:** Fragment omitted. `buildChronicEffects :3484-3491 (prefix match H360x/H361x); chronic_repr_1/2, chronic_lactation (en.php:424-426)`
Chronic effects: STOT repeated exposure:	TEXT CHOSEN BY HAZARD One sentence for H372, otherwise H373, naming the target organs (all entered, in a fixed order) when every source CAS has organs and no hazard override or trade-secret declaration carries the code, plus the route only when all those CAS share one. Precedence: organ sentence > generic sentence that refers to Section 2. *On a sheet: Specific target organ toxicity, repeated exposure (Category 1): causes damage… (lungs)…* **You maintain:** STOT RE data on ingredient CAS. Target organs and route on CAS determinations, STOT Target Organs tab (cas_master.stot_re_organs / stot_re_route). **If missing:** No H372/H373: fragment omitted. No organs, or organs missing on any source CAS, or a source in a hazard override or trade-secret declaration: the generic '…(H372). Refer to Section 2 for the affected organs and route of exposure where specified.' Route left out when blank, 'all' or mixed. `buildChronicEffects :3493-3503; StotTargetOrgans::resolve :163, chronicText :238 (stot.chronic_re_1/2[_route], en.php:36-39); generic chronic_stot_re_1/2 en.php:427-428`
Chronic effects: repeated skin / eye contact:	TEXT CHOSEN BY HAZARD Skin sentence for H314/H315; eye sentence for H318/H319. *On a sheet: Repeated or prolonged skin contact may cause irritation, dryness, or dermatitis.* **You maintain:** Skin/eye irritation data on ingredient CAS. **If missing:** Fragment omitted. `buildChronicEffects :3506-3511; chronic_repeated_skin / chronic_repeated_eye (en.php:429-430)`
Chronic effects: none known:	TEXT CHOSEN BY HAZARD A fixed sentence that prints only when none of the chronic sentences above applies and there is no override. *On a sheet: None known. Based on available data for the mixture and its components, the…* **You maintain:** Nothing **If missing:** This is itself the fallback. `buildChronicEffects :3514-3516; section11.chronic_none (en.php:431)`
Carcinogenicity:	FIXED TEXT (TRANSLATED) Fixed label, always printed, with the text on the lines below. *On a sheet: Carcinogenicity:* **You maintain:** Nothing **If missing:** Sheet language > English > the key name. `labels.carcinogenicity (en.php:773); PDFService::renderSection11 :940-947`

Carcinogenicity: no listed components:

REGULATORY LIST A fixed sentence that prints when no reportable registry listing (IARC 1/2A/2B, NTP Known/RAHC, OSHA) at 0.1% or more survives filtering. Precedence: per-product Carcinogenicity override > this sentence.

On a sheet: No components present at or above 0.1% are listed as carcinogens by IARC, NTP, or OSHA.

You maintain: Carcinogen registry kept current (storage/data/seed/carcinogens.csv + load-seed-data.php).

If missing: This is itself the default text.

`text_overrides(11,carcinogenicity) > section11.carcinogenicity (en.php:388; :threshold = CarcinogenService::LISTING_THRESHOLD_PCT 0.1); section11() :3942-3966`

Carcinogenicity: listed-components intro:

REGULATORY LIST Prints when any ingredient at 0.1% or more has a reportable registry listing after the physical-state filters. Precedence: override > registry result.

On a sheet: The following component(s), present at or above 0.1%, are listed as carcinogens or…

You maintain: Carcinogen registry; ingredient CAS and %; physical state on Raw materials; Settings: sds.inhalation_only_cas.

If missing: The negative sentence prints instead.

`CarcinogenService::buildSummaryText :285; analyse :146; applySolidPowderLiquidFiltering :5898; section11.carcinogenicity_listed_intro (en.php:389)`

Carcinogenicity: listed component line:

REGULATORY LIST One line per listed ingredient: name, CAS, Section 3 band and each reportable agency classification in the sheet language. Name: Prop 65 name > CAS master name > constituent name > registry name; trade secrets are masked.

On a sheet: Ethylbenzene (CAS 100-41-4, 1 - 5%) — IARC: Group 2B

You maintain: Carcinogen registry (one correct row per CAS+agency in the seed CSV); constituent name, CAS and exact or min/max % on Raw materials.

If missing: IARC Group 3 and unrecognised classifications are never reported, so a CAS with only those gets no line.

`CarcinogenService::buildComponentTexts :233; section11.carcinogenicity_listed_line/_listing (en.php:391-392); formatConcentration :5292`

Carcinogen inhalable-dust note:

FORMULA / RAW MATERIALS Added after the listing of an inhalation-only substance (Settings list; default carbon black and TiO2) in the Carcinogenicity line and the component block, when any raw material is not Solid/Powder. That listing adds no H350/H351, exposure limit or Prop 65 entry.

On a sheet: (The listing applies to inhalable dust only; in this product the substance is bound…

You maintain: Settings: Airborne/Unbound Particles Override (sds.inhalation_only_cas); physical state on Raw materials.

If missing: Not printed for all-powder/solid products, which get Carc. 2 (H351) unless H350/H351 is already present or a 'replace' override applies. A blank raw-material state counts as not powder.

`removeInhalationOnlyFromResults :5956 (inhalable_dust_only); section11.carcinogenicity_inhalable_dust_note (en.php:393); PDFService.php:979-981`

Component Toxicological Data:

FIXED TEXT (TRANSLATED) Fixed sub-heading that prints only when at least one component row qualifies.

On a sheet: Component Toxicological Data:

You maintain: Nothing

If missing: Not printed when no component at 0.1% or more has an exposure limit or a reportable carcinogen listing.

`labels.component_tox_data (en.php:774); PDFService::renderSection11 :950-955`

Component row (name, CAS, band):

FORMULA / RAW MATERIALS Each ingredient at 0.1% or more that has an exposure limit or a reportable carcinogen listing, banded as in Section 3. Name: Prop 65 name > CAS master name > constituent name.

On a sheet: Toluene (CAS 108-88-3) — 10 - 30%

You maintain: Constituent CAS, name and pct_exact / pct_min / pct_max on Raw materials; formula lines on Finished goods.

If missing: Row omitted. Solid/powder-only ingredients in a liquid product lose their limits and listings, so they have no row. Settings inhalation-only substances keep their listing and row.

`section11() :3872-3933 over calcResult.composition; formatConcentration :5292 (summed min/max bounds); literals 'CAS ', ' – ' PDFService.php:960-964`

Trade-secret component (masked name and CAS):

FORMULA / RAW MATERIALS A constituent flagged trade secret, or a no-CAS trade-secret raw material, prints its trade-secret description and 'TRADE SECRET' in place of its name and CAS, in the component block and the Carcinogenicity lines. Trade-secret description > translated 'Trade Secret'. The band still prints, and English registry descriptions are dropped.

On a sheet: Acrylate monomer (CAS TRADE SECRET) — 5 - 10%

You maintain: Trade-secret flag and trade-secret description on each constituent on Raw materials, or the no-CAS trade-secret raw material and its hazard declaration.

If missing: A blank description prints 'Trade Secret (CAS TRADE SECRET)'. A CAS flagged on any one raw material is masked on the whole sheet.

`maskTradeSecretRow :3092; tradeSecretName :3072; labels.trade_secret / trade_secret_cas (en.php:835-836); section11() :3916-3930, :3954`

Component carcinogen listing line:

REGULATORY LIST Each reportable registry listing for the CAS: agency and translated classification. English sheets add the registry description; other languages add a translated gloss.

On a sheet: IARC: Group 1 — Carcinogenic to humans.

You maintain: Carcinogen registry rows (agency, classification, description).

If missing: The ' — description' part is omitted when the description is empty or the component is a masked trade secret. Group 3 and unknown classes are not listed.

`CarcinogenService::localiseListings :113; carcinogen_list.agency/classification/description; PDFService.php:969-977`

Component exposure limit line:

HAZARD ENGINE All exposure limits found for the CAS, combined and printed as stored (untranslated): the federal datasets, plus CPD limits for a CAS with no federal hazard data, plus trade-secret declarations.

On a sheet: PEL-TWA: 435 mg/m3 (29 CFR 1910.1000 Table Z-1)

You maintain: Federal OEL dataset refresh (OSHA PEL / ACGIH TLV seed, NIOSH connector); CPD exposure limits for CAS without federal data; physical state on Raw materials.

If missing: No line. '(notes)' is omitted when notes are blank. An inhalation-only CAS in a non-powder product and a solid-only CAS in a liquid product print no limits.

`hazardResult.exposure_limits by CAS: exposure_limits+hazard_source_records.is_current  CPD exposure_limits  manual JSON; PDFService.php:984-995`

UV Acrylate Information:

FIXED TEXT (TRANSLATED) Fixed label that prints only with the UV note, so only on UV/LED-family products.

On a sheet: UV Acrylate Information:

You maintain: Nothing

If missing: Not printed without a note.

`labels.uv_acrylate_note (en.php:837); PDFService::renderSection11 :999-1004`

UV acrylate note text:

PRODUCT FAMILY Prints on every product whose resolved family is an active UV/LED family while the UV acrylate rule pack is on; it does not depend on acrylate detection or the CAS acrylate flag. The sensitizer wording prints when the product has H317, otherwise the unclassified wording. Precedence: per-product UV Acrylate Information override > that text.

On a sheet: Contains acrylate monomers that are known skin sensitizers. Repeated exposure may…

You maintain: Family on the Finished good (manual pick, rule match or content share) and the family's UV flag and active status on Product families; Settings: UV acrylate rule pack.

If missing: Not printed, override included, when the pack is off or no active UV family resolves. A missing settings row counts as enabled.

`UVAcrylateRulePack::isApplicable :105 (isEnabled + familyIsUv), section11Note :261; SDSGenerator.php:231, :282-285; section11.uv_acrylate_note(_unclassified) en.php:433-435`

Preview warning: trade secret on one raw material, disclosed on another:

FORMULA / RAW MATERIALS Shown only in the preview Warnings box, never on the SDS. It appears when one raw material flags a CAS as trade secret and another discloses it; Section 11 then masks that CAS.

On a sheet: Trade-secret warning: CAS 123-45-6 is marked as a trade secret on raw material(s) …

You maintain: Trade-secret flags kept consistent across raw materials on Raw materials.

If missing: No warning.

`SDSGenerator::tradeSecretDisclosureWarnings :6262 (contributing_materials.is_trade_secret); added to sds.warnings :336-338, :758-760`

Preview warning: STOT target organs incomplete:

HAZARD ENGINE Shown only in the preview Warnings box, never on the SDS and never blocking, when at least one source CAS of an H370-H373 code has organs entered but another source CAS has none or the code also comes from a hazard override or trade-secret declaration. Sections 2, 4 and 11 then print the generic statement.

On a sheet: Section 2 warning: H372 comes from CAS 14808-60-7 (target organs entered) and also…

You maintain: Target organs on every CAS behind the code (CAS determinations, STOT Target Organs tab).

If missing: No warning. With no organs entered on any source CAS, or a code carried only by a hazard override or trade-secret declaration, the generic statement prints silently.

`StotTargetOrgans::warnings :249 (hazardResult.stot_incomplete from resolve :163-207); added to sds.warnings in generate() :355-359 and generateFromBase() :777-781; Views/sds/preview.php :670`

SECTION 12: ECOLOGICAL INFORMATION

Section 12 is built from each constituent's aquatic classifications and M-factors, as summed by the hazard engine or set by a finished-good hazard override, and from the EPA PBT designation on the SARA 313 list. The component table lists only constituents that Section 3 lists, and vendor trade-secret identities are masked. Fixed translation sentences fill anything with no data, and a per-product text override can replace any of the four paragraphs. Batch F did not change Section 12's logic; the only visible change is the French wording of H410-H412.

Ecotoxicity:

TEXT CHOSEN BY HAZARD Per-product text override > the product's aquatic hazard statements (lead-in + H statements + release warning) > a not-classified sentence when aquatic data exists or a replace-mode override is set > the standard no-data sentence. The aquatic H codes come from the M-factor summation of the constituents' aquatic categories, or from a finished-good hazard override.

On a sheet: Classified as hazardous to the aquatic environment by the GHS summation method ... H410

You maintain: Aquatic categories and M-factors on each constituent CAS (federal hazard data, or a CAS determination when no federal data exists). Constituent percentages on raw materials and formula lines. The finished-good hazard override if used. Section 12 text overrides in the SDS editor.

If missing: "No data available on the mixture. Avoid release to the environment."

`text_overrides[12]['ecotoxicity'] (blank ignored) > H4(0[0-2]|1[0-3]) in h_statements (lead-in+codes+environmental_warning) > *not_classified* > section12.ecotoxicity; SDSGenerator:4064-4110`

Ecotoxicity: lead-in when classified:

TEXT CHOSEN BY HAZARD Fixed sentence chosen by where the printed codes came from. The manufacturer's-assessment version prints under a replace-mode override or when a printed code came only from the override. The summation version ("listed below") prints when every code came from the summation and the table prints; otherwise a neutral version that names no method.

On a sheet: Classified as hazardous to the aquatic environment (GHS Rev. 7, Chapter 4.1):

You maintain: Nothing beyond the aquatic data and the finished-good hazard override (mode + JSON).

If missing: Not printed when the product has no aquatic H code; a not-classified or no-data sentence is used instead.

`hazard aquatic_basis (override_mode, override_codes, summation_codes) → ecotoxicity_classified_manufacturer > ecotoxicity_classified > _no_table; SDSGenerator:4072-4092`

Ecotoxicity: aquatic H statements:

HAZARD ENGINE The aquatic hazard statements the product resolved to, in code order, in standard GHS wording for the sheet's language (French H410-H412 now read "effets néfastes à long terme"). Only the most severe code per route prints. Summation alone gives only H400, H410, H411, H412; H401, H402, H413 come from a finished-good override, or rarely a CAS determination or no-CAS hazard JSON.

On a sheet: H410: Very toxic to aquatic life with long lasting effects.

You maintain: Constituent aquatic categories and M-factors. Finished-good hazard override (mode + JSON) where used.

If missing: Not printed when the product has no aquatic H code. All seven aquatic codes have English, Spanish, French and German wording, so a printed code always carries its full statement.

`h_statements ^H4(0[0-2]|1[0-3])$ ksort, one per route (HazardEngine::dropLessSevereAquaticCodes :2136); EN GHSStatements, non-EN ghs_{lang}.php > EN; SDSGenerator:4004-4011`

Ecotoxicity: release warning:

TEXT CHOSEN BY HAZARD Fixed warning sentence added after the H statements, only when the product has an aquatic H code.

On a sheet: Avoid release to the environment. Prevent entry into waterways, sewers, and soil.

You maintain: Nothing

If missing: Not printed on unclassified products. The not-classified and no-data sentences carry their own "Avoid release" clause.

`section12.environmental_warning (en.php:449); SDSGenerator.php:4093-4095`

Ecotoxicity: not-classified sentence:

TEXT CHOSEN BY HAZARD Fixed sentence printed when the product has no aquatic H code but aquatic data or a replace-mode override exists. Replace mode gives the manufacturer's-assessment version; a printed component table gives "Based on the component data below"; aquatic data only on constituents Section 3 does not list gives a version with no table reference.

On a sheet: Based on the component data below, the mixture does not meet the GHS criteria ...

You maintain: Constituent aquatic classifications; finished-good hazard override mode.

If missing: Falls back to the no-data sentence when no constituent has aquatic data.

`no aquatic code: replace override → ecotoxicity_not_classified_manufacturer; table → ecotoxicity_not_classified; else aquatic_components → _no_table; SDSGenerator:4096-4107`

Ecotoxicity: no-data sentence:

FIXED TEXT (TRANSLATED) Fixed sentence from the translation file, printed only when no constituent has aquatic data, no replace-mode override is set and there is no text override. The data decides whether it prints.

On a sheet: No data available on the mixture. Avoid release to the environment.

You maintain: Nothing

If missing: This is the fallback itself.

`section12.ecotoxicity; SDSGenerator.php:4108; en.php:439; es/fr/de.php:417`

Component Aquatic Hazard Data:

FIXED TEXT (TRANSLATED) Fixed sub-heading, printed only when at least one constituent that Section 3 lists has an aquatic classification. It never prints under a replace-mode override.

On a sheet: Component Aquatic Hazard Data:

You maintain: Nothing

If missing: The heading and table are omitted when there are no rows.

`labels.component_ecotox_data; printed when component_aquatic non-empty; PDFService.php:1046 (adds ':'), preview.php:437 (no colon)`

Component table headers: Chemical Name | CAS Number | Conc% | Aquatic Acute | Aquatic Chronic:

FIXED TEXT (TRANSLATED) Fixed column labels from the translation file, printed only with the component table. Conc% is the same label used in Section 8.

On a sheet: Chemical Name | CAS Number | Conc% | Aquatic Acute | Aquatic Chronic

You maintain: Nothing

If missing: Not printed when the table is empty. Long French and German headers wrap inside their cells.

`labels.chemical_name, cas_number, el_conc_pct, aquatic_acute, aquatic_chronic; PDFService.php:1020-1080 (68/25/17/32/33 mm; headers wrap in one shared-height row)`

Component table: Chemical Name:

FORMULA / RAW MATERIALS Prop 65 list name > CAS master preferred name > the constituent name entered on the raw material. A trade-secret constituent shows its trade-secret description instead (see the trade-secret row).

On a sheet: Zinc oxide

You maintain: CAS master preferred names and raw-material constituent names (Raw materials screen).

If missing: Blank cell when all name sources are empty.

`aquatic_components[].name = composition chemical_name: COALESCE(NULLIF(p65.chemical_name,''),NULLIF(cm.preferred_name,''),rmc.chemical_name) Formula.php:293`

Component table: CAS Number:

FORMULA / RAW MATERIALS The constituent's CAS number from the raw material. Trade-secret constituents are masked.

On a sheet: 1314-13-2

You maintain: CAS numbers and trade-secret flags on raw-material constituents.

If missing: A constituent with no CAS prints no row of its own. No-CAS trade-secret constituents with H codes and no-CAS hazardous raw materials share one "TRADE SECRET" row; a flagged constituent with a real CAS gets its own row showing "TRADE SECRET".

`component_aquatic[].cas_number = aquatic_components[].cas (rmc.cas_number or TRADE_SECRET sentinel) → labels.trade_secret_cas when masked; SDSGenerator:4021-4045`

Component table: Conc%:

FORMULA / RAW MATERIALS The same prescribed concentration band that Section 3 prints for that constituent. The exact percentage is never printed.

On a sheet: 10 - 30%

You maintain: Constituent percentages (exact or min/max) on raw materials and formula line percentages.

If missing: Always a band, because rows under 0.1% are not listed. Only a Substance sheet's identity constituent could print "<0.1%".

`formatConcentration(compByCas[cas]) on concentration_min/max → widest PRESCRIBED_RANGES band containing it; SDSGenerator:4029-4036, :5292`

Component table: Aquatic Acute / Aquatic Chronic:	HAZARD ENGINE Each constituent's aquatic category per route: current federal hazard data (an unreadable row's category can be inferred from its H codes) > the CAS determination, used only when no federal data exists > the trade-secret hazard list. Category 1 also shows its M-factor. *On a sheet: Category 1 (M = 1, default) | Category 1 (M = 10)* **You maintain:** Aquatic acute/chronic categories and M-factors on each constituent CAS: the federal hazard data import, or the CAS determinations screen for any CAS with no federal data. **If missing:** An em dash (—) when there is no class on that route. Category 1 with no M-factor on file prints "(M = 1, default)" (translated). `aquatic_components[].{acute,chronic}_category + _m_factor/_source (HazardEngine::resolveMFactor :1429) → formatAquaticCategory :4175; default → section12.m_factor_default`
Component table: trade-secret constituent:	FORMULA / RAW MATERIALS A vendor-claimed trade secret prints its trade-secret description in place of the name, "TRADE SECRET" in place of the CAS, and the Section 3 band of its total. The real identity never prints. Its aquatic categories count toward the summation at the raw material's real share of the product. *On a sheet: Biocide blend | TRADE SECRET | 1 - 5% | — | Category 1 (M = 1, default)* **You maintain:** Trade-secret flag, trade-secret description and trade-secret H codes on the constituent, or the manual hazard JSON on a no-CAS hazardous raw material (Raw materials screen). **If missing:** A blank description (or one that just says "Trade Secret") prints the translated "Trade Secret". A no-CAS trade secret whose only declared hazards are aquatic gets no row, because Section 3 does not list it, but it still counts in the summation. `cas TRADE_SECRET or composition is_trade_secret → tradeSecretName(trade_secret_description) :3072, labels.trade_secret_cas, Section 3 band; SDSGenerator:4027-4033`
Component table: which rows print:	HAZARD ENGINE One row for each constituent that has an aquatic class and that Section 3 lists (0.1% or more and classified, including as a contributor to a summation that fired, or with an exposure limit, plus a Substance sheet's identity). Sorted by concentration, highest first; dropped entirely under a replace-mode override. *On a sheet: Zinc oxide | 1314-13-2 | 10 - 30% | Category 1 (M = 1) | Category 1 (M = 1)* **You maintain:** Constituent aquatic data and percentages; current flags on hazard source records; the finished-good hazard override mode. **If missing:** Table omitted when there are no rows. Inhalation-only CAS (e.g. carbon black) bound in a non-powder product never appear. A no-CAS trade secret with only aquatic hazards never gets a row. `HazardEngine::buildAquaticComponentSummary :1487 (conc desc, then CAS) ∩ section3ListedCas :4206; emptied when aquatic_basis.override_mode = 'replace' (SDSGenerator:4061)`
Persistence and Degradability:	REGULATORY LIST Per-product override > a sentence naming every PBT-flagged constituent at any concentration above 0 > "No data available.". The flag comes from the SARA 313 list or the TRI lead/mercury compound categories. TRI PFAS (special concern but not PBT) are not named here. *On a sheet: Contains component(s) identified as persistent, bioaccumulative and toxic (PBT) ...* **You maintain:** SARA 313 list kept current with its PBT flag (Regulatory lists import). CAS master molecular formula or names for category matching. Constituent CAS numbers. **If missing:** "No data available." `text_overrides[12]['persistence'] > section12.pbt_components (SARA313Service::evaluate is_pbt: sara313_list or regulatory_categories, conc>0) > section12.persistence; SDSGenerator:4127-4147`
PBT component item:	REGULATORY LIST For each PBT constituent: the SARA list name, or the TRI category name such as "Lead compounds", with footnote markers stripped, falling back to the composition name, followed by the CAS number. A trade-secret constituent shows its description and "TRADE SECRET". *On a sheet: Lead compounds (CAS 1314-41-6)* **You maintain:** SARA 313 list chemical names and PBT flags; seeded TRI categories (migration 058). **If missing:** Not printed when there is no PBT constituent. `section12.pbt_component_item ':name (CAS :cas)'; name = sara_name (list row > TRI category name, †‡* stripped) > composition name; TS masked; SDSGenerator:4128-4143`
Bioaccumulative Potential:	REGULATORY LIST Per-product override > "See Persistence and Degradability above." when the PBT sentence prints (the full PBT sentence if Persistence is overridden) > "No data available." *On a sheet: See Persistence and Degradability above.* **You maintain:** Same SARA 313 PBT flags as Persistence. **If missing:** "No data available." `No PBT: text_overrides[12]['bioaccumulation'] > section12.bioaccumulation. PBT: override > (persistence overridden ? pbtLine : section12.pbt_see_persistence); SDSGenerator:4148-4153`

Mobility in Soil:

FIXED TEXT (TRANSLATED) Per-product override > fixed "No data available.". No data source exists for this line, so the automatic text is the same on every sheet.

On a sheet: No data available.

You maintain: Nothing (an override only).

If missing: "No data available." On old snapshots the PDF leaves the line out, while the HTML preview prints an empty label.

`text_overrides[12]['mobility'] > section12.mobility; SDSGenerator.php:4161; PDFService.php:1089; preview.php:462`

SECTION 13: DISPOSAL CONSIDERATIONS

Section 13 prints three lines. Disposal Methods is a standard sentence. On a UV/LED-family sheet whose RCRA check found nothing and whose composition is complete, a cure-residue sentence is added to it. Typed per-product text replaces the whole Disposal Methods line. Contaminated Packaging is the same fixed sentence on every sheet. The RCRA Hazardous Waste Classification line is built when the sheet is generated, from the sheet's final hazard codes, the formula flash point, the resolved physical state (product > product family default > largest raw material > Liquid), and a match of each component's CAS against the admin-maintained EPA RCRA waste-code list. That match is extended to metal compounds through each CAS's RCRA metals flags.

Disposal Methods:

PRODUCT OVERRIDE Per-product text > standard sentence (plus the UV cure-residue sentence when it applies). Typed text replaces the whole line, cure-residue sentence included. Finished goods and resale raw materials each keep their own text.

On a sheet: Dispose of in accordance with all applicable federal, state, and local regulations. Do not…

You maintain: Nothing for the standard text. Optional wording per language on SDS > Edit, Section 13 (finished goods), or SDS Creation Readiness Check > 'Edit SDS text' (resale)

If missing: The standard sentence: "Dispose of in accordance with all applicable federal, state, and local regulations. Do not dump into sewers, drains, or waterways." Blank text, or text equal to the automatic text, stores no override

`TextOverrideService::effective(text_overrides 13/'methods', FG or resale RM, lang) > section13.methods [+ ' ' + section13.uv_cure_residue]; SDSGenerator.php:4432-4439; en.php:462`

Disposal Methods: UV cure-residue sentence:

PRODUCT FAMILY Fixed wording added after the standard sentence only when all three hold: the product family is flagged UV/LED, the RCRA line found no item and no listed component, and the composition is complete. The rule-pack switch is ignored.

On a sheet: Small amounts of non-hazardous residue may be cured before disposal where permitted.

You maintain: 'UV / LED curable family' box on the product family (Admin > Product Families). Complete constituent CAS numbers and percentages on every raw material

If missing: Not printed: non-UV family, any RCRA item or listed match, a constituent with no CAS (or a no-CAS trade secret), a raw with no constituents or no %, or a per-product Disposal Methods text

`section13.uv_cure_residue (en.php:466) if items===[] && listed===[] && uvFamily (family_is_uv) && compositionIsComplete() (:4498, TSCAService::incompleteRawMaterials); :4433`

Contaminated Packaging:

FIXED TEXT (TRANSLATED) One fixed sentence from the language file, printed on every sheet. It cannot be edited per product.

On a sheet: Empty containers retain product residue; follow all precautions on this safety data sheet…

You maintain: Nothing

If missing: Always prints. Sheets published before batch F (A5) have no packaging entry, so nothing prints for them

`section13.packaging (en.php:464), label labels.contaminated_packaging (en.php:790); SDSGenerator.php:4442; PDFService.php:64; preview.php:635`

RCRA Hazardous Waste Classification (lead-in sentence):

TEXT CHOSEN BY HAZARD Fixed wording that prints only when at least one as-sold item applies: D001, D002, D003, a D-code component, or a trade-secret toxicity-characteristic constituent. The items follow, joined by '; ' and ending with '.'. In the PDF the long label sits on its own line.

On a sheet: If discarded as sold, this product may be regulated as hazardous waste under US EPA RCRA (40 CFR 261):

You maintain: Nothing directly. It depends on the hazard, flash point, physical state and RCRA-list inputs below

If missing: Replaced by one of the two 'No RCRA hazardous waste characteristic…' sentences

`section13.rcra_intro + '; '-joined items + '.' (SDSGenerator.php:4416-4417); label labels.rcra_classification (en.php:789); PDFService::labelValue :1369`

Ignitable (D001) item:	HAZARD ENGINE Prints when any D001 reason applies: a Flammable Liquids Category 1-3 code (H224-H226) on the sheet, or an oxidizer, pyrophoric/self-heating, flammable gas/aerosol or flammable solid code. Reasons are joined with commas. *On a sheet: ignitable (D001) — flash point below 60 °C (140 °F)* **You maintain:** Flash points and physical state on raw materials, physical state on the finished good or its product family, CAS determination hazard data, finished-good hazard override **If missing:** No D001 item. With no other D001 code, none prints for a Category 4 (H227) product, a Category 3 product at exactly 60 °C, or a product with no flash point data (unless the hazard override sets H224-H226) `section13.rcra_d001 :reasons; d001FlashReasonKey (SDSGenerator.php:4592) + H270-H272, H250-H252, H220-H223, H228 checks (:4307-4331)`
D001 reason: flash point below 60 °C / minimum value below 60 °C:	FORMULA / RAW MATERIALS Prints only when the sheet carries H224-H226 and the product is a liquid (not Solid/Powder/Paste) with a flash point below 60 °C. That flash point is the wt%-weighted average over raw materials that have one, rounded to 0.1 °C; a '>' on any of them prints the 'reported only as a minimum value' wording. *On a sheet: flash point below 60 °C (140 °F)* **You maintain:** Flash point (°C) and '>' flag on raw materials (Raw materials screen), including a high value for water and similar raws. Physical State on the finished good, its product family default, or its main raw material **If missing:** No reason when no raw material has a flash point (treated as not flammable), when the state is Solid/Powder/Paste, or at 60.0 °C or above. Neither a Section 9 edit nor the whole-degree '(calculated)' Section 9 value is used `hazardResult['flammability'] (flammabilityInputs :4546, state FG > family default > formula_props); FormulaCalcService::weightedFlashPoint round(,1); fp<60 && !gt > rcra_reason_flash_point; gt > _gt`
D001 reason: classified as a flammable liquid (Category 1, 2 or 3):	PRODUCT OVERRIDE Prints when H224-H226 are on the sheet but no formula flash point below 60 °C on a liquid supports them. In practice this means the finished-good hazard override set the class. *On a sheet: classified as a flammable liquid (Category 1, 2 or 3)* **You maintain:** Finished-good hazard override (Finished goods screen). Enter the tested flash point as a Section 9 edit so Sections 5 and 9 agree **If missing:** Not printed `d001FlashReasonKey (:4592): H224-H226 present, no liquid fp<60, engine flammability category not 1-3 > section13.rcra_reason_flammable_liquid (en.php:483)`
D001 reasons: oxidizer / pyrophoric or self-heating material:	HAZARD ENGINE Prints when the final hazard codes contain H270-H272 (oxidizer) or H250-H252 (pyrophoric or self-heating). These codes also make Section 14 'Not determined', which blocks publishing. *On a sheet: oxidizer* **You maintain:** Hazard classification inputs for oxidizing or self-heating constituents (CAS determinations, vendor data) **If missing:** Not printed `section13.rcra_reason_oxidizer / rcra_reason_ignitable_solid; SDSGenerator.php:4314-4319; TransportClassifier::UNSUPPORTED_CODES`
D001 reasons: ignitable compressed gas / flammable solid:	HAZARD ENGINE Prints for H220-H223 (flammable gas or aerosol) or H228 (flammable solid). It is a safety net only: the company sells none, and these codes also make Section 14 'Not determined', which blocks publishing. *On a sheet: ignitable compressed gas (flammable gas or aerosol)* **You maintain:** Hazard classification inputs (CAS determinations, vendor data) **If missing:** Not printed `section13.rcra_reason_flammable_gas / rcra_reason_flammable_solid (en.php:486-487); SDSGenerator.php:4323-4328`
Corrosive (D002) item and reasons:	HAZARD ENGINE Prints when the product is H314 and/or H290, naming each as a reason, unless its physical state is Solid, Powder or Paste. State: finished good's own > product family default > highest-wt% raw material with one > 'Liquid'. *On a sheet: corrosive (D002) — classified as corrosive to skin (H314)* **You maintain:** Skin- and metal-corrosion classification inputs (CAS determinations, vendor data, finished-good hazard override). Physical State on the finished good, its product family's SDS profile, or raw materials **If missing:** No D002 item. pH is never considered (not even the family typical pH) `rcraIsNonLiquidState (:4483) < resolvePhysicalState (:2980); H314 > rcra_reason_skin_corrosion, H290 > rcra_reason_metal_corrosion; rcra_d002 (:4333-4346)`

Reactive (D003) item and reasons:	HAZARD ENGINE Prints when the product has a water-reactive (H260/H261), explosive (H200-H205) or self-reactive/organic peroxide (H240-H242) code, with the reasons joined by commas. All of these codes also make Section 14 'Not determined', which blocks publishing. *On a sheet: reactive (D003) — reacts with water, explosive* **You maintain:** Hazard classification inputs (CAS determinations, vendor data) **If missing:** No D003 item `section13.rcra_d003 + rcra_reason_water_reactive | rcra_reason_explosive | rcra_reason_unstable; SDSGenerator.php:4348-4361`
Toxicity-characteristic component (D code) item:	REGULATORY LIST Each disclosed component's CAS is matched against the RCRA list; a compound also inherits the D codes of each metal in its RCRA metals flags. D codes print at any concentration with the TCLP limit; name: Prop 65 name > CAS preferred name > constituent name > CAS. *On a sheet: contains Lead chromate (D007 if the toxicity characteristic regulatory level of 5 mg/L…* **You maintain:** RCRA Waste Codes list (Regulatory Data > RCRA Waste Codes). RCRA metals box per CAS (CAS Determinations > CAS Descriptions, or 'Seed RCRA metal flags'). Constituent CAS numbers and percentages **If missing:** No item. A constituent with no percentage, or a compound with neither its own row nor metal flags, is not reported. A blank TCLP limit prints '…if the toxicity characteristic regulatory level is exceeded' `RCRAService::analyse/match: rcra_waste_codes kind D + cas_master.tc_metals > TC_METAL_ELEMENT_CAS rows; formatRcraCode rcra_code_d/_nolimit; name Formula.php:293`
Trade-secret component fragments:	REGULATORY LIST A matched trade-secret constituent is never identified: no waste code, no TCLP level, only its trade-secret description. A D match prints in the as-sold list. An F/K/P/U match prints '…a listed hazardous waste constituent (code withheld as a trade secret)' in the reference sentence. *On a sheet: contains a toxicity characteristic constituent (Trade Secret) and may exhibit the…* **You maintain:** Trade-secret flag and description on the raw-material constituent (Raw materials screen) **If missing:** A blank description, or one reading 'Trade Secret', prints the translated 'Trade Secret'. Vendor trade-secret materials with no CAS are never matched `is_trade_secret on composition row > section13.rcra_component_ts_tc / rcra_component_ts_listed; tradeSecretName() :3072; duplicates printed once (:4381-4400)`
'No characteristic identified' sentence:	TEXT CHOSEN BY HAZARD Prints when no as-sold item applies. If only listed (F/K/P/U) components matched, the shorter version prints, without the clause 'no component has been identified as…'. *On a sheet: No RCRA hazardous waste characteristic (40 CFR 261 Subpart C) has been identified…* **You maintain:** A complete RCRA waste-code list, RCRA metals flags, and complete constituent CAS and percentage data, because this sentence is a claim based on finding no match **If missing:** This sentence is itself the default when nothing matches `section13.rcra_none (nothing matched) / rcra_none_characteristic (only F/K/P/U matched); SDSGenerator.php:4418-4421; en.php:474-475`
Listed-waste codes of components, for reference:	REGULATORY LIST Prints when a component matches an F, K, P or U code on the RCRA list and makes up 0.1 % or more of the formula in total. Each constituent counts at its exact %, else its range maximum. Compounds never inherit a metal element's F/K/P/U rows. *On a sheet: Listed-waste codes of components, for reference (they do not apply to this product as sold):…* **You maintain:** F/K/P/U rows in the RCRA Waste Codes list (Regulatory Data > RCRA Waste Codes). Constituent CAS numbers and percentages **If missing:** Sentence omitted `section13.rcra_listed_intro + rcra_component + rcra_code_f/k/p/u; RCRAService::MIN_CONCENTRATION_PCT 0.1 on summed concentration_pct; SDSGenerator.php:4422-4424`
Generator-responsibility sentence:	FIXED TEXT (TRANSLATED) Fixed wording from the language file. It is always the last sentence of the RCRA line and cannot be edited per product. *On a sheet: The waste generator is responsible for the final hazardous waste determination at the time…* **You maintain:** Nothing **If missing:** The English text, or the raw key `section13.rcra_generator (en.php:477) appended unconditionally; SDSGenerator.php:4425`

SECTION 14: TRANSPORT INFORMATION

Section 14 is worked out for each product by the transport classifier. Class 3 follows the product's Flammable Liquids category: the wt%-weighted flash point and lowest boiling point of the raw materials, on a liquid. So Section 14 always agrees with Sections 2, 5, 7, 9, 10 and 13. Classes 8, 6.1 and 9 come from the hazard engine's corrosion, toxicity and aquatic statements. The UN entry depends on the transport product type (or the description) and the physical state. Physical state comes from the product, else its product family's default physical state, else the largest raw material. Special Precautions for User is the standard sentence plus a fragment for each case that applies: Class 3 or Class 8 on the printed hazard class, "Marine pollutant: Yes", or a UV/LED product family. Per-product overrides win line by line. They are entered for each language, resale sheets included, and Special Precautions can now be overridden too. Every language is checked before publishing. Fixed wording (entry names, notes, bulk sentence, fragments, labels) comes from the translation files.

UN Number:

TRANSPORT RULES Worked out from the product's Flammable Liquids category (raw flash points and boiling points), its corrosion, toxicity and aquatic hazard statements, its physical state and its transport product type (or description keywords). Per-product override > classifier result > "Not regulated" / "Not determined".

On a sheet: UN1210

You maintain: Flash point on every raw that has one, boiling point on volatile raws (Raw materials). Hazard data on each CAS (CAS determinations). Transport product type, description, physical state (Finished goods), or the family's default physical state (Product families).

If missing: No flash point on any raw = not flammable, so "Not regulated" unless another class applies. "Not determined" (publish blocked) only for gas/aerosol/oxidizer-type codes, Gas state, or 3+8+6.1 precedence.

`text_overrides(14,un_number,lang) > TransportClassifier::classify() un_number (TC.php:182-372); inputs SDSGenerator::section14() :4811-4839 via flammabilityOf() :4565`

Proper Shipping Name:

TRANSPORT RULES The official DOT name for the chosen UN entry, from the translation file. ES/FR/DE keep the English name and add a local translation in brackets. Override > classifier entry name > "Not regulated" / "Not determined".

On a sheet: Printing ink, flammable

You maintain: Same inputs as UN Number. The names themselves are kept in the translation files.

If missing: "Not regulated" / "Not determined". A name missing in one language falls back to the English text.

`text_overrides(14,proper_shipping_name) > section14.psn_{psn_key} (en.php:511-531; es/fr/de:478-500) + technical names; SDSGenerator.php:4855-4858`

Proper Shipping Name: technical names in brackets (n.o.s. entries only):

FORMULA / RAW MATERIALS For n.o.s. entries, up to two disclosed ingredients are added in brackets, highest wt% first, picked in this order: those whose CAS carries the primary hazard's codes > (Class 3 only) ingredients of raws with a flash point of 60 °C or below, never water > ingredients carrying any transport hazard. The name used is the Prop 65 list name > CAS preferred name > raw-material constituent name. Trade secrets are never named, and names are not translated.

On a sheet: Flammable liquids, n.o.s. (Toluene)

You maintain: Ingredient names (CAS determinations > preferred name; Raw materials > constituent name; a Prop 65 list name wins on finished goods). Flash point on solvent raws. Hazard determinations on each CAS. Trade-secret flags.

If missing: No brackets are printed, and the preview Warnings box shows an English transport warning (publishing is not blocked). Never added to UN1210/UN1263 or when the PSN is overridden.

`TC::technicalNames() > technicalNamesFromCas(flammable_cas) > any-hazard pass (TC.php:377-396,606-641); flammableConstituentCas() SDSGenerator.php:4760; Formula.php name COALESCE`

Hazard Class:

TRANSPORT RULES Class 3 when the product is Flammable Liquids Cat 1-3 (flash point 60 °C or below on a liquid, or set by the finished-good hazard override). Paste, Solid and Powder, including a family default, are not liquids. Class 8 for Skin Corr. 1 or for H290 on a liquid, Division 6.1 for acute toxicity Cat 1-3, and Class 9 for aquatic H400/H410/H411 alone. Primary and secondary classes follow 49 CFR 173.2a. Override > classifier > "Not regulated" / "Not determined".

On a sheet: 3 (8)

You maintain: Flash point and boiling point on the raws. Skin-corrosion, metal-corrosion and acute-toxicity categories on each CAS. Physical state (Finished goods, else the family's default physical state), which decides liquid vs not and whether 8 or 6.1 comes first.

If missing: "Not regulated" / "Not determined"

`text_overrides(14,hazard_class) > TC primary + ' (' + subsidiaries + ')' (TC.php:291-375); UNSUPPORTED_CODES TC.php:138-147; SDSGenerator.php:4819-4825,4859`

Packing Group:	TRANSPORT RULES The most stringent of the Class 3 group (I when the lowest raw boiling point is 35 °C or below, or Cat 1 with no boiling point; II for Cat 1/2; III for Cat 3), the Class 8 group (1A = I, 1B or 1 = II, 1C = III, H290 alone = III) and the 6.1 group (Cat 1/2/3 = I/II/III). Class 9 alone is III. Override > classifier > "Not applicable" / "Not determined". *On a sheet: II* **You maintain:** Boiling point on every volatile raw (Raw materials). Flash point. Skin-corrosion sub-category and acute-toxicity category on each CAS. A Section 9 boiling-point edit is not used here. **If missing:** With no boiling point, a flash point below 23 °C gives PG II (with a preview warning). "Not applicable" when not regulated; "Not determined" when undetermined. `text_overrides(14,packing_group) > mostStringent(pg3, pg8, pg61) (TC.php:256-287,502-559); IBP = FormulaCalcService lowest raw boiling_point_c; SDSGenerator.php:4860`
Environmental Hazards:	TRANSPORT RULES "Marine pollutant: Yes" when the product carries H400, H410 or H411, otherwise "No". Override > engine aquatic codes > "Not determined". *On a sheet: Environmental Hazards: Marine pollutant: No* **You maintain:** Aquatic toxicity data and M-factors on each CAS (CAS determinations). **If missing:** "Marine pollutant: No" when no aquatic code is present. "Not determined" while Section 14 is undetermined, unless a core line (UN, name, class, packing group) is overridden; then Yes/No comes from the codes. `text_overrides(14,environmental_hazards) > section14.marine_pollutant_yes|no (en.php:532-533) on H400/H410/H411; SDSGenerator.php:4845,4847-4861,4883-4885`
Transport in Bulk:	FIXED TEXT (TRANSLATED) Fixed sentence from the translation file, printed on every sheet. It cannot be overridden. *On a sheet: Not intended for transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code.* **You maintain:** Nothing **If missing:** Always printed (skipped only on snapshots made before the line existed). `section14.transport_in_bulk_text (en.php:538; es/fr/de:503) set at SDSGenerator.php:4929; PDFService.php:1101; preview.php:473-475`
Special Precautions for User: standard sentence:	FIXED TEXT (TRANSLATED) Fixed sentence from the translation file. It starts the line on every sheet unless a per-product Special Precautions override replaces the whole line. *On a sheet: Read this safety data sheet before handling. Keep containers closed, upright and secured…* **You maintain:** Nothing **If missing:** Always printed when there is no override (skipped only on snapshots made before the line existed). `section14.special_precautions_text (en.php:539; es/fr/de:504) first in $special, SDSGenerator.php:4902-4918; PDFService.php:1102; preview.php:476-478`
Special Precautions for User: flammable-liquid fragment (conditional):	TRANSPORT RULES Added after the standard sentence when the printed Hazard Class line contains class 3, as the primary or a secondary class. It reads the printed line, so a Hazard Class override counts. *On a sheet: Flammable liquid (Class 3): keep away from heat, sparks and open flames; no smoking…* **You maintain:** Same inputs as Hazard Class, or a Hazard Class override (SDS editor). **If missing:** Not printed (Not regulated, Not determined, or no class 3). `transportClassTokens(ov hazard_class) has '3' > section14.special_precautions_flammable (en.php:541); SDSGenerator.php:4904-4908; transportClassTokens() :4963`
Special Precautions for User: corrosive fragment (conditional):	TRANSPORT RULES Added when the printed Hazard Class line contains class 8, as the primary or a secondary class, including through a Hazard Class override. *On a sheet: Corrosive (Class 8): keep packages upright and protected against damage; avoid contact…* **You maintain:** Skin-corrosion and metal-corrosion (H290) data on each CAS, physical state, or a Hazard Class override. **If missing:** Not printed `transportClassTokens(ov hazard_class) has '8' > section14.special_precautions_corrosive (en.php:542); SDSGenerator.php:4909-4911`
Special Precautions for User: marine-pollutant fragment (conditional):	TRANSPORT RULES Added when the printed Environmental Hazards line is exactly "Marine pollutant: Yes" in the sheet's language. That comes from H400/H410/H411, or from an override typed with that exact text. *On a sheet: Marine pollutant: prevent leaking product from entering drains, sewers or waterways.* **You maintain:** Aquatic toxicity data and M-factors on each CAS (CAS determinations), or the Environmental Hazards override. **If missing:** Not printed `ov(environmental_hazards,$env) === section14.marine_pollutant_yes > section14.special_precautions_marine (en.php:543); SDSGenerator.php:4912-4914`

Special Precautions for User: UV/EB-curable fragment (conditional):	PRODUCT FAMILY Added last when the product's family (a resale raw material's family on resale sheets) is flagged UV/LED curable. It ignores the UV rule-pack on/off setting and does not depend on the transport class. *On a sheet: UV/EB-curable product: protect from heat and direct sunlight during transport…* **You maintain:** Product families > Edit > "UV / LED curable family" checkbox; family assignment on the finished good or raw material. **If missing:** Not printed (no family, an inactive family, or not flagged UV). `$this->uvFamily (setUvState, familyIsUv = product_families.is_uv) > section14.special_precautions_uv (en.php:544; es/fr/de:508); SDSGenerator.php:4915-4917`
Special Precautions for User: per-product override:	PRODUCT OVERRIDE Text typed in the SDS editor for that language replaces the whole line: the standard sentence and all fragments. It does not change the classification or the publish gate. *On a sheet: Keep from freezing. Transport upright in closed containers.* **You maintain:** SDS editor: finished goods SDS > Edit; resale SDS Creation Readiness Check > Edit SDS text, Special Precautions for User, per language tab. **If missing:** Blank = the automatic standard sentence plus fragments prints. `text_overrides(14,special_precautions,lang) (TextOverrideService.php:40-41) > automatic line; SDSGenerator.php:4902-4903`
Note: combustible-liquid sentence (conditional):	TRANSPORT RULES Printed when the product is not regulated, is not Solid/Powder/Paste, and carries H227 without a flash point of 93 °C or more. That means a flash point above 60 °C and below 93 °C (a "> n" value counts as just above n), or H227 from the hazard override with no flash point. Dropped when any Section 14 core line is overridden. *On a sheet: Combustible liquid; regulated in bulk packaging (> 450 L) only.* **You maintain:** Flash point on the raws (Raw materials). Physical state (Finished goods, else the family's default physical state). **If missing:** Not printed `TC combustible (TC.php:203,248-250) > section14.note_combustible (en.php:534); SDSGenerator.php:4890-4896; engine Cat 4 HazardEngine.php`
Note: viscous Packing Group III sentence (conditional):	TRANSPORT RULES Printed only when the derived result is Class 3, Packing Group II, with no secondary class. Dropped when any Section 14 core line is overridden. *On a sheet: If viscous, may be assigned to Packing Group III under 49 CFR 173.121(b)(1)…* **You maintain:** Same inputs as Packing Group. **If missing:** Not printed `TC notes[]='viscous' (TC.php:397-401) > section14.note_viscous (en.php:536); SDSGenerator.php:4890-4896`
Note: carrier verification sentence:	FIXED TEXT (TRANSLATED) Fixed sentence from the translation file, always printed last on the Note line. *On a sheet: Transport classification should be verified with the carrier and against current 49 CFR…* **You maintain:** Nothing **If missing:** Always printed `section14.note (en.php:508) appended unconditionally at SDSGenerator.php:4896; PDFService.php:1104; preview.php:479-481`
Outcome wording: Not regulated / Not applicable / Not determined:	TRANSPORT RULES Fixed words from the translation file, chosen by the classification. With no class, the UN, name and class lines say "Not regulated" and Packing Group says "Not applicable". When the classifier cannot decide, all five lines say "Not determined". A per-line override still wins. *On a sheet: Not regulated* **You maintain:** Nothing directly (the outcome depends on flash point, physical state and hazard data) **If missing:** n/a `labels.not_regulated / labels.not_applicable / labels.not_determined (en.php:830-832); SDSGenerator.php:4842-4862`
Preview Warnings box: Section 14 messages (operator only, not printed):	TRANSPORT RULES English messages in every preview (finished good, resale, private label). They cover: the publish-block reason, missing n.o.s. technical names, PG II assumed for a missing boiling point, and raws with a flammable ingredient but no flash point. They also cover a hazard override that disagrees with the flash point, and a Section 9 flash-point or boiling-point edit dropped for contradicting the class. The last two are physical-state notes: the default "Liquid", or a family default state that differs from the largest raw. *On a sheet: Publishing blocked: the Section 14 transport classification for 12345 (EN sheet) is…* **You maintain:** Nothing directly; each message names the data to enter. **If missing:** No message when nothing applies. `generate():139-141,311-351 / generateFromBase():733-773; familyPhysicalStateWarning:1243; vetFlashPointEdit:4616; flammabilityWarnings:4692; SDSReadinessService:632`

SECTION 15: REGULATORY INFORMATION

Section 15 is built in SDSGenerator::section15(). It checks the product's composition against the regulatory lists you maintain: the TSCA inventory, SARA 313 (with its metal-compound and glycol-ether categories), HAP, SNUR and Prop 65. It also uses the Section 2 hazard result, which now includes OSHA Combustible Dust and the product family's default physical state, plus one per-product editor note (State Regulations) and fixed translated wording. OSHA and TSCA Status are automatic and cannot be edited per product. Trade-secret ingredient identities are masked on every line. Batch F changed two things here: the Prop 65 warning grammar ("is"/"are" by count, cancer clause first) and the English HAP heading, which now uses the Clean Air Act 112(b) title like ES/FR/DE.

OSHA Status:

HAZARD ENGINE Follows the same classification Section 2 prints: 'classified' when there is a signal word, pictogram, hazard class (including OSHA Combustible Dust) or H-statement, otherwise 'not classified'. It cannot be edited per product.

On a sheet: This product is not classified as hazardous under OSHA HazCom 2024 (29 CFR 1910.1200).

You maintain: The hazard data behind Section 2: constituent classifications and CPDs (CAS Determinations); flash and boiling points on raw materials; Physical State on the product or the family default physical state (Product families > SDS profile); Combustible Dust on raw materials (Raw materials > Edit; a Yes counts on any raw in a Powder/Solid product); the finished-good hazard override.

If missing: Nothing classified: 'This product is not classified as hazardous under OSHA HazCom 2024 (29 CFR 1910.1200).' No flash point on any raw material counts as not flammable; Combustible Dust left Not assessed adds no class.

`resolveOshaStatus (SDSGenerator.php:5080) → isClassified (:1474); applyCombustibleDust (:5657); resolvePhysicalState (:2980) own > family > largest raw > Liquid`

TSCA Status:

REGULATORY LIST Every ingredient CAS is checked against the EPA TSCA inventory and the per-CAS TSCA Review, and every formula raw material must have constituents with a CAS. Sentence: a CAS marked Not listed > all listed/exempt > not verified.

On a sheet: One or more components of this product are not listed on the TSCA inventory.

You maintain: TSCA Inventory imported from EPA (Regulatory lists > TSCA Inventory). TSCA Review on CAS Determinations for exempt, confidential, crossover, absent or INACTIVE CAS (INACTIVE: Listed + NOA note). Constituents with a CAS on every raw material.

If missing: 'TSCA inventory status has not been verified for all components.' prints when a CAS is unchecked, a trade secret has no CAS, or a raw material has no constituents or a blank-CAS constituent. A TSCA warning also shows (not printed).

`TSCAService::analyse(composition, formulaRawMaterialIds) (SDSGenerator.php:5058): cas_master.tsca_status<>'auto' > tsca_inventory > not_verified; statusKey (:5062)`

SARA 313 / TRI Reporting:

FIXED TEXT (TRANSLATED) Fixed heading from the translation file. Always printed on new sheets.

On a sheet: SARA 313 / TRI Reporting:

You maintain: Nothing

If missing: Older saved sheets without SARA data leave out the whole block.

`labels.sara_313_title; printed when sections[15].sara_313.reportable is an array (PDFService.php:1119-1121; preview.php:494)`

SARA 313 supplier-notification statement / 'none' sentence:

REGULATORY LIST Fixed wording chosen by the data: the statement when any ingredient is reportable (at or above its de minimis, or a PBT or TRI PFAS chemical at any amount), otherwise the 'none' sentence.

On a sheet: This product contains the following toxic chemical(s) subject to ... Section 313 ...

You maintain: SARA 313 list (Regulatory lists > SARA 313) and ingredient CAS numbers

If missing: Nothing reportable: 'This product does not contain any toxic chemicals subject to ... Section 313 ... at or above the applicable de minimis concentration.'

`labels.sara_313_statement when reportable non-empty, else labels.sara_313_none (PDFService.php:1123-1146; preview.php:496-510)`

SARA 313 chemical lines (name, CAS, band, threshold or no-de-minimis note):

REGULATORY LIST Ingredients on the SARA 313 list by CAS, or in a listed metal-compound or glycol-ether category, at or above the de minimis; PBT and TRI PFAS at any amount. Name: list name > category name > ingredient name; band from Section 3.

On a sheet: • Toluene (CAS 108-88-3) — 10 - 30% (de minimis threshold: 1%)

You maintain: SARA 313 list rows: CAS, name, de minimis % (0.1 for carcinogens), PBT flag, special-concern flag (CSV column 7 or migration 058). Category definitions from migration 058. Ingredient CAS and percentages on raw materials.

If missing: PBT prints 'PBT chemical; no de minimis exemption'; PFAS prints 'Chemical of special concern (40 CFR 372.28); no de minimis exemption'. Blank list name: category or ingredient name. CSV with no de minimis: 1%. Old sheets: empty band.

`SARA313Service::evaluate (sara313_list + RegulatoryCategoryService LIST_SARA; is_pbt|is_special_concern → threshold 0); bandRegulatoryEntries; PDFService.php:1125-1139`

SARA 313 range note:

FIXED TEXT (TRANSLATED) Fixed italic note that prints only when SARA chemicals are listed.

On a sheet: Concentrations are stated as the prescribed concentration ranges of 29 CFR 1910.1200(i)(1)...

You maintain: Nothing beyond the SARA 313 data above

If missing: Not printed when no SARA chemical is listed.

`labels.sara_313_range_note (PDFService.php:1141-1143; preview.php:508)`

Clean Air Act Section 112(b) — Hazardous Air Pollutants (HAP) (heading):

FIXED TEXT (TRANSLATED) Fixed heading, now the same Clean Air Act 112(b) title in English, Spanish, French and German.

On a sheet: Clean Air Act Section 112(b) — Hazardous Air Pollutants (HAP):

You maintain: Nothing

If missing: Older saved sheets without HAP data leave out the block. Sheets published before batch F keep 'EPA HAPs' (stored labels) until republished.

`labels.hap_title (en.php:810, es/fr.php:773, de.php:769); PDFService.php:1150-1188; preview.php:518, :539`

HAP table: Triggering HAP Chemical | Wt% in Formula:

REGULATORY LIST Ingredients at 0.01% or more whose CAS is on the HAP list or in a HAP compound category, plus HAPs typed on raw materials. Name: HAP list name > category name. Amount: Section 3 range band.

On a sheet: Toluene | 10 - 30%

You maintain: HAP List (Regulatory lists > HAPs) with a name for every CAS. HAPs typed on the raw material (name and weight %) for any HAP not matched by CAS. A molecular formula or recognisable name on the CAS so category members are found.

If missing: Blank HAP-list name: empty name cell. Typed HAP with a blank name or 0%: left out. No HAP at all: 'This product does not contain any EPA HAPs listed under Clean Air Act Section 112(b).'

`HAPService::evaluate (hap_list + RegulatoryCategoryService LIST_HAP, ≥0.01%) + getManualHaps (haps_data × summed RM %); bandRegulatoryEntries; sorted by exact % desc`

Total HAP Content:

FORMULA / RAW MATERIALS The exact percentages of all HAP rows are added up and printed as the prescribed-range band that contains the total.

On a sheet: Total HAP Content: 10 - 30%

You maintain: Same as the HAP rows

If missing: Printed only when there are HAPs. Older saved sheets that stored an exact total print no total row.

`total_hap_range = formatConcentration(total_hap_pct) (SDSGenerator.php:5012-5015); labels.hap_total; PDFService.php:1174-1181; preview.php:532`

EPA Significant New Use Rules (SNUR):

REGULATORY LIST Ingredients whose CAS is on the SNUR list, at any concentration. Name, CAS, rule citation and description come from the SNUR list exactly as typed (English).

On a sheet: • Chemical X (CAS 123-45-6) — 40 CFR 721.1234

You maintain: SNUR List (Regulatory lists > SNUR List, /admin/snur-list) with name, CAS, rule citation and description

If missing: No match: no SNUR block at all. Blank name: the line starts '(CAS ...)'. Blank citation or description: that part is left out.

`SDSGenerator::analyseSnur (:6442) on snur_list.cas_number; chemical_name, rule_citation, description; labels.snur_title; PDFService.php:1190-1210; preview.php:547`

Trade-secret ingredients in SARA 313, HAP and SNUR lines:

FORMULA / RAW MATERIALS A constituent flagged trade secret prints its trade-secret description (else 'Trade Secret') and 'TRADE SECRET' instead of its name and CAS; its band still prints. A SNUR line also drops the citation and prints a withheld-identity sentence.

On a sheet: • Acrylic monomer (CAS TRADE SECRET) — 1 - 5% (de minimis threshold: 1%)

You maintain: Trade-secret tick and trade-secret description on the raw-material constituent (Raw materials).

If missing: Blank description: 'Trade Secret' prints. Trade-secret constituents with no CAS are never checked against SARA 313, HAP or SNUR.

`maskTradeSecretRow (SDSGenerator.php:3092; called :5021-5036): name → trade_secret_description ?? labels.trade_secret; CAS → labels.trade_secret_cas; section15.snur_trade_secret`

California Proposition 65 (heading):

FIXED TEXT (TRANSLATED) Fixed heading, always printed.

On a sheet: California Proposition 65:

You maintain: Nothing

If missing: Uses the English wording if the sheet language lacks it.

`labels.prop65_title (PDFService.php:1216, :1253; preview.php:573, :586)`

Prop 65 warning symbol:

ADMIN SETTING Shown only when a Prop 65 warning prints. Admin-uploaded PROP65 image > image shipped with the system.

On a sheet: (yellow warning triangle beside the warning)

You maintain: Optional PROP65 image upload (Regulatory lists > Pictograms); otherwise the shipped image must exist

If missing: No image file: no symbol prints, but the warning text still prints. No warning: no symbol.

`PictogramHelper getPngPath/getWebPath('PROP65'): uploads/pictograms/PROP65.(png|jpg|gif) > assets/pictograms/png/PROP65.png; PDFService.php:1220-1231; preview.php:569-572`

Prop 65 WARNING text:

REGULATORY LIST Cancer and reproductive chemicals from ingredient CAS (0.01% or more) on the Prop 65 list plus raw-material Prop 65 entries, minus inhalation-only chemicals unless every formula line is Powder or Solid. Safe-harbor text: cancer clause first, then reproductive; 'is'/'are' by name count.

On a sheet: WARNING: This product can expose you to chemicals including Toluene, which is known...

You maintain: Prop 65 list (Regulatory lists > Prop 65: CAS, name, toxicity types). Prop 65 entries on raw materials (a listed CAS, or Override with name and types). The inhalation-only CAS setting. Physical state on every raw material.

If missing: No cancer or reproductive chemical left: 'This product is not known to contain any chemicals listed under California Proposition 65.' Types match in any case. Old name-only entries are used as typed.

`Prop65Service::analyse; removeInhalationOnlyFromResults (SDSGenerator.php:5956); rebuildProp65Warning (:6044) → Prop65Service::safeHarborWarning (:492)`

Prop 65 '(trace)' marker:

ADMIN SETTING Added to a chemical name in the warning when every occurrence is below the admin trace threshold (CAS path) or ticked 'trace' on the raw material.

On a sheet: Benzene (trace)

You maintain: Prop 65 trace threshold in Admin Settings; trace tick-boxes on raw-material Prop 65 entries

If missing: Setting blank or 0: 0.1% is used.

`setting prop65.auto_trace_threshold_pct (default 0.1; Prop65Service.php:127); prop65_data[].is_trace (:283); section15.prop65_trace_name (:441; SDSGenerator.php:6051)`

Listed chemicals:

REGULATORY LIST One line per Prop 65 chemical with its listing types, printed under the warning. The types come from the Prop 65 list or the raw-material override.

On a sheet: • Toluene (CAS 108-88-3) — developmental toxicity, female reproductive toxicity

You maintain: Prop 65 list toxicity types: cancer, developmental, female reproductive, male reproductive or reproductive. Any case is accepted, as are a 'toxicity' suffix and 'female'/'male'.

If missing: No CAS: the line prints without '(CAS ...)'. A type outside that list is skipped; a chemical with no recognised type gets no line. No warning: no list.

`buildProp65ListedLines (SDSGenerator.php:3813-3869); Prop65Service::normaliseTypes; labels.prop65_listed; section15.prop65_listed_line / _no_cas / prop65_type_*`

State Regulations:

PRODUCT OVERRIDE Only from the note the operator types in the SDS editor for this product or resale raw material, separately for each language. Prints whenever present.

On a sheet: State Regulations: New Jersey Right-to-Know: Toluene (CAS 108-88-3)

You maintain: SDS editor > Section 15 > Additional State Regulations (/sds/{fg}/edit or /sds/resale/{rm}/edit), per language

If missing: Line left out. There is no fallback from another language.

`trim(overrides[15][state_regs]) via getOverrides(fg | resale rm, lang) (SDSGenerator.php:4996); labels.state_regulations; PDFService.php:1263-1266; preview.php:591-595`

Sections 12–15 footnote:

ADMIN SETTING Fixed translated note, switched on or off for all sheets by one admin checkbox (on by default).

On a sheet: Sections 12–15 are included as required by OSHA HazCom (29 CFR 1910.1200(g)(2))...

You maintain: Admin Settings checkbox 'Print the Sections 12–15 footnote'

If missing: Checkbox off: nothing prints.

`settings sds.show_ghs_section_note (missing = on, '0' = off) → ghsSectionNote() (SDSGenerator.php:5189) → ghs_note; PDFService::renderGhsSectionNote (:1298-1310); preview.php:657`

Preview Warnings box: TSCA, incomplete composition, trade-secret Prop 65, combustible dust (not printed):

REGULATORY LIST Operator messages shown above the preview and never printed. They cover a TSCA gap or INACTIVE entry, incomplete constituent data, a trade-secret constituent on the Prop 65 list (also blocks publishing) and powder/solid raws whose combustible dust is not assessed.

On a sheet: TSCA warning: Section 15 prints "One or more components ... not listed ..." — ...

You maintain: The same data as TSCA Status, Prop 65 and OSHA Status: TSCA inventory and reviews, constituents with CAS and percentage, trade-secret ticks, Combustible Dust on powder/solid raw materials.

If missing: No message when everything is covered.

`TSCAService::warningText; SDSReadinessService::incompleteCompositionWarning, tradeSecretProp65Error, combustibleDustNotAssessedWarning (SDSGenerator.php:298-366, :720-788)`

SECTION 16: OTHER INFORMATION

Section 16 is built entirely from publish metadata (the version number and effective date stamped when a sheet is published), the per-language translation files (labels, draft text, abbreviation master table) and the Admin or manufacturer disclaimer text. Nothing in it can be typed per product. Batch F (0c9207b / 072cd54) changed only the abbreviation tables: SDS (ES HDS, FR FDS), ASTM, ISEA and PBT are now defined wherever they print.

Version:

PUBLISH RECORD The number the system assigns at publish: the next number for that finished good or resale raw material, alias (when a sender publishes it, pack variants share one counter), or private-label item. Unpublished previews print the translated draft text.

On a sheet: Version: 3 (unpublished: Version: Draft (not yet published))

You maintain: Nothing to enter. Never hard-delete published SDS rows (soft-deleted rows still count toward the next number).

If missing: Draft or preview: 'Draft (not yet published)' (translated). A pre-version snapshot, now re-rendered only in a private-label record preview, prints no Version line.

`section16() :5099 = section16.draft; stampPublishedVersion() :1073 sets it; MAX(sds_versions.version)+1 per base / alias id / alias group (AliasPublisher :216-227), or private_label_sds`

Effective Date:

PUBLISH RECORD The publish date (today in the admin time zone) written to the version record. It prints as month/day/year in every language.

On a sheet: Effective Date: 10/12/2026

You maintain: Nothing

If missing: Draft or preview: the line is not printed. A pre-version private-label snapshot in its record preview has no line; the old 'Revision Date' line prints instead.

`stampPublishedVersion :1079 format_date(...,'m/d/Y'); publishers pass PublishClock::todayLocal() (AliasPublisher :220, PrivateLabelPublisher :370; SDSController/SDSUpdateController)`

Abbreviations:

FIXED TEXT (TRANSLATED) A master list of abbreviations for each language, filtered to the terms that print elsewhere on that sheet (including the alias or private-label code and the disclaimer) and printed in file order.

On a sheet: Abbreviations: ASTM = ASTM International (formerly …); CAS = Chemical Abstracts Service …; SDS = Safety Data Sheet; …

You maintain: Developers keep the four translation tables (en/es/fr/de section16.abbreviation_table) complete and alphabetical. Any new abbreviation in generated text, labels, disclaimers or overrides must be added to all four.

If missing: No term matched: the line is not printed. A printed term missing from the sheet language's table is left undefined. An empty table falls back to the English one.

`AbbreviationService::build :46 (generate :371, generateFromBase :792, alias :884, PL :1026); section16.abbreviation_table[lang] > EN; PATTERNS :27; collectCorpus :107-213`

DOCUMENT HEADER, FOOTER AND DISCLAIMER

Document chrome is the page-1 header, section banners, footer, disclaimer, PDF properties, file names and the preview page around the PDF. It comes mainly from company or manufacturer settings, the per-language translation files and the version and date stamped at publish, plus a few fixed layout constants. Alias sheets print the alias base code, and senders always use an alias's own published PDF. In this scope, batch F (0c9207b, 072cd54) changed only the preview Warnings box, which gained STOT target-organ, combustible-dust and family physical-state messages. Nothing printed on the PDF chrome changed.

Header logo (page 1) - standard, alias and resale SDS:

COMPANY / MANUFACTURER The company logo uploaded in Admin Settings. It prints on page 1 only.

On a sheet: Company logo top-left of page 1, about 51 mm wide

You maintain: Upload the company logo (PNG/JPG/GIF, 2 MB max) in Admin > Settings > Company Logo. The file must stay on the server.

If missing: No logo prints, but the title and rule still print. Re-renders of older snapshots use today's logo file, or none if the logo was removed or its file type changed.

`settings.company.logo_path (AdminController::processLogoUpload) > meta.company_logo_path (SDSGenerator :245/:668) > PDFService::resolveLogoPath :1395-1402 > SDSTcpdf::Header :97-111`

Header logo (page 1) - private label SDS:

COMPANY / MANUFACTURER The private label manufacturer's own logo. The company logo is never used in its place.

On a sheet: Manufacturer's logo top-left of page 1

You maintain: Upload a logo on each private label manufacturer's form.

If missing: No logo prints. The company logo is not used instead.

`manufacturers.logo_path ('/uploads/manufacturers/mfg-logo-<id>.<ext>') > SDSGenerator::createManufacturerVariant :999 sets meta.company_logo_path`

SAFETY DATA SHEET (header title):

FIXED TEXT (TRANSLATED) Fixed title from the translation file for the sheet language, printed bold and right-aligned on page 1. Re-renders of older snapshots that lack it get the sheet-language title.

On a sheet: SAFETY DATA SHEET / HOJA DE DATOS DE SEGURIDAD

You maintain: Nothing

If missing: "SAFETY DATA SHEET" (the English default) only if no translation file defines it.

`meta.document.title > translation document.title (sheet lang; SDSDocumentStrings::forLanguage) > DEFAULTS; SDSTcpdf::Header :88-95`

Header rule; no header on pages 2+:

FIXED TEXT A navy rule sits under the page-1 header. Pages 2 onward have no header and the text starts near the top edge.

You maintain: Nothing

If missing: n/a

`SDSTcpdf::Header :70-72 returns when page > 1, rule :113-117; PDFService MARGIN_TOP 30 / MARGIN_TOP_CONTINUATION 15 (:19-21), SetTopMargin :300`

SECTION N: TITLE banners (all 16):

FIXED TEXT (TRANSLATED) The word 'SECTION' and each section title come from the translation file and print in capitals, accented letters included. Older snapshots without a stored prefix get the sheet-language word.

On a sheet: SECTION 1: IDENTIFICATION / SECCIÓN 1: IDENTIFICACIÓN

You maintain: Nothing

If missing: The sheet-language "SECTION" plus the translation title

`meta.document.section_prefix (forLanguage) + sections[N].title (translation sectionN.title); mb_strtoupper in PDFService::renderSection :316-327; preview.php :18/:45`

Sections 12-15 footnote (after Section 15):

ADMIN SETTING One shared footnote in the sheet language, printed once after Section 15 when the admin toggle is on (the default). The words are the same on every sheet; the toggle only turns it on or off.

On a sheet: Sections 12–15 are included as required by OSHA HazCom (29 CFR 1910.1200(g)(2))...

You maintain: Admin > Settings: tick 'Print the Sections 12–15 footnote' to print it (on by default).

If missing: Toggle off: nothing prints.

`settings.sds.show_ghs_section_note ('0' = off) gates document.ghs_section_note > sections[15].ghs_note (SDSGenerator :5070, ghsSectionNote :5189-5199); PDFService :372 > renderGhsSectionNote :1299-13…`

DISCLAIMER (bar):

FIXED TEXT (TRANSLATED) A navy bar after Section 16, labelled from the translation file. It prints only when there is disclaimer text.

On a sheet: DISCLAIMER / DESCARGO DE RESPONSABILIDAD / HAFTUNGSAUSSCHLUSS

You maintain: Nothing

If missing: Newly generated sheets always print it. A re-render of a pre-#34 snapshot with no stored disclaimer prints no bar and no text.

`meta.labels.disclaimer > translation labels.disclaimer (sheet lang, then EN) > key; PDFService::renderLegalDisclaimer :1407-1426; preview.php :663-666`

Disclaimer text - standard, alias and resale SDS:

ADMIN SETTING Admin per-language disclaimer > built-in translation text for that language. It is frozen into the snapshot at publish.

On a sheet: The information provided in this Safety Data Sheet is correct to the best of our...

You maintain: Optional. Admin > Settings > Legal / Disclaimer Statement, one box per language (EN was seeded by migration 052).

If missing: The built-in translation paragraph for the sheet language (never blank on new sheets). Re-renders of pre-#34 snapshots keep the old value, which may be blank or English.

`settings.sds.legal_disclaimer.<lang> > translation section16.disclaimer; SDSGenerator::resolveLegalDisclaimer :5166-5173 (meta :274/:697); migration 052 seeded .en only`

Disclaimer text - private label SDS:

COMPANY / MANUFACTURER Manufacturer's disclaimer for that language > admin per-language disclaimer > built-in translation text.

On a sheet: Manufacturer's own disclaimer paragraph in the sheet language

You maintain: Optional. A per-language disclaimer on the manufacturer form.

If missing: The admin text for that language, else the built-in translation text.

`manufacturers.disclaimer_json[lang] (blank = inherit) > settings.sds.legal_disclaimer.<lang> > section16.disclaimer; createManufacturerVariant :1005-1007`

Bold field labels throughout the sheet:

FIXED TEXT (TRANSLATED) Labels stored with the sheet > translation label in the sheet language > English translation > a capitalized key name.

On a sheet: Product Identifier: / Contaminated Packaging: / Identificador del producto:

You maintain: Nothing

If missing: The translation label (sheet language, then English)

`meta.labels (SDSGenerator::getLabels :5347) > labels.<key> via SDSDocumentStrings::translate > key; PDFService::label :1349-1354`

Footer left: product code (every page):

FORMULA / RAW MATERIALS The code of the product the sheet is issued for. Alias and resale sheets print the code without its pack extension; private label custom codes print exactly as typed.

On a sheet: UVNG009 / BK1080 (alias BK1080-5G) / AC-100 (private label custom code)

You maintain: The product code on the finished good, the customer code on each alias, the internal code on resale raw materials, and the custom code on private label items.

If missing: The cell is empty.

`meta.product_code = FG code; alias/resale stripPack(code); PL custom_code > stripped alias > FG (PrivateLabelPublisher::resolveIdentity :51); SDSTcpdf::Footer :144-146`

Footer centre: Page X of Y:

FIXED TEXT (TRANSLATED) The words come from the translation file for the sheet language (also on re-renders of older snapshots) and the numbers from the PDF engine.

On a sheet: Page 2 of 5 / Página 2 de 5 / Page 2 sur 5 / Seite 2 von 5

You maintain: Nothing

If missing: "Page X of Y" in English only if no translation file defines the words.

`meta.document.page / page_of (SDSDocumentStrings::forLanguage) > DEFAULTS; SDSTcpdf::Footer :148-153`

Footer right: Rev. version and effective date (published):

PUBLISH RECORD The version number and publish date stamped automatically at publish, the same values as Section 16. The date is today in the admin time zone and always prints as m/d/Y.

On a sheet: Rev. 3 — 10/09/2026 / Rév. 3 — 10/09/2026

You maintain: Nothing. It is automatic at publish. The date follows the time zone set in Admin > Settings.

If missing: Only the version prints: "Rev. 3".

`PDFService::footerRevision :1320-1335: revision_prefix + meta.sds_version + (s16.effective_date > meta.effective_date > s16.revision_date); PublishClock::todayLocal`

Footer right: Draft (unpublished):

FIXED TEXT (TRANSLATED) Prints on previews and ad-hoc renders that have no version stamp. A published private label record re-rendered because its snapshot is missing prints its own version instead.

On a sheet: Draft (not yet published) / Borrador (aún no publicado)

You maintain: Nothing

If missing: n/a

`sections[16].version = translation section16.draft (SDSGenerator :5099) when meta.sds_version is 0/absent; footerRevision :1340; PrivateLabelController preview :975-979 stamps`

Footer right: legacy generation date (old snapshots):

PUBLISH RECORD Re-renders of snapshots made before #32 print the old generation date instead of a version.

On a sheet: Rev. 10/01/2026

You maintain: Nothing

If missing: The Draft text

`footerRevision :1336-1338 uses sections[16].revision_date when meta.sds_version is absent`

Footer rule and typography; page size and font:

FIXED TEXT Every sheet in every language uses a grey rule 15 mm from the bottom, 8 pt Helvetica in three equal cells, and US Letter portrait pages. The core font prints Latin-1 and Windows-1252 characters only.

You maintain: Nothing

If missing: Any other character (for example '≤' or '→') prints as "?".

`SDSTcpdf::Footer :123-158; new SDSTcpdf('P','mm','LETTER') PDFService :261; core font helvetica; MARGIN_BOTTOM 20; tests/Services/CoreFontCharsetTest.php`

PDF property: Title:

FIXED TEXT (TRANSLATED) A translated prefix plus the code printed in the footer.

On a sheet: SDS - UVNG009 / HDS - BK1080 / FDS - ... / SDB - ...

You maintain: Nothing beyond the product code

If missing: English "SDS - <code>" only if no translation file defines the prefix. Re-renders of older non-English snapshots get their own-language prefix.

`meta.document.pdf_title (forLanguage) with :code = meta.product_code; PDFService :271`

PDF property: Author:

COMPANY / MANUFACTURER The supplier name printed in Section 1: the company name from Admin Settings, or the private label manufacturer's name.

On a sheet: Acme Inks, Inc.

You maintain: Company name in Admin > Settings, and the name on each manufacturer (required; a blank name cannot be saved).

If missing: "SDS System", with a preview warning. If the company name was never saved, the config value (placeholder "Your Ink Company, Inc.") is used, also with a warning.

`trim(sections[1].manufacturer_name): settings.company.name (config company.name only if never stored) / manufacturers.name; PDFService :268-270`

PDF property: Subject:	FIXED TEXT (TRANSLATED) Fixed text from the translation file for the sheet language. *On a sheet: Safety Data Sheet / Hoja de datos de seguridad* **You maintain:** Nothing **If missing:** English "Safety Data Sheet" only if no translation file defines it. `meta.document.pdf_subject (SDSDocumentStrings::forLanguage) > DEFAULTS; PDFService :272`
PDF properties: Creator, Producer, dates, XMP, language:	FIXED TEXT Fixed program names and the render time, with an embedded XMP copy of Title, Author and Subject. No document language is declared. *On a sheet: Creator 'SDS System'; Producer 'TCPDF 6.11.3 (http://www.tcpdf.org)'* **You maintain:** Nothing **If missing:** n/a `PDFService :265 SetCreator; TCPDF _putinfo / _putXMP; no /Lang (setLanguageArray never called); no Keywords`
Stored PDF file name (published):	PUBLISH RECORD The code as printed in the footer (characters other than letters, digits, '.', '-' and '_' become '_'), then the PL tag, the version and the language (the default language has no suffix). A name clash adds _2, _3. *On a sheet: UVNG009_v3.pdf / BK1080_v2_es.pdf / AC-100_PL_Acme_Coatings_v2.pdf* **You maintain:** Product and alias codes, private label custom codes, and the manufacturer name for private label. **If missing:** An empty code gives "unnamed_file". A blank manufacturer name gives the bare tag "PL", but such a sheet is refused at publish. `PDFService::generate :113-124 (sanitize_filename), versionedBaseName :160, reserveUniquePath :179; meta.filename_tag (name cut to 40 chars) createManufacturerVariant :1016-1018`
Stored PDF file name (unversioned render):	FIXED TEXT Code, an optional PL tag, the language and a timestamp in the admin time zone, used when the sheet has no version. *On a sheet: UVNG009_SDS_en_20261009_143012.pdf* **You maintain:** Nothing beyond the product code **If missing:** An empty code gives "unnamed_file_SDS_...". `PDFService::generate :122 ($stem . '_SDS_' . lang . '_' . date('Ymd_His'))`
Download file names:	PUBLISH RECORD Built when a user downloads; standard and Lookup names strip the pack extension, and only Lookup leaves out the language. Preview names use the printed code. *On a sheet: SDS_UVNG009_v3_en.pdf / SDS_BK1080_v3.pdf / PL_SDS_AC_100_Acme_v2_en.pdf* **You maintain:** Nothing **If missing:** No code: "SDS_v3_en.pdf" / Lookup "SDS_SDS_v3.pdf" / private label "PL_SDS_SDS_..." / preview "SDS_draft_en.pdf" `SDSController::download :852-866; LookupController :87-95 (no lang); PrivateLabelController :912-915; SDSPreviewResponse::filename :56-68 (v{n} or draft)`
Email attachment and report ZIP names:	PUBLISH RECORD Item code + '_SDS' (+ language when not English), with no version. Auto-send cuts an alias item code at its first hyphen (R1055-84 becomes R1055); the send queue (English only) and report ZIP keep the code as shipped. *On a sheet: R1055_SDS.pdf / R1055_SDS_ES.pdf / BK1080-5G_SDS.pdf* **You maintain:** Nothing **If missing:** Auto-send skips a language with no published PDF on disk and attaches nothing for an alias that cannot be published. The report ZIP logs the item in _MISSING_ITEMS.csv. `SDSAutoSendService::sendOrderEmail :384-432 (alias: code up to first '-'); SDSSendQueueController::sendEmail :267-273 (item_name or item_code); ReportController ZIP :972-1084`
Alias PDFs in sends (send queue, auto-send, report ZIP):	PUBLISH RECORD The alias's own published PDF; pack variants of one alias share one sheet. If a wanted language is missing or its PDF is gone from disk, the alias is published first from current data. *On a sheet: Footer 'BK1080' with the alias's own 'Rev. 2 — 10/09/2026'* **You maintain:** Nothing beyond the alias code and description. **If missing:** If the alias cannot be published: the queue shows the reason, auto-send attaches nothing for it, and the report ZIP lists it in _MISSING_ITEMS.csv. `AliasPublisher::ensurePublished :112-146 > publish :164-270 (createAliasVariant(stripPack code) + stampPublishedVersion); never a snapshot re-render`
Preview page: browser title, back links, status line, buttons:	FIXED TEXT English application text around the embedded real PDF. The browser title names the product code, the status line shows the language and the version or Draft, and the private label back link names the manufacturer. *On a sheet: EN — Draft: this is the PDF Publish will produce...* **You maintain:** Nothing **If missing:** Links and buttons that do not apply are left out. `layouts/main.php:7 ('SDS Preview: ' . product_code); SDSController::preview :58-111, previewResale :121-200; PrivateLabelController preview; preview-pdf.php; SDSPreviewResponse`

Preview page: Warnings box:

FORMULA / RAW MATERIALS Data-quality and publish-block messages from the generator on every preview, plus supplier checks per path. They never print on the PDF; the private label stored preview shows those frozen at publish.

On a sheet: Combustible dust: this product's physical state is Powder and its formula contains ...

You maintain: The underlying data: emergency phones, company name and phone, raw material data (incl. Combustible Dust), STOT target organs, family default physical state, transport data, TSCA list, product status.

If missing: The box is not shown.

`sds.warnings: calc (SG/VOC, physical state, family state), UV, TSCA, composition, S14, trade secret, flash point, STOT organs, combustible dust; FG adds SDSController :76-90`

CALCULATED VALUES AND THEIR INPUTS

In this scope, every number, range and derived classification on the sheet is worked out when the sheet is generated. The inputs are the formula lines and each raw material's own data: constituent percentages, flash and boiling points, specific gravity, VOC, solubility, physical state and combustible-dust status. Next come the CAS data (classifications, exposure limits, and the element, acrylate and target-organ flags), the product family's UV flag and SDS profile, and the admin regulatory lists. These pass through the hazard engine and the transport classifier. The product flash point is the wt%-weighted average of the raw materials that have one, and a product with no flash point is not flammable. Batch F made these changes. The calculated flash point prints rounded to a whole degree with "(calculated)", and a measured edit prints as typed. A product family can set a default physical state, storage temperatures, water-based status and a typical pH. Section 5 names combustion products from the CAS element flags, and H370-H373 can name target organs. A Powder or Solid product that contains a raw material marked combustible dust gets the OSHA Combustible Dust class and its Section 5 and 7 sentences. Section 8 adds a skin-notation line and a ventilation sentence for products with no exposure limits. Section 13 adds contaminated packaging and a UV cure-residue sentence. Section 14 Special Precautions adds Class 3, Class 8, marine and UV sentences. Missing data raises preview warnings, which are never printed. Per-product text overrides still win where they exist.

Flammable Liquids class line and H224-H227 (Section 2; drives Sections 5-8, 10, 13, 14):

FORMULA / RAW MATERIALS A Flammable Liquids class or H224-H227 in the finished-good hazard override > the category from the formula flash point (unrounded, 0.1 °C) and initial boiling point: Cat 1 below 23 °C with IBP 35 °C or less, Cat 2 below 23 °C, Cat 3 up to 60 °C, Cat 4 up to 93 °C. Ingredient flammable classes are ignored, and a product whose resolved state (its own, the family default or the largest raw material's) is Solid, Powder or Paste is never classified.

On a sheet: Flammable Liquids, Category 3 — H226 Flammable liquid and vapor

You maintain: Flash Point (°C) and Boiling Point (°C) on raw materials; Physical State on the product, its family's default, or its raw materials; a Flammable Liquids class in the finished good's hazard override only for a tested product

If missing: No flash point on any raw material: not classified as flammable. Flash point below 23 °C with no boiling point: Category 2 is assumed and a preview warning is shown

`HazardEngine::applyFlammableLiquidsFromFlashPoint :2868, flammableLiquidCategory :2781, FG override :1969-2009; SDSGenerator::flammabilityInputs :4546`

Combustible Dust hazard class line (Section 2):

FORMULA / RAW MATERIALS Added when the resolved physical state is Powder or Solid and any raw material in the formula (at any depth, above 0%) is marked Combustible Dust = Yes, unless a replace-mode finished-good hazard override is set. It sets the signal word to Warning when there is none and adds no pictogram or H-code.

On a sheet: Combustible Dust: May form combustible dust concentrations in air.

You maintain: Combustible Dust (Not assessed / Yes / No) on each Powder or Solid raw material, from the vendor SDS Section 2 (Raw Materials > Edit)

If missing: Not assessed or No: no line. A Powder/Solid product with not-assessed Powder/Solid raw materials gets a preview and readiness warning only; publishing is never blocked

`SDSGenerator::combustibleDustStatus :5600, applyCombustibleDust :5657, localizeCombustibleDustClass :5709; raw_materials.combustible_dust (062)`

Target-organ hazard statements H370-H373 (Section 2; also Section 4/11 symptoms and the Section 11 STOT RE line):

HAZARD ENGINE The organs are named when every CAS behind the code has organs entered and no hazard override or trade-secret entry also carries the code. Organs from several CAS are combined, the route prints only when all give the same single route, and otherwise the standard GHS statement prints.

On a sheet: Causes damage to organs (lungs) through prolonged or repeated exposure by inhalation

You maintain: Target organs and route per CAS (CAS Determinations > STOT Target Organs tab), from the vendor SDS or the CLP Annex VI entry

If missing: The standard statement, e.g. 'Causes damage to organs through prolonged or repeated exposure'. Organs on only some of the CAS, or the code also coming from a hazard override or trade-secret entry: standard statement plus a preview warning

`StotTargetOrgans::resolve :163, statementText :227, chronicText :238; HazardEngine stot_details :1125; SDSGenerator::applyStotTargetOrgans :1446, buildChronicEffects :3436`

UV polymerization wording (Section 2 Other Hazards; Section 10 Reactivity; Sections 7/10 incompatible materials):	PRODUCT FAMILY Printed when the product's active family is flagged UV/LED (the UV rule-pack switch is not needed): Other Hazards uses the polymerization sentence instead of 'None known.', Reactivity adds the hazardous-polymerization sentence, and 'peroxides and free-radical initiators' joins the incompatible materials that Section 7 storage repeats. A per-product override of a field replaces it. *On a sheet: Uncured product may polymerize when exposed to UV light, elevated temperatures…* **You maintain:** UV/LED flag and membership rules on the product family (Product Families) **If missing:** Not a UV family (or inactive family): 'None known.', the standard Reactivity sentence and no peroxides item `SDSGenerator::section2 :1529, section10 :3314, incompatibleMaterialItems :2520; section2.other_hazards_uv, section10.reactivity_uv_polymerization`
Carbon black / titanium dioxide cancer hazard (Sections 2, 3, 8, 11 and 15 Prop 65):	FORMULA / RAW MATERIALS When every raw material line is Solid or Powder, each inhalation-only CAS at 0.1% or more adds Carcinogenicity Cat 2 (H351) unless H350/H351 is already present or a replace-mode override is set, and Prop 65 keeps it. Otherwise these CAS are kept out of classification and exposure limits, Prop 65 drops them, and Section 11 keeps the listing with an inhalable-dust note. *On a sheet: H351 added only when every raw material in the formula is Solid or Powder* **You maintain:** Physical State on every raw material (Raw Materials form); the inhalation-only CAS list (Admin > Settings; default carbon black 1333-86-4 and TiO2 13463-67-7) **If missing:** A raw material with a blank Physical State counts as bound (not dry), so H351 and the Prop 65 listing are suppressed `SDSGenerator::isBoundMixture :5549, inhalationOnlyCasToExclude :5568, applyCarbonBlackLogic :5754, removeInhalationOnlyFromResults :5956; setting sds.inhalation_only_cas`
Section 3 Type (Substance / Mixture):	FORMULA / RAW MATERIALS Substance/Mixture setting on the finished good > on Auto, 'Substance' only for a single-ingredient formula whose raw material is marked Substance, otherwise 'Mixture'. Resale sheets use the raw material's own setting. *On a sheet: Mixture* **You maintain:** Substance/Mixture on the finished good, or on the raw material for single-ingredient and resale products **If missing:** 'Mixture' `SubstanceMixtureResolver; finished_goods.substance_mixture > raw_materials.substance_mixture (single line); SDSGenerator :147, :496, section3 :1701`
Section 3 component rows (which ingredients are listed):	FORMULA / RAW MATERIALS A constituent is listed when its amount (the exact %, else the top of its range) is 0.1% or more and it is classified, has an exposure limit (ignored for an ingredient supplied only by Solid or Powder raws when the formula has another state), or (at 1% or more) is a flammable-liquid ingredient behind the flash-point class. A Substance sheet always lists its largest constituent; inhalation-only CAS with no remaining hazard are dropped, and trade-secret rows are merged by description. *On a sheet: Toluene | 108-88-3 | 10 - 30% | H225, H304, H315* **You maintain:** Constituent CAS and % on each raw material (Raw Materials > Constituents); classifications per CAS (CAS determinations) **If missing:** Rows below 0.1% are left out, even when several trade-secret rows with one description together reach 0.1% `SDSGenerator::section3 :1533-1729; HazardEngine flash-point contributors :2942; exposure_limits solid-in-liquid filter :5898; sds.inhalation_only_cas`
Section 3 Concentration column:	FORMULA / RAW MATERIALS Each constituent's lower and upper % (an exact value counts as both) times its line % are summed per CAS and placed in the widest prescribed HazCom range that contains that span. If no range contains it, the range containing the maximum with the lowest lower end is used, so the upper end is never below the maximum present. *On a sheet: 10 - 30%* **You maintain:** Constituent % (exact, or min and max) on each raw material (Raw Materials > Constituents) and the line % on each formula **If missing:** A constituent with no % counts as 0 and is dropped. '<0.1%' prints only when the maximum is below 0.1%; above 100% prints '80 - 100%' `SDSGenerator::formatConcentration :5292, PRESCRIBED_RANGES :5278; Formula::constituentBounds :821, resolveConstituentPct :791; resale FormulaCalcService :355`

Trade-secret ingredient identity (Sections 3, 8, 11, 12, 13 and 15):	FORMULA / RAW MATERIALS A constituent marked trade secret on any raw material prints its trade-secret description (else 'Trade Secret') and 'TRADE SECRET' for the CAS, with the band only, in every section, including the Section 8 skin-notation line. Section 13 uses generic RCRA wording with no code, and the SARA category code and SNUR citation are blanked. *On a sheet: Acrylic resin | TRADE SECRET | 10 - 30%* **You maintain:** Trade-secret flag and description (plus declared H-codes on blank-CAS rows) on the raw material's constituents (Raw Materials > Constituents) **If missing:** A blank description prints 'Trade Secret'. A trade-secret CAS on the Prop 65 list blocks publishing `SDSGenerator::maskTradeSecretRow :3092, tradeSecretName :3072; section3 :1644; section8 :2797, :2811; section13 :4382-4400; section15 :5019-5040`
Section 5 Specific Hazards (flammable sentence and 'Flash point:' line); flammable media and firefighter advice:	FORMULA / RAW MATERIALS Per-product override, with its 'Flash point:' sentence refreshed each time > the sentence for the sheet's Flammable Liquids category (H224-H227) followed by the same flash point Section 9 prints. Extinguishing media and firefighter advice switch to the flammable wording on the same test. *On a sheet: Flammable liquid and vapor… Flash point: 45 °C (113 °F) (calculated).* **You maintain:** Flash and boiling points on raw materials (they set the category); Physical State **If missing:** No flammable sentence and no 'Flash point:' line when the product is not a flammable liquid (including when there is no flash point data); a stale 'Flash point:' sentence in an edit is removed `SDSGenerator::section5 :2158-2289, refreshFlashPointSentence :2129, resolveFlashPointDisplay :2034; section5.specific_hazards_flammable_cat1-4, flash_point_line`
Section 5 Specific Hazards: combustion products sentence:	FORMULA / RAW MATERIALS Per-product Specific Hazards override > carbon monoxide and carbon dioxide always, plus nitrogen oxides, sulfur oxides and hydrogen halides for each element flag on any CAS in the composition (the same items as Section 10 Decomposition). Oxidizers and organic peroxides get their own lead sentence before it. *On a sheet: Combustion may produce carbon monoxide, carbon dioxide, nitrogen oxides, and other toxic gases.* **You maintain:** Nitrogen, sulfur and halogen flags per CAS (CAS Determinations > Element flags) **If missing:** No flagged CAS: 'Combustion may produce carbon monoxide, carbon dioxide, and other toxic gases.' `SDSGenerator::section5 :2247-2252, combustionProductsSentence :2299, decompositionElements :2476; cas_master has_nitrogen/has_sulfur/has_halogen; section5.combustion_products`
Combustible dust sentences (Section 5 Unsuitable Media and Specific Hazards; Section 7 Handling and ignition sentences):	FORMULA / RAW MATERIALS Printed when Section 2 carries the Combustible Dust class. Section 5 adds the no-high-pressure-jet sentence and the dust-explosion sentence, and Section 7 adds its ignition-source sentences and the dust-handling sentence (without the grounding sentence when the product is also flammable); a per-product override of a field replaces it. *On a sheet: Fine dust dispersed in air in sufficient concentration can form an explosive mixture…* **You maintain:** Combustible Dust on Powder/Solid raw materials; Physical State on the product, family default or raw materials **If missing:** Not printed `SDSGenerator::section5 :2214-2216, :2256-2258; section7 :2631-2654, :2715; hasCombustibleDust :5637; section5.*combustible_dust, section7.handling_combustible_dust(_no_ground)`
Section 6 Methods and Material for Containment and Cleaning Up:	FORMULA / RAW MATERIALS Per-product override > wording for the resolved physical state: Solid or Powder sweep, Gel or Paste scrape, Gas ventilate, anything else the liquid absorbent text. On a UV rule-pack sheet the UV containment sentence follows. *On a sheet: Stop leak if safe to do so. Dam or contain spill. Absorb with inert absorbent material…* **You maintain:** Physical State on the finished good, the Default physical state of its family, or its highest-weight raw material that has one **If missing:** Liquid wording `SDSGenerator::section6 :2420-2437; text_overrides(6,'containment') > section6.containment_solid|_paste|_gas|_liquid (+ section6.uv_containment); resolvePhysicalState :2980`
Storage temperature (Section 7 Storage sentence; Section 10 Conditions to Avoid item):	PRODUCT FAMILY Per-product Storage or Conditions to Avoid override > the active family's storage minimum and maximum: Section 7 prints the range sentence and Section 10 adds 'temperatures above X °C (Y °F)', which takes the place of the generic storage-limit item self-reactive and self-heating products get. °C prints without decimals when whole; °F is rounded to a whole degree. *On a sheet: Store between 5 and 30 °C (41–86 °F). … temperatures above 30 °C (86 °F)* **You maintain:** Storage temperature minimum and maximum (°C, both or neither) in the family's SDS profile (Product Families > Edit > SDS profile) **If missing:** No storage-temperature sentence and the generic Section 10 items; no family or an inactive family prints nothing `SDSGenerator::setProfileState :1311, storageTemperatureSentence :1364, section7 :2708, section10 :3349 (cond_storage_max); product_families.storage_temp_min_c/_max_c (061)`

Protect from freezing (Section 7 Storage; 'freezing' in Section 10 Conditions to Avoid):	PRODUCT FAMILY Per-product Storage or Conditions to Avoid override > printed when the product is water-based: the family's Water-based setting Yes or No > on Auto (or with no active family), water CAS 7732-18-5 at or above the water-based threshold (default 20%) of the expanded formula. Water held only in a raw material's water wt% does not count. *On a sheet: Protect from freezing. … Heat, freezing, sources of ignition, …* **You maintain:** Water-based (Auto / Yes / No) in the family's SDS profile; water CAS 7732-18-5 as a constituent on water-containing raws; Water-Based Threshold (%) in Admin > Settings **If missing:** Auto with no water constituent: not water-based, so no freezing wording `WaterBasedRule::forProduct, isWaterBased; SDSGenerator::applyFamilyProfile :1229, section7 :2712, section10 :3353; setting sds.water_based_threshold_pct`
Section 8 exposure-limit table, Conc. % column:	FORMULA / RAW MATERIALS The same band as the CAS row in Section 3 > the limit's own concentration when the CAS is not in the composition. Trade-secret rows are masked. *On a sheet: 10 - 30%* **You maintain:** As Section 3 (constituent %) **If missing:** '<0.1%' for a component whose maximum is below 0.1% `SDSGenerator::section8 :2788-2799 formatConcentration(compByCas[cas]) else exposure_limits[].concentration_pct; maskTradeSecretRow`
Section 8 skin notation line (under the exposure-limit table):	HAZARD ENGINE Names each Section 8 limit row whose Notes carry a Skin notation (ACGIH 'Skin', NIOSH '[skin]'; not 'skin sensitizer', 'skin irritant' or similar wording), in table order without repeats. Trade-secret components print by their description. *On a sheet: Components with a skin notation (absorption through the skin…): Toluene.* **You maintain:** Nothing on the sheet; the Notes come with each CAS's exposure-limit data (federal source refresh, determinations or a raw material's manual hazard data) **If missing:** No line `SDSGenerator::section8 :2811-2819, skinNotationNames :2898, SKIN_NOTATION_REGEX :2887; section8.skin_notation; PDFService::renderSection8`
Section 8 Engineering Controls opening sentence:	HAZARD ENGINE Per-product override > the no-limits ventilation sentence when the Section 8 table has no rows > otherwise the standard sentence that refers to exposure limits. *On a sheet: Use adequate general ventilation. Use local exhaust ventilation where mist, vapor…* **You maintain:** Nothing on the sheet; exposure limits per CAS decide which sentence prints **If missing:** No limit rows: the no-limits sentence `SDSGenerator::section8 :2821, engineeringControls :2851; section8.engineering_no_limits | section8.engineering`
Section 8 Engineering Controls (dust and explosion-proof sentences):	FORMULA / RAW MATERIALS Per-product override (whole field) > the opening sentence plus: the dust sentence when the resolved physical state is Powder; the explosion-proof sentence when the sheet carries H224-H226; and the eyewash sentence for H314/H318. *On a sheet: Adds the dust sentence for powders and explosion-proof ventilation for Flam. Liq. 1-3* **You maintain:** Physical State on the finished good, its family default or raw material; raw-material flash and boiling points **If missing:** No dust sentence (Solid and blank states do not get it); no explosion-proof sentence without a flash point `SDSGenerator::engineeringControls :2846-2862; section8.engineering_dust, engineering_flammable, engineering_corrosive`
Physical State:	FORMULA / RAW MATERIALS The finished good's Physical State (the raw material's own on resale sheets) > the active family's default physical state > the state of the highest-weight raw material that has one > 'Liquid'. The same value drives Sections 6, 8, 13 (D002) and 14, the flammability and combustible-dust tests and the ATE dust route. *On a sheet: Paste* **You maintain:** Physical State on the finished good (Finished Goods form), a Default physical state in the family's SDS profile, or Physical State on its raw materials **If missing:** 'Liquid', plus a preview warning. When a family default differs from the largest raw material's state, a preview note says the family default was used `SDSGenerator::resolvePhysicalState :2980, physicalStateIsDefault :2999, familyPhysicalStateWarning :1243; product_families.default_physical_state; FormulaCalcService :607-626`
Color:	FORMULA / RAW MATERIALS Only the finished good's Color dropdown. Resale raw-material sheets never show a color. *On a sheet: Black* **You maintain:** Color on the finished good (Finished Goods form) **If missing:** The line is omitted `finished_goods.color (options from setting sds.color_options); SDSGenerator::section9 :3124; resale :507 reads raw_materials.color, which does not exist`

Appearance:	FORMULA / RAW MATERIALS Per-product override > color plus state when the finished good has a color > the physical state in lower case. Resale sheets use the raw material's Appearance text before falling back to the state. *On a sheet: Black liquid* **You maintain:** Color on the finished good; Appearance on the raw material for resale sheets **If missing:** The lower-case physical state, e.g. 'liquid' `SDSGenerator::section9 :3163-3181; text_overrides(9,'appearance') > section9.appearance_color_state > (resale) formula_props.appearance > section9.appearance_state`
Odor:	FORMULA / RAW MATERIALS Per-product override > Odor of the raw material with the highest weight % > 'Not determined'. *On a sheet: Mild solvent* **You maintain:** Odor on the highest-weight raw material of each formula (Raw Materials form) **If missing:** 'Not determined' `SDSGenerator::section9 :3187-3193; text_overrides(9,'odor') > formula_props.odor (raw_materials.odor of dominant RM) > labels.not_determined`
Odor Threshold:	PRODUCT OVERRIDE Always 'Not determined'. The system holds no data for this property; only a value typed as a per-product Section 9 edit replaces it. *On a sheet: Not determined* **You maintain:** Optionally a tested value in SDS > Edit Text, Section 9 (per language) **If missing:** 'Not determined' `SDSGenerator::SECTION9_APPENDIX_D_FIELDS :3013, section9 :3206-3219; text_overrides(9,'odor_threshold')`
pH:	PRODUCT FAMILY Per-product Section 9 edit > the active family's typical pH range marked '(typical)' (one value when min equals max) > 'Not determined'. Display only: never used for RCRA D002 or any classification. *On a sheet: 8.0 – 9.5 (typical)* **You maintain:** Typical pH minimum and maximum (both or neither, 0-14) in the family's SDS profile, intended for water-based families; optionally a tested value in SDS > Edit Text **If missing:** 'Not determined' `SDSGenerator::typicalPhText :1382, section9 :3207-3213; product_families.ph_min/ph_max (061); section9.ph_typical_range, ph_typical_value`
Melting Point/Freezing Point:	PRODUCT OVERRIDE Always 'Not determined'. The system holds no data for this property; only a value typed as a per-product Section 9 edit replaces it. *On a sheet: Not determined* **You maintain:** Optionally a tested value in SDS > Edit Text, Section 9 (per language) **If missing:** 'Not determined' `SDSGenerator::SECTION9_APPENDIX_D_FIELDS :3013, section9 :3206-3219; text_overrides(9,'melting_point')`
Initial Boiling Point and Boiling Range:	FORMULA / RAW MATERIALS Per-product override, if it reads as a temperature and does not move the product between Flammable Liquids Category 1 and 2 > the lowest boiling point of any raw material in the formula, including sub-formulas, whatever its amount > 'Not determined'. *On a sheet: 78.4 °C (173.1 °F)* **You maintain:** Boiling Point (°C) on each volatile raw material (Raw Materials form) **If missing:** 'Not determined'. An edit that is not a temperature, or that contradicts the classification, is not printed (preview warning) `SDSGenerator::resolveBoilingPointDisplay :2068, vetBoilingPointEdit :4658; TemperatureParser; FormulaCalcService :549-557 MIN(raw_materials.boiling_point_c)`
Flash Point:	FORMULA / RAW MATERIALS A per-product edit that reads as a temperature and gives the same flammable category and D001 outcome, printed exactly as typed > the wt%-weighted average flash point of the raw materials that have one ('>' if any is flagged), rounded to a whole degree but never across the 23, 60 or 93 °C limits, with '(calculated)' > 'Not determined'. Classification uses the unrounded 0.1 °C value; Section 5 prints the same string. *On a sheet: 45 °C (113 °F) (calculated)* **You maintain:** Flash Point (°C) and the '>' flag on every raw material, including a high value for water and other non-flammable raws (Raw Materials form); a measured value only as a Section 9 edit **If missing:** 'Not determined', and the product is treated as not flammable everywhere `SDSGenerator::resolveFlashPointDisplay :2034, flashPointDisplay :2010, roundFlashPointC :1977, vetFlashPointEdit :4616; FormulaCalcService::weightedFlashPoint :693`

Evaporation Rate:	PRODUCT OVERRIDE Always 'Not determined'. There is no formula data for this property; only a value typed as a per-product Section 9 edit replaces it. *On a sheet: Not determined* **You maintain:** Optionally a tested value in SDS > Edit Text, Section 9 (per language) **If missing:** 'Not determined' `SDSGenerator::SECTION9_APPENDIX_D_FIELDS :3013, section9 :3206-3219; text_overrides(9,'evaporation_rate')`
Flammability (Solid, Gas):	PRODUCT OVERRIDE Per-product Section 9 override > 'Not applicable' when the resolved physical state is Liquid > 'Not determined'. *On a sheet: Not applicable* **You maintain:** Physical State (product, family default or raw materials); optionally a value in SDS > Edit Text, Section 9 **If missing:** 'Not determined' for every state other than Liquid (Paste and Powder included) `SDSGenerator::section9 :3214-3215 (flammability_solid_gas; labels.not_applicable when state = Liquid)`
Upper/Lower Flammability or Explosive Limits; Vapor Pressure; Vapor Density:	PRODUCT OVERRIDE Always 'Not determined'. There is no formula data for these properties; only a value typed as a per-product Section 9 edit replaces it. *On a sheet: Not determined* **You maintain:** Optionally a tested value in SDS > Edit Text, Section 9 (per language) **If missing:** 'Not determined' `SDSGenerator::SECTION9_APPENDIX_D_FIELDS :3013, section9 :3206-3219; text_overrides(9,'flammability_limits'|'vapor_pressure'|'vapor_density')`
Specific Gravity:	FORMULA / RAW MATERIALS A volume-additive blend of each raw material's specific gravity (total % divided by the sum of % / SG), rounded to 3 decimals. *On a sheet: 1.053* **You maintain:** Specific Gravity on every raw material (Raw Materials form) **If missing:** Each raw material with no value counts as 1.000, and a preview warning names it. The line never prints 'Not determined' `VOCCalculator::getMixtureSG :248-281, applyDefaults :612-622 (else 1.0); FormulaCalcService::dataGapWarnings :661; SDSGenerator::section9 :3237`
Solubility:	FORMULA / RAW MATERIALS Per-product override > a band from the weighted share of raw materials marked soluble (1), partially soluble (0.5) or negligible (0.03), among the raws that have any value: 90% or more soluble, 5% partial, 1% negligible, else not soluble. *On a sheet: Partially soluble in water* **You maintain:** Solubility on each raw material (Raw Materials form). Water, glycols and other water-soluble raws must read 'Soluble in water' **If missing:** 'Not determined' when no raw material has a value `FormulaCalcService SOLUBILITY_WEIGHTS :30-35, thresholds :38-40, solubilityKeyForFraction :639; SDSGenerator::solubilityText :3024, section9 :3149`
Partition Coefficient: n-Octanol/Water; Auto-Ignition Temperature; Decomposition Temperature; Viscosity:	PRODUCT OVERRIDE Always 'Not determined'. There is no formula data for these properties; only a value typed as a per-product Section 9 edit replaces it. *On a sheet: Not determined* **You maintain:** Optionally a tested value in SDS > Edit Text, Section 9 (per language) **If missing:** 'Not determined' `SDSGenerator::SECTION9_APPENDIX_D_FIELDS :3013, section9 :3206-3219; text_overrides(9,'partition_coefficient'|'auto_ignition_temp'|'decomposition_temp'|'viscosity')`
VOC (lb/gal) (EPA Method 24):	FORMULA / RAW MATERIALS The VOC weight fraction (each raw's VOC % times its line %, divided by the formula's total line %) times the mixture specific gravity times 8.345. Exempt compounds and water are not subtracted. *On a sheet: 2.15* **You maintain:** VOC wt% (or the 'less than 1%' flag) and Specific Gravity on every raw material **If missing:** A missing VOC counts as 0 and a missing SG as 1.0. A preview warning names each raw material; nothing is marked on the sheet `VOCCalculator::getVOCLbPerGal :289, getTotalVOCWeightPercent :181 (÷ totalLinePct :555); VOC_i = 0.99 if voc_less_than_one else voc_wt else 0; SDSGenerator::section9 :3243`
VOC (wt%):	FORMULA / RAW MATERIALS '<1' when every raw material is flagged less than 1% VOC > otherwise each raw's VOC % times its line %, divided by the formula's total line %. *On a sheet: 25.71 (or '<1')* **You maintain:** VOC wt% or the 'less than 1%' flag on every raw material **If missing:** A missing VOC counts as 0, with a preview warning `SDSGenerator::section9 :3141-3144, :3244; FormulaCalcService all_voc_less_than_one; VOCCalculator::getTotalVOCWeightPercent :181`

Solids (wt%):	FORMULA / RAW MATERIALS Each raw's solids % times its line %, divided by the formula's total line %. A raw with no solids % is estimated as 100 minus its VOC, exempt VOC and water. *On a sheet: 74.3* **You maintain:** Solids wt% (or VOC, exempt VOC and water wt%) on each raw material; the Exempt VOC list (admin) **If missing:** Estimated. A raw material with no VOC or solids data counts as 100% solids `VOCCalculator::getSolidsWeightPercent :390, applyDefaults :706-720; raw_materials.solids_wt else 100-voc-exempt-water; exempt_voc_list (FormulaCalcService :457-470)`
Section 10 Chemical Stability: acrylate inhibitor sentence:	FORMULA / RAW MATERIALS Appended on a UV/LED family product when an acrylate is present at 0.1% or more; a per-product Stability override replaces the field. A component counts as an acrylate by its CAS acrylate flag (Yes, or a manual No) > when the flag is not set, the legacy acrylate CAS list or an acrylate-like name. *On a sheet: Contains acrylates stabilized with a polymerization inhibitor that requires oxygen…* **You maintain:** Acrylate flag per CAS (CAS Determinations > CAS Descriptions, or the 'Seed acrylate flags' preview); UV/LED flag on the family **If missing:** Flag not set: the legacy list and name rule decide. No acrylate found, or not a UV family: no sentence `SDSGenerator::section10 :3335; UVAcrylateRulePack::detectAcrylates, effectiveFlag, LEGACY_ACRYLATE_CAS; cas_master.is_acrylate (060) via Formula / FormulaCalcService`
Likely Routes of Exposure (Section 11):	HAZARD ENGINE Skin and eye contact always print. Inhalation is added for H330-H332 or H334-H336, or when any component has an exposure limit; Ingestion is added for H300-H302, H304 or H305. *On a sheet: Inhalation, Skin Contact, Eye Contact, Ingestion.* **You maintain:** Classifications per CAS (CAS determinations / CPD); exposure limits **If missing:** 'Skin Contact, Eye Contact.' `SDSGenerator::deriveRoutesOfExposure :3722; labels.inhalation, skin_contact, eye_contact, ingestion`
Most Important Symptoms/Effects, Acute and Delayed (Section 11):	HAZARD ENGINE Section 4 symptoms override > the sheet's H3xx statements (H370-H373 with target organs where entered), split into acute and delayed (H317, H334, H340-H362, H372, H373). The same text prints in Section 4. *On a sheet: Acute: Causes skin irritation. Delayed: Suspected of causing cancer.* **You maintain:** Classifications per CAS (CAS determinations / CPD); STOT target organs per CAS **If missing:** 'No significant symptoms or effects are known or expected under normal conditions of use.' `SDSGenerator::section11 :3976, deriveSymptoms :1916, DELAYED_EFFECT_H_PREFIXES :107; text_overrides(4,'symptoms')`
Acute Toxicity (oral, dermal, inhalation lines):	HAZARD ENGINE Per-product override > for each route, the most severe Category 1-4 from the engine's classes or H-codes (cut-off or ATEmix), with its H-statement. *On a sheet: Acute toxicity (oral): Category 4 — Harmful if swallowed (H302).* **You maintain:** Acute toxicity categories per CAS (CAS determinations / CPD) **If missing:** 'Acute toxicity (oral): Not classified based on available data.' `SDSGenerator::buildAcuteToxicity :3538-3720; text_overrides(11,'acute_toxicity'); section11.acute_route_line(_no_category); h_statements`
ATEmix value or 'Classified based on ingredient concentration.' (after each acute toxicity line):	HAZARD ENGINE The ATEmix (GHS additivity from each component's LD50/LC50, else its category's default point estimate) prints only when it produced the printed category. Otherwise the line ends 'Classified based on ingredient concentration.', unless the route came only from the finished-good override or a bare H-code. *On a sheet: ATEmix = 1255 mg/kg bw.* **You maintain:** Acute toxicity categories per CAS, and optionally the oral/dermal/inhalation LD50/LC50 values (CAS determinations). Solid and Powder products use the dust/mist route **If missing:** Category default ATEs are used silently. Nothing follows an override-only or H-code-only route `HazardEngine::applyATECalculation :2322, CATEGORY_DEFAULT_ATES :207; SDSGenerator::buildAcuteToxicity :3538-3720, formatAte :3744`
Section 11 Carcinogenicity:	FORMULA / RAW MATERIALS Per-product override > one line per component at 0.1% or more with an IARC 1/2A/2B, NTP or OSHA listing, with its band (plus an inhalable-dust note for bound carbon black or TiO2) > the 'none listed' sentence. IARC Group 3 is not reported, and listings of an ingredient supplied only by Solid or Powder raws are dropped when the formula has another state. *On a sheet: No components present at or above 0.1% are listed as carcinogens by IARC, NTP, or OSHA.* **You maintain:** Carcinogen list (admin); constituent %; raw material physical state (carbon black / TiO2) **If missing:** The 'none listed at or above 0.1%' sentence `SDSGenerator::section11 :3936-3970; CarcinogenService::buildSummaryText :285, isReportableListing, LISTING_THRESHOLD_PCT 0.1; carcinogen_list; solid-in-liquid filter :5898`

Section 11 component heading (name, CAS, range):	FORMULA / RAW MATERIALS Printed for each component at 0.1% or more that has an exposure limit or a reportable carcinogen listing, with the same band as Section 3. Trade-secret components print their description and 'TRADE SECRET'. *On a sheet: Toluene (CAS 108-88-3) — 10 - 30%* **You maintain:** As Section 3; exposure limits and the carcinogen list **If missing:** No band on legacy snapshots `SDSGenerator::section11 :3882-3935; formatConcentration(compositionByCas); maskTradeSecretRow; PDFService::renderSection11`
Section 12 component aquatic table (Conc. %, Aquatic Acute, Aquatic Chronic with M-factor):	HAZARD ENGINE Every component listed in Section 3 that has an aquatic classification, with its Section 3 band and, for Category 1, its M-factor; trade-secret identities are masked. The table is dropped under a replace-mode finished-good override. *On a sheet: Zinc oxide | 1314-13-2 | 1 - 5% | Category 1 (M = 10) | Category 1 (M = 1)* **You maintain:** Aquatic categories and M-factors per CAS (CAS determinations / CPD) **If missing:** An empty cell for no class. A missing M-factor prints '(M = 1, default)' `HazardEngine aquatic_components, resolveMFactor :1429; SDSGenerator::section12 :4014-4060, section3ListedCas :4206, formatAquaticCategory :4175`
Persistence and Degradability / Bioaccumulative Potential:	REGULATORY LIST Per-product override > a line naming every SARA 313 PBT chemical present at any concentration, by its EPA list name > 'No data available.' Bioaccumulation then reads 'See Persistence and Degradability above.' *On a sheet: Contains component(s) identified as … (PBT) under SARA 313 (40 CFR 372.28): Lead (CAS 7439-92-1).* **You maintain:** SARA 313 list with the PBT flag (admin) **If missing:** 'No data available.' on both lines `SDSGenerator::section12 :4114-4152; SARA313Service reportable + below_threshold (sara313_list.is_pbt); section12.pbt_components, pbt_see_persistence`
Section 13 Disposal Methods: UV cure-residue sentence:	PRODUCT FAMILY Appended to the automatic disposal methods on a UV/LED family product when no RCRA characteristic or listed code matched at all and the composition is complete (every composition row has a CAS number, so no trade-secret or hazardous no-CAS bucket, and no raw material has incomplete constituent data). A per-product Disposal Methods override replaces the field. *On a sheet: Small amounts of non-hazardous residue may be cured before disposal where permitted.* **You maintain:** UV/LED flag on the family; complete constituent CAS and % on every raw material; the RCRA list **If missing:** Not printed `SDSGenerator::section13 :4432-4435, compositionIsComplete :4498; TSCAService::incompleteRawMaterials; section13.uv_cure_residue`
Contaminated Packaging:	FIXED TEXT (TRANSLATED) A fixed sentence on every sheet (HazCom Appendix D, Section 13); it cannot be overridden. *On a sheet: Empty containers retain product residue; follow all precautions on this safety data sheet…* **You maintain:** Nothing **If missing:** Always printed (snapshots made before batch F have no line) `SDSGenerator::section13 :4442; section13.packaging, labels.contaminated_packaging; PDFService section 13 field map`
Section 13 RCRA ignitable (D001):	FORMULA / RAW MATERIALS Printed when the sheet carries H224-H226. The reason is the formula flash point when below 60 °C (never the Section 9 edit), else a hazard-based reason for an override class; oxidizer, pyrophoric, flammable gas and flammable solid H-codes add their own reasons. *On a sheet: ignitable (D001) — flash point below 60 °C (140 °F)* **You maintain:** Raw material flash and boiling points; Physical State; constituent classifications **If missing:** No D001 when the product has no flash point, is not a liquid, or is at exactly 60 °C, unless a finished-good hazard override sets Flammable Liquids 1-3 `SDSGenerator::section13 :4307-4331, d001FlashReasonKey :4592, flammabilityOf :4565; section13.rcra_reason_flash_point(_gt)`
Section 13 RCRA corrosive (D002) / reactive (D003):	HAZARD ENGINE D002 for H314 and/or H290, never on a Solid, Powder or Paste product (resolved state, including a family default). D003 from water-reactive, explosive, self-reactive or organic-peroxide H-codes. *On a sheet: corrosive (D002) — classified as corrosive to skin (H314)* **You maintain:** Constituent classifications (CAS determinations); Physical State **If missing:** Omitted `SDSGenerator::section13 :4333-4361, rcraIsNonLiquidState :4483; H314/H290 -> D002; H260/H261, H200-H205, H240-H242 -> D003`

Section 13 RCRA toxicity-characteristic and listed-waste components:	REGULATORY LIST Composition CAS numbers are matched to the admin RCRA list, and metal compounds to their element's D004-D011 codes. D codes print at any amount in the 'as sold' sentence; F/K/P/U codes print at 0.1% or more in a separate reference sentence. *On a sheet: contains Lead (D008 if the … regulatory level of 5 mg/L (TCLP) is exceeded)* **You maintain:** RCRA waste code list (Admin > RCRA); TC metals per CAS (CAS determinations > TC metals) **If missing:** The 'No RCRA hazardous waste characteristic (40 CFR 261 Subpart C) has been identified…' sentence `RCRAService::match :131-200 (rcra_waste_codes, cas_master.tc_metals), MIN_CONCENTRATION_PCT 0.1; SDSGenerator::section13 :4363-4426`
UN Number / UN Proper Shipping Name:	TRANSPORT RULES Per-field override > classifier: Class 3 from H224-H226 on the sheet, 8 from H314 (or H290 on a liquid), 6.1 from acute Cat 1-3, 9 from H400/H410/H411, ranked by 49 CFR 173.2a. Class 3 alone uses UN1210/UN1263 (UN1993 for wash/cleaner types) by Transport Product Type, else description keywords; other entries are n.o.s. with up to two constituent names. *On a sheet: UN1210 / Printing ink, flammable; or 'Flammable liquids, n.o.s. (Toluene, Xylene)'* **You maintain:** Raw material flash and boiling points; constituent classifications; Transport Product Type on the finished good; Physical State (product, family default or raw materials) **If missing:** 'Not regulated' when no class applies, including when there is no flash point. 'Not determined' (publishing blocked) for gas, aerosol, oxidizer-type codes, Gas state, or Class 3+8+6.1 where 8 or 6.1 outranks 3 `SDSGenerator::section14 :4811-4862; TransportClassifier::classify :182-404, resolveProductType :414; override completeness :4872-4882`
Transport Hazard Class(es):	TRANSPORT RULES Per-product override > primary class plus subsidiaries ranked by the 49 CFR 173.2a precedence table. *On a sheet: 3 (6.1, 8)* **You maintain:** As UN Number **If missing:** 'Not regulated' / 'Not determined' `SDSGenerator::section14 :4925; text_overrides(14,'hazard_class') > TransportClassifier::classify :291-375`
Packing Group:	TRANSPORT RULES Per-product override > the most stringent of: Class 3 PG I at IBP 35 °C or less, II for H224/H225, III for H226; Class 8 from the skin corrosion sub-category (H290 alone III); 6.1 from the acute category; Class 9 III. *On a sheet: II* **You maintain:** Raw material boiling and flash points; constituent classifications **If missing:** 'Not applicable' when not regulated; 'Not determined'. Class 8 with no sub-category defaults to II `TransportClassifier::classify :258-287, corrosivePackingGroup :502, toxicPackingGroup :526; IBP from the engine flammability block (Section 9 edit not read)`
Environmental Hazards (Marine pollutant):	TRANSPORT RULES Per-product override > 'Marine pollutant: Yes' when the sheet carries H400, H410 or H411, otherwise 'Marine pollutant: No'. *On a sheet: Marine pollutant: Yes* **You maintain:** Aquatic classifications per CAS **If missing:** 'Marine pollutant: No'; 'Not determined' while the classification is not determined and there is no override `SDSGenerator::section14 :4845, :4927; text_overrides(14,'environmental_hazards') > section14.marine_pollutant_yes|_no`
Transport in Bulk:	FIXED TEXT (TRANSLATED) A fixed sentence on every sheet; it cannot be overridden. *On a sheet: Not intended for transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code.* **You maintain:** Nothing **If missing:** Always printed (older snapshots without it skip the line) `SDSGenerator::section14 :4929; section14.transport_in_bulk_text`
Special Precautions for User:	TRANSPORT RULES Per-product Special Precautions override > the standard sentence plus a Class 3 sentence when the printed hazard class contains 3, a Class 8 sentence when it contains 8, a marine sentence when Environmental Hazards prints 'Marine pollutant: Yes', and a UV/EB sentence for a UV/LED family. The fragments read the printed (overridden if edited) class and environmental lines. *On a sheet: Read this safety data sheet before handling… Flammable liquid (Class 3): keep away from heat…* **You maintain:** As UN Number and Environmental Hazards; UV/LED flag on the family; optionally an override in SDS > Edit Text, Section 14 **If missing:** Only the standard sentence `SDSGenerator::section14 :4902-4919, transportClassTokens :4963; text_overrides(14,'special_precautions'); section14.special_precautions_text|_flammable|_corrosive|_marine|_uv`

Transport notes:	TRANSPORT RULES A combustible note for a non-regulated liquid with H227 and a flash point below 93 °C (or none), and a viscous-liquid note for Class 3 PG II with no subsidiary, then the fixed carrier sentence. Under a Section 14 override only the carrier sentence prints. *On a sheet: Combustible liquid; regulated in bulk packaging (> 450 L) only. …verify with the carrier…* **You maintain:** Raw material flash points; Physical State **If missing:** Only the carrier sentence `SDSGenerator::section14 :4888-4896; TransportClassifier :249 (combustible), :401 (viscous); section14.note_combustible, note_viscous, section14.note`
SARA 313 reportable components:	REGULATORY LIST Composition CAS numbers on the SARA 313 list, or members of a TRI category (metal compounds, certain glycol ethers), at or above their de minimis; PBT chemicals and TRI PFAS are reported at any concentration. Each prints with the EPA list name, the Section 3 band, and its threshold or 'no de minimis exemption'. *On a sheet: • Ethylene glycol (CAS 107-21-1) — 1 - 5% (de minimis threshold: 1%)* **You maintain:** SARA 313 list with PBT and special-concern flags, and TRI categories (admin); constituent % **If missing:** The 'does not contain any toxic chemicals subject to … Section 313' sentence `SARA313Service::evaluate :77-140 (sara313_list deminimis_pct, is_pbt, is_special_concern; RegulatoryCategoryService); SDSGenerator::section15 :5010, :5023; PDFService::renderSection15`
SARA 313 range note:	FIXED TEXT (TRANSLATED) A fixed sentence printed under the list only when at least one SARA 313 chemical is reportable. *On a sheet: Concentrations are stated as the prescribed concentration ranges…; the upper end … is the maximum* **You maintain:** Nothing (it follows the SARA 313 matches) **If missing:** Not printed when nothing is reportable `labels.sara_313_range_note; PDFService::renderSection15`
Hazardous Air Pollutants (rows and Total HAP Content):	REGULATORY LIST Composition CAS numbers on the HAP list or in a 112(b) category (metal compounds, glycol ethers) at 0.01% and up, plus HAPs entered by hand on raw materials, each with a band. The total is the summed % shown as a band. *On a sheet: Toluene | 10 - 30% … Total HAP Content: 10 - 30%* **You maintain:** HAP list and 112(b) categories (admin); manual HAP entries on raw materials **If missing:** 'This product does not contain any EPA HAPs listed under Clean Air Act Section 112(b).' `HAPService::evaluate :102-160 (hap_list, RegulatoryCategoryService) + SDSGenerator::getManualHaps :6368 (raw_materials.haps_data); section15 :5011-5017`
Preview Warnings box (operator only; never printed on the SDS):	FORMULA / RAW MATERIALS Raised by missing or conflicting data: blank specific gravity or VOC, no physical state anywhere, a family default state that differs from the largest raw, flash point problems, non-temperature Section 9 edits, n.o.s. without technical names, a Substance with no CAS, organs entered for only some CAS behind H370-H373, and not-assessed combustible-dust raws in a Powder/Solid product. None of these blocks publishing; the box also shows TSCA, incomplete-composition and storage-override notes and repeats the publish-blocking transport and trade-secret Prop 65 reasons. *On a sheet: Raw material R-123: no specific gravity entered; … count it as 1.0.* **You maintain:** The raw-material, family, CAS-determination or SDS editor field each warning names **If missing:** No box when nothing is missing `FormulaCalcService::dataGapWarnings :661; SDSGenerator::generate :291-366, familyPhysicalStateWarning :1243, flammabilityWarnings :4692; StotTargetOrgans::warnings :249`

PRODUCT FAMILIES AND UV PRODUCTS

In this scope, text comes from the product family resolved for each item. The family supplies its UV/LED flag, its active flag, its per-language Section 1 text and, new in batch F, its SDS profile (storage temperature range, water-based setting, default physical state, typical pH). A product gets its family from a manual pick, a rule or the content vote, in which all UV/LED families count together. The UV/LED flag together with the UV Acrylate Rule Pack switch drives the UV sentences in the Section 2/8 PPE fields, Sections 4, 5 and 6, Section 7 Handling and the Section 11 note. The flag alone drives Section 2 Other Hazards, Section 7 Storage, Section 10, the Section 13 cure-residue sentence and the Section 14 UV fragment. The Acrylate flag on each CAS registry entry supplies the acrylate names in Section 4 and the Section 10 inhibitor sentence. On UV sheets with the pack on, the UV glove and eyewear sentences replace the hazard-tier sentences. Gloves are always nitrile. Eyewear keeps goggles and a face shield for H314/H318. Resale raw-material sheets ignore the family for Section 1 only, and the family name prints nowhere. Combustion products, Skin notation, ventilation, contaminated packaging, STOT organs and combustible dust are not driven by the family; they are covered in the other scopes.

Recommended Use (Section 1, finished good / intermediate / alias / private-label sheet):

PRODUCT OVERRIDE Per-product, per-language SDS editor text > the product's own Recommended Use field > the family's text in the sheet language > canned translated sentence. Inactive families are ignored, and a blank language never falls back to the family's English text.

On a sheet: Recommended Use: UV-curable offset ink for packaging.

You maintain: Family text in every language on an active family (Settings > Product Families), and the product must resolve to that family. Keep the product's Recommended Use field blank, because it has no language. Put per-language product wording in the SDS editor.

If missing: "Printing ink for commercial and industrial applications." (translated on ES/FR/DE). If the family has English text only, ES/FR/DE sheets print their canned translated sentence.

`SDSGenerator::resolveUseText (:1183): text_overrides(1,'recommended_use',fg,lang) > finished_goods.recommended_use > family_defaults[lang] (familyFields, active only) > section1.recommended_use`

Recommended Use (Section 1, resale raw-material sheet):

FIXED TEXT (TRANSLATED) Per-raw-material, per-language text from the resale SDS editor > canned resale sentence. The raw material's family is not used.

On a sheet: Recommended Use: Raw material for industrial formulation.

You maintain: Nothing normally. For product-specific wording, use the resale SDS editor (/sds/resale/{rm_id}/edit).

If missing: "Raw material for industrial formulation." (translated per language)

`resolveUseText resale branch (is_resale; family tier skipped, Q15): text_overrides(raw_material_id,1,'recommended_use') via getOverrides :5208 > section1.recommended_use_resale`

Restrictions on Use (Section 1, finished good / intermediate / alias / private-label sheet):

PRODUCT OVERRIDE Per-product, per-language SDS editor text > the product's own Restrictions field > the family's restriction text in the sheet language > canned translated sentence. There is no English family fallback, and inactive families are ignored.

On a sheet: Restrictions on Use: For professional/industrial use only. Not for household consumer use.

You maintain: Family restriction text in every language (Settings > Product Families). Keep the product's Restrictions field blank: it has no language and masks the family text.

If missing: "For professional/industrial use only. Not for household consumer use." (translated per language)

`resolveUseText(field='restrictions') :1183: text_overrides(1,'restrictions') > finished_goods.restrictions_on_use > family_defaults['restrictions'][lang] > section1.restrictions`

Restrictions on Use (Section 1, resale raw-material sheet):

FIXED TEXT (TRANSLATED) Per-raw-material, per-language text from the resale SDS editor > canned resale restriction sentence. The family is not used.

On a sheet: Restrictions on Use: For professional/industrial use only. Not for household consumer use.

You maintain: Nothing normally; exceptions are entered in the resale SDS editor.

If missing: "For professional/industrial use only. Not for household consumer use." (resale key, translated)

`resolveUseText resale branch: text_overrides(raw_material_id,1,'restrictions') > section1.restrictions_resale (SDSGenerator.php:1183-1205)`

Recommended Personal Protective Equipment (PPE): Respiratory / Hand Protection / Eye Protection / Skin/Body Protection (Section 2):

TEXT CHOSEN BY HAZARD Only fields with a hazard-driven tier print, with the same text as Section 8: Section 8 override > hazard-tier sentence. On UV rule-pack products the UV glove and eyewear sentences replace the tier sentence, except the H314/H318 goggles-and-face-shield sentence, which stays; the UV respirator and clothing sentences are added after the tier sentence.

On a sheet: Hand Protection: For uncured acrylate product, wear chemical-resistant gloves (nitrile, ...

You maintain: Nothing in this scope beyond the family UV/LED flag and the pack switch (the tiers come from H-codes)

If missing: Unclassified product: no PPE block in Section 2. With the pack off or a non-UV family, the tier sentences print alone (e.g. butyl/Viton gloves for H310/H311/H314).

`section2 (:1500) → resolvePPE($uvPack) (:2936): text_overrides(8,field) > section8.ppe.<field>.<tier>; uv_hand/uv_eye replace (UV_PPE_TIER_FLOOR: eye only), others appended`

Other Hazards (Section 2):	PRODUCT FAMILY Per-product, per-language Section 2 override > the UV polymerization sentence when the product's active family is flagged UV/LED > "None known." The pack switch plays no part. *On a sheet: Other Hazards: Uncured product may polymerize when exposed to UV light, elevated temperatures ...* **You maintain:** UV/LED flag on an active family (Settings > Product Families); product resolved to that family. Per-product wording in the SDS editor (Section 2 Other Hazards). **If missing:** "None known." (translated) `section2() :1529: text_overrides(2,'other_hazards') ?? (this->uvFamily (UVAcrylateRulePack::familyIsUv) ? section2.other_hazards_uv : section2.other_hazards)`
Type (Section 3: Substance / Mixture):	FORMULA / RAW MATERIALS Product's Substance/Mixture setting > Auto: 'Substance' only if the formula has exactly one line and it is a raw material marked Substance > 'Mixture'. Resale sheets use the raw material's setting; the product family has no influence. *On a sheet: Type: Mixture* **You maintain:** Mark single-substance raw materials as Substance (raw-material form). Set the finished-good setting only when the automatic rule is wrong. **If missing:** "Mixture" `SubstanceMixtureResolver::resolveForFinishedGood / resolveForRawMaterial; labels.type / labels.substance / labels.mixture`
Skin Contact (Section 4, UV sentence):	PRODUCT FAMILY Added after the hazard-driven skin text when the rule pack is on and the product's active family is flagged UV/LED. Names the acrylates at 0.1 % or more, recognised by the registry Acrylate flag (if it is not set: the old 14-CAS list or an acrylate-like name). Each name is taken from the Prop 65 name > registry description > constituent name, and trade secrets print by description. *On a sheet: ... product containing acrylate monomers/oligomers (Hexanediol Diacrylate): remove ...* **You maintain:** UV/LED flag on the family; pack switch (Admin Settings); Acrylate flag and description per CAS (CAS Determinations > CAS Descriptions, or the acrylate-flag seed); trade-secret descriptions on secret acrylates **If missing:** With the pack off or a non-UV family there is no UV sentence. If no acrylate is recognised, the sentence prints without names, and a CAS marked not-acrylate is never named. A Section 4 Skin override replaces the whole field. `section4() :1799 + UVAcrylateRulePack::section4SkinFragment (uv_skin / uv_skin_names); detectAcrylates :198 effectiveFlag(cas_master.is_acrylate, LEGACY_ACRYLATE_CAS, name)`
Specific Hazards (Section 5, UV sentence):	PRODUCT FAMILY The same gate (pack on and a UV/LED family) adds a fixed translated sentence after the combustion sentence and before any combustible-dust sentence. Acrylate detection plays no part. *On a sheet: ... uncured UV/EB-curable product may polymerize uncontrollably, releasing heat ...* **You maintain:** As the Section 4 UV sentence (family flag + pack switch) **If missing:** Omitted. A Specific Hazards override replaces the whole field. `section5() :2254 appends section5.uv_specific_hazards after combustionProductsSentence(); text_overrides(5,'specific_hazards') replaces the field`
Containment and Cleanup (Section 6, UV sentence):	PRODUCT FAMILY Same gate. A fixed translated sentence follows the physical-state containment sentence. *On a sheet: ... Keep collected uncured product in closed containers away from UV light ...* **You maintain:** As the Section 4 UV sentence **If missing:** Omitted. A Containment override replaces the whole field. `section6() :2439 appends section6.uv_containment unless text_overrides(6,'containment')`
Handling (Section 7, UV sentence):	PRODUCT FAMILY Same gate. A fixed translated sentence is added after the hazard-driven handling fragments. *On a sheet: ... Avoid prolonged or repeated skin contact with uncured product.* **You maintain:** As the Section 4 UV sentence **If missing:** Omitted. A Handling override replaces the whole field. `section7() :2697 appends section7.uv_handling when uvPack; skipped if text_overrides(7,'handling')`
Storage (Section 7, storage temperature sentence):	PRODUCT FAMILY Printed right after the base storage sentence when the product's active family has a storage temperature range in its SDS profile. The °C values print as entered (at most one decimal) and the °F values are rounded to a whole degree. *On a sheet: ... Store in the original container. Store between 5 and 30 °C (41–86 °F).* **You maintain:** Minimum and maximum storage temperature on the family SDS profile (Settings > Product Families > Edit), both or neither; enter only verified QC ranges **If missing:** No temperature sentence; the rest of the storage text prints `section7() :2706 storageTemperatureSentence() :1364 → section7.storage_temp_range (profile min+max; ProductFamily::profile drops half pairs); skipped if text_overrides(7,'storage')`

Storage (Section 7, Protect from freezing):	PRODUCT FAMILY Printed when the product is water-based. The family's Water-based Yes or No setting decides; on Auto, or with no active family, the product is water-based when water (CAS 7732-18-5, through intermediates) is at or above the Water-Based Threshold (default 20 %) of the formula. *On a sheet: ... Store between 5 and 30 °C (41–86 °F). Protect from freezing.* **You maintain:** Water-based setting on the family SDS profile; water as a CAS 7732-18-5 constituent on raw materials; Water-Based Threshold (%) in Settings > SDS Configuration **If missing:** No freezing sentence. Water held only in a raw material's water % field (no 7732-18-5 constituent) does not count. `section7() :2712 profileWaterBased ← WaterBasedRule::forProduct(family_profile.water_based ?? 'auto', composition, threshold()) via applyFamilyProfile :1229`
Storage (Section 7, UV light sentence):	PRODUCT FAMILY Added before the locked-storage and incompatible-materials sentences whenever the product's active family is flagged UV/LED. It does not depend on the pack switch (same condition as the Section 10 UV item). *On a sheet: ... Store away from UV light sources and direct sunlight to prevent premature polymerization.* **You maintain:** UV/LED flag on the family (Settings > Product Families); product resolved to that family **If missing:** Omitted; the rest of the storage text prints `section7() :2751 $this->uvFamily (UVAcrylateRulePack::familyIsUv) → section7.uv_storage; skipped if text_overrides(7,'storage')`
Respiratory Protection (Section 8, UV sentence):	PRODUCT FAMILY Prints the hazard-tier sentence, followed by the UV respirator sentence when the pack gate fires and the tier is hazard-driven or the mixture carries H317. A per-product override replaces the whole field. *On a sheet: ... Use local exhaust ventilation when handling uncured product; if mist or vapor is ...* **You maintain:** As the Section 4 UV sentence **If missing:** Only the tier sentence prints. An unclassified UV product without H317 keeps its 'No special ... required' sentence. `resolvePPE (:2936-2966): section8.uv_respiratory appended if $uvPack && (tier not in PPE_BASELINE_TIERS || H317); text_overrides(8,'respiratory') wins`
Hand Protection (Section 8, UV sentence):	PRODUCT FAMILY When the pack gate fires and the glove tier is hazard-driven or H317 is present, the UV nitrile glove sentence replaces the tier sentence, also on corrosive or dermally toxic inks (H310/H311/H314), because acrylates can permeate butyl. A per-product override replaces the field. *On a sheet: For uncured acrylate product, wear chemical-resistant gloves (nitrile, minimum 0.4 mm ...* **You maintain:** As the Section 4 UV sentence **If missing:** With the pack off, a non-UV family, or a baseline tier without H317, the tier sentence prints (e.g. butyl/Viton for H310/H311/H314 on non-UV inks). `resolvePPE :2936-2966: uv_hand_protection replaces the tier text (UV_PPE_REPLACE_FIELDS :2865; UV_PPE_TIER_FLOOR['hand_protection'] = [] :2875); text_overrides(8,'hand_protection') wins`
Eye Protection (Section 8, UV sentence):	PRODUCT FAMILY Same condition. The UV goggles-or-face-shield sentence replaces the tier sentence, except on H314/H318 products, whose 'goggles and a face shield' tier sentence prints alone. An override replaces the field. *On a sheet: Chemical splash goggles or a face shield are required when handling uncured acrylate product.* **You maintain:** As the Section 4 UV sentence **If missing:** Only the tier sentence prints `resolvePPE :2936-2966: uv_eye_protection replaces the tier text unless tier is goggles_faceshield (UV_PPE_TIER_FLOOR :2875); text_overrides(8,'eye_protection') wins`
Skin Protection (Section 8, UV sentence):	PRODUCT FAMILY Tier sentence, followed by the UV skin sentence when the gate fires and the tier is hazard-driven or H317 is present; an override replaces the field. *On a sheet: ... Avoid prolonged or repeated skin contact with uncured product. A barrier cream is ...* **You maintain:** As the Section 4 UV sentence **If missing:** Only the tier sentence prints `resolvePPE: section8.uv_skin_protection appended under the same condition; text_overrides(8,'skin_protection') wins`
Physical State / Appearance (Section 9; the same state drives Sections 6, 8, 13 and 14):	PRODUCT FAMILY Product's own Physical State > the active family's default physical state > the largest raw material's state > 'Liquid'. The resolved state also feeds Sections 6, 8, 13 (RCRA D002) and 14 and the flammable-liquid and combustible-dust checks. *On a sheet: Physical State: Paste* **You maintain:** Default physical state on the family SDS profile (e.g. offset and UV paste inks = Paste); Physical State on the product where the family default is wrong for it **If missing:** The largest raw material's state prints; if no raw material has one, "Liquid" prints and the preview shows a warning `resolvePhysicalState :2980 (fg.physical_state > family_profile.default_physical_state > formula_props.physical_state > 'Liquid'); resale: RM state > its family default`

pH (Section 9):	**PRODUCT FAMILY** Per-product, per-language Section 9 pH edit > the active family's typical pH range (one value with "(typical)" when min equals max) > 'Not determined'. Display only, never used for classification or RCRA D002. *On a sheet: pH: 8.0 – 9.5 (typical)* **You maintain:** Typical pH minimum and maximum on the family SDS profile (both or neither, 0-14); a tested value in the SDS editor (Section 9 pH) **If missing:** "Not determined" `section9() Appendix D loop: text_overrides(9,'ph') > typicalPhText :1382 (section9.ph_typical_range / ph_typical_value from family_profile ph_min/ph_max) > labels.not_determined`
Reactivity (Section 10, UV polymerization sentence):	**PRODUCT FAMILY** On products whose active family is flagged UV/LED, the polymerization sentence follows any reactive-class fragments; alone, it replaces the 'No dangerous reaction known' default. The pack switch plays no part, and an override replaces the field. *On a sheet: Reactivity: Hazardous polymerization may occur when uncured product is exposed to UV light ...* **You maintain:** UV/LED flag on an active family (Settings > Product Families) **If missing:** "No dangerous reaction known under conditions of normal use." (or the reactive-class fragments alone) `section10() :3305-3320 $isUv (familyIsUv) → section10.reactivity_uv_polymerization appended to reactivity_<class> parts; text_overrides(10,'reactivity') wins`
Chemical Stability (Section 10, acrylate inhibitor sentence):	**PRODUCT FAMILY** Added after the stability sentence when the family is flagged UV/LED and at least one acrylate at 0.1 % or more is recognised (same detection as the Section 4 names). The pack switch plays no part, and an override replaces the field. *On a sheet: ... Contains acrylates stabilized with a polymerization inhibitor that requires oxygen ...* **You maintain:** UV/LED family flag; Acrylate flags on CAS Descriptions (or the acrylate-flag seed); constituent CAS numbers and names on raw materials **If missing:** Only "Stable under recommended storage conditions." (or the unstable-class sentence) `section10() :3333-3337 $isUv && UVAcrylateRulePack::detectAcrylates($composition) !== [] → section10.stability_uv_inhibitor; text_overrides(10,'stability') wins`
Conditions to Avoid (Section 10, storage-maximum and freezing items):	**PRODUCT FAMILY** A family storage maximum adds 'temperatures above X °C (Y °F)', which replaces the generic storage-limit item of self-reactive and self-heating products. A water-based product adds 'freezing' (same rule as Section 7); an override replaces the whole list. *On a sheet: Conditions to Avoid: Excessive heat, temperatures above 30 °C (86 °F), freezing, ...* **You maintain:** Storage temperature range and Water-based setting on the family SDS profile; Water-Based Threshold (%) in Settings **If missing:** Items omitted; the rest of the list prints `section10() :3347-3355 profileStorageMaxC → cond_storage_max (:max_c/:max_f), profileWaterBased → cond_freezing; COND_ORDER :2458; text_overrides(10,'conditions_avoid')`
Conditions to Avoid (Section 10, UV light item):	**PRODUCT FAMILY** Added whenever the product's active family is flagged UV/LED; the pack switch and acrylate detection play no part. A per-product Conditions to Avoid override replaces the whole list. *On a sheet: Conditions to Avoid: Excessive heat, ..., exposure to UV light and direct sunlight (...)* **You maintain:** UV/LED flag on an active family (Settings > Product Families); product resolved to that family **If missing:** UV item omitted; the rest of the list (heat / ignition / oxidizers ...) prints `section10($isUv = UVAcrylateRulePack::familyIsUv) :3378 adds 'uv' → section10.cond_uv (last in COND_ORDER); skipped if text_overrides(10,'conditions_avoid')`
Incompatible Materials (Section 10, UV item; repeated in the Section 7 Storage list):	**PRODUCT FAMILY** A UV/LED family flag adds 'peroxides and free-radical initiators' at the end of the list, and the Section 7 'Store away from incompatible materials' sentence repeats the same list. A Section 10 Incompatible Materials override replaces both. *On a sheet: ... strong acids, strong bases, peroxides and free-radical initiators.* **You maintain:** UV/LED flag on an active family (Settings > Product Families) **If missing:** Item omitted; the hazard-driven list prints `incompatibleMaterialItems :2520 ($uv ?? uvFamily → incompat_peroxides_initiators, last in INCOMPAT_ORDER); resolveIncompatibleMaterials / incompatibleMaterialsInline :2589`
UV Acrylate Information (Section 11, after the component table):	**PRODUCT FAMILY** Prints when the pack is on and the product's active family is flagged UV/LED, even if no acrylate is recognised. It uses the sensitizer wording with H317 and the unclassified wording otherwise; a per-product override replaces it. *On a sheet: UV Acrylate Information: Contains acrylate monomers that are known skin sensitizers...* **You maintain:** As the Section 4 UV sentence; per-product wording in the SDS editor (Section 11) **If missing:** Note and label omitted. An override cannot add the note when the gate does not fire. `SDSGenerator :231/:635 section11Note → section11.uv_acrylate_note(_unclassified); text_overrides(11,'uv_acrylate_note') :282/:704; PDFService :1000, preview :427`

UV Acrylate Information (Sections 4-7, stored private-label snapshots):

FIXED TEXT The private-label preview renders the stored snapshot the customer received. Snapshots from before batch E hold separate UV notes in Sections 4-7 (English-only before batch C), and the renderers still print them.

On a sheet: UV Acrylate Information: (old note text stored in the snapshot)

You maintain: Republish the private-label variant to replace the snapshot

If missing: Nothing prints: new sheets never carry the key in Sections 4-7, and alias sends and reports use the alias's own published PDF

`PrivateLabelController::preview snapshot_json → PDFService FIELD_LABEL_MAP / preview genericLabelMap 'uv_acrylate_note' (Sections 4-7, preview.php :625)`

Disposal Methods (Section 13, UV cure-residue sentence):

PRODUCT FAMILY Added after the disposal sentence when the family is flagged UV/LED, no component matched any RCRA characteristic or listed waste, and the composition is complete (every row has a CAS and no raw material has incomplete constituent data). A Disposal Methods override replaces the field.

On a sheet: ... Small amounts of non-hazardous residue may be cured before disposal where permitted.

You maintain: UV/LED family flag; complete constituent CAS numbers and percentages on every raw material (raw-material form)

If missing: Only the standard disposal sentence prints

`section13() :4430-4435 items===[] && listed===[] && uvFamily && compositionIsComplete :4498 (TSCAService::incompleteRawMaterials) → section13.uv_cure_residue; text_overrides(13,'methods')`

Special Precautions for User (Section 14, UV fragment):

PRODUCT FAMILY The standard transport sentence prints, then the Class 3, Class 8 and marine-pollutant fragments, then the UV fragment when the family is flagged UV/LED (pack switch ignored). A per-product, per-language Special Precautions override replaces the whole line.

On a sheet: ... UV/EB-curable product: protect from heat and direct sunlight during transport to ...

You maintain: UV/LED family flag; exceptions in the SDS editor (Section 14 Special Precautions for User)

If missing: Fragment omitted; the standard sentence and any class fragments print

`section14() :4900-4918: special_precautions_text + _flammable/_corrosive/_marine + special_precautions_uv if uvFamily; text_overrides(14,'special_precautions') wins`

Abbreviations (Section 16, UV-related entries):

FIXED TEXT (TRANSLATED) A master-table entry prints when its term appears in the printed text, such as the UV sentences in Sections 2, 4-8, 10, 13 and 14, the Section 11 note or Section 1 family text. Alias and private-label sheets are re-filtered against what they print.

On a sheet: UV: Ultraviolet (energy-curable); EB: Electron Beam (energy-curable); NIOSH: ...

You maintain: Nothing

If missing: Entry omitted

`AbbreviationService::build over the printed corpus (re-run for alias and private-label variants); section16.abbreviation_table (en.php: 'UV', 'EB', 'LED', NIOSH, CFR)`

DATA YOU MUST KEEP POPULATED

Grouped by the screen where the information is entered. Blank entries do not stop most sheets from publishing; they change what prints, as shown in the last column.

Where	Keep populated	Feeds	If left blank
Raw materials	Ingredient CAS numbers and percentages (exact, or min and max) on every raw material, including water as 7732-18-5	Sections 2-15: hazard classification, Section 3 rows, exposure limits, regulatory list matches, water-based rule, missing-data check	Ingredient counts as 0% and is dropped; hazards under-reported; Section 15 prints TSCA 'not verified'; Composition warning in preview
Raw materials	Trade-secret tick, trade-secret description and declared hazard codes; manual hazard data on hazardous raws with no CAS	Masked identity in Sections 3, 8, 11, 12, 13, 15; trade-secret hazards in every hazard-driven section	'Trade Secret' prints; hazards not counted. No manual hazard data at or above the threshold: Publishing is blocked. Trade secret on Prop 65 list blocks too
Raw materials	Flash point (°C, '>' flag) and initial boiling point; enter a high flash point for water and similar raws	Product flash point and Flammable Liquids category: Sections 2, 5, 7, 8, 9, 10, 13 (D001), 14 (Class 3, Packing Group)	Raw left out of the average. None on any raw: 'Not determined', product not flammable. No boiling point: Category 2 assumed, preview warning
Raw materials	Physical state on every raw material	Product state fallback; carbon black/TiO2 rule (Sections 2, 3, 8, 11, 15); combustible-dust check	Counts as bound: carbon black/TiO2 left out and lose limits. No state anywhere: 'Liquid' prints with a preview warning
Raw materials	Combustible Dust (Not assessed / Yes / No) from the vendor SDS Section 2, on every powder or solid raw	Section 2 Combustible Dust line and Warning signal word; Sections 5 and 7 dust sentences; Section 15 OSHA status (Powder/Solid products)	Not assessed: no dust wording; preview and readiness warning on Powder/Solid products. Publishing is never blocked

Where	Keep populated	Feeds	If left blank
Raw materials	Review the raw material (any user save)	Bulk and scheduled publishing eligibility; readiness verdict; resale Publish button	Products using it, their aliases and private-label items are left out of bulk publishing with no reason given
Raw materials	Specific gravity, VOC % (or '<1% VOC'), solids/exempt/water %, odor, solubility, appearance (resale)	Section 9 Specific Gravity, VOC, Solids, Odor, Solubility; Appearance on resale sheets	SG 1.0 and VOC 0% with preview warning; solids back-calculated; 'Not determined'; Appearance prints the state
Raw materials	Supplier product name (resale); manual Prop 65 and HAP entries	Section 1 identifier and Section 14 shipping entry on resale sheets; Section 15 Prop 65 and HAP	Code alone prints, usually defaulting to printing ink (UN1210); only list matches print in Section 15
CAS determinations	Competent-person determination for every CAS without federal data: classes, H/P codes, pictograms, signal word, ATE values, aquatic M-factors, exposure limits	Section 2 and every hazard-driven sentence in Sections 4-15; Sections 8 and 11 exposure limits; Section 12 aquatic table	Publishing is blocked at or above the missing-data threshold (default 1%); below it the ingredient adds nothing
CAS determinations	STOT Target Organs tab: organs and route for single and repeated exposure (H370-H373)	Organ-named H370-H373 statements in Section 2, Section 4 and 11 symptoms, Section 11 Chronic Effects	Standard statement without organs. Organs on only some sources of a code: standard statement plus a preview warning
CAS determinations	Acrylate flag (Acrylate box on CAS Descriptions, or Seed acrylate flags preview and Apply)	Acrylate names in the Section 4 UV skin sentence; Section 10 inhibitor sentence on UV products	Not set: old CAS list (one more release) or the name rule decides. Unticked: never treated as an acrylate
CAS determinations	Nitrogen/sulfur/halogen element flags and RCRA metal flags; re-seed and check name-guessed flags after new CAS are added	Section 5 combustion products; Section 10 decomposition products; Section 13 D004-D011 codes for metal compounds	Only carbon monoxide and carbon dioxide named; a metal compound without its own RCRA list row prints no D code
CAS determinations	TSCA Review status (listed / exempt / not listed); preferred chemical name and molecular formula	Section 15 TSCA sentence; chemical names in Sections 3, 4, 8, 11-15; SARA 313 and HAP compound-category matching	'Not verified' sentence with warning; constituent name used; lead or cobalt compounds may go unreported
Finished goods	Current formula, line percentages (including component products) and Active status	Every section	Publishing is blocked ('No current formula found', or inactive product). A component with no formula adds no ingredients
Finished goods	Hazard override mode and contents (needs hazard-override permission)	Section 2 and every hazard-driven sentence; a Flammable Liquids class here replaces the flash-point category; replace mode suppresses Combustible Dust	Calculated classification used. Override without a mode is ignored; preview warns when it disagrees with the flash point
Finished goods	Physical State, Color and Substance/Mixture	Sections 6, 8, 9, 13 (D002), 14; flammable-liquid and combustible-dust checks; Section 3 Type	Family default, then largest raw's state, else 'Liquid' with preview warning; Color line omitted; Type 'Mixture' unless one Substance raw
Finished goods	Product code, description, alias code and description (CMS import); Transport Product Type	Section 1 identifier, footer code, PDF title, file names; Section 14 UN number and shipping name	Code alone prints ('BK1080'); Section 14 uses description keywords, else printing ink (UN1210)
Finished goods	Own Recommended Use / Restrictions (keep blank unless product-specific); family pick Auto vs manual	Section 1 use text; which family (UV flag, SDS profile) applies	Blank is correct: family text in the sheet language. If filled, prints as typed in every language
Product families	UV/LED flag and Active flag	UV text in Sections 2, 4-8 (nitrile gloves, UV eyewear), 10, 11, 13, 14	No UV text anywhere; Other Hazards prints 'None known.' An inactive family is ignored everywhere, including its profile
Product families	Membership rules, Recompute, 'Reset legacy manual picks to Auto'	Which family each product and raw material gets, and so all family-driven text	Wrong or no family: standard Section 1 text, no UV text, no profile. Old family prints until recompute or reset
Product families	SDS profile: default physical state, storage temperature min/max, water-based (Auto/Yes/No), typical pH min/max	Product state (Sections 6, 8, 9, 13, 14, flammability, dust); Section 7 storage range; Sections 7 and 10 freezing; Section 9 pH	Largest raw's state, else 'Liquid'; no storage sentence; water-based by threshold; pH 'Not determined'

Where	Keep populated	Feeds	If left blank
Product families	Recommended Use and Restrictions on Use text per language (EN, ES, FR, DE)	Section 1 use text (not resale sheets)	That language's standard sentence prints, never the family's English text
Manufacturers	Name, emergency phone, address and contacts, logo, disclaimer per language, private-label custom code/description	Private-label Section 1, header logo, disclaimer, footer code, PDF Author and file names	Blank name or emergency phone: Publishing is blocked. Other lines omitted (no company fallback); Admin disclaimer used
Admin settings	Emergency Phone (Admin > Settings)	Section 1 Emergency line on standard, resale and alias SDSs	Publishing is blocked
Admin settings	Missing-data publish block and Missing Data Threshold %	Missing-hazard-data gate on every publish path	Block on at 1.0%
Admin settings	Company name, address, phone, email, website, logo	Section 1 supplier block; PDF Author; page 1 header logo	Line omitted; Author 'SDS System'; preview warning for name and phone. Never saved: placeholder 'Your Ink Company, Inc.' prints
Admin settings	UV acrylate rule pack; Water-Based Threshold %; inhalation-only CAS list; Prop 65 trace threshold	UV PPE and handling sentences (Sections 2, 4-8, 11); freezing wording (Sections 7, 10); carbon black/TiO2 rule; Prop 65 '(trace)'	Rule pack on; 20%; carbon black and TiO2; 0.1%
Admin settings	Legal disclaimer per language; Sections 12-15 footnote; time zone; pictogram artwork	Disclaimer after Section 16; footnote after Section 15; effective date; GHS and PPE images	Built-in translated disclaimer; footnote printed; America/Chicago; shipped images
Regulatory lists	Federal hazard data and exposure-limit refresh, including ATE values, M-factors and Skin notations	Sections 2-4, 8, 11, 12 hazard classification and exposure limits	CAS counts as missing data: Publishing is blocked at or above threshold. Section 8 prints the no-limits sentence
Regulatory lists	Carcinogen list	Section 2 carcinogenicity, Section 3 rows, Section 4 symptoms, Section 11 Carcinogenicity	No carcinogen line unless hazard data has it; Section 11 prints 'No components … listed'
Regulatory lists	SARA 313, HAP, SNUR, Prop 65, TSCA inventory, RCRA waste codes, exempt VOC list	Section 15 regulatory lines; Section 12 PBT; Section 13 RCRA codes; Section 9 Solids	'None', 'not verified' or 'No RCRA hazardous waste characteristic' sentences; trade-secret Prop 65 block never fires
SDS editor	Section 14 overrides per language: UN number, shipping name, hazard class, packing group (plus environmental hazards, special precautions)	Section 14 lines and special-precautions fragments; clears the 'Not determined' transport block	Automatic value. Incomplete core four, or missing where needed: Publishing is blocked
SDS editor	Per-language text overrides (Sections 1-13, 15), including measured Section 9 flash point and State Regulations	Replaces the whole field, UV text included; cross-feeds Section 4 to 11, 10 to 7, 8 to 2	Automatic text; State Regulations omitted; an English edit does not carry to other languages
SDS editor	Review kept old overrides, 'Reset to automatic', then republish after any data change	Every section on published PDFs (they print the stored snapshot)	Old frozen wording keeps printing (e.g. 'Stable' on a reactive product); published sheets keep publish-time text

PUBLISH GATES

Conditions checked when an SDS is published. A block stops that publish; a warning lets it through and tells the operator.

Condition	Effect	Applies to	How to clear it
Missing federal hazard data: an ingredient CAS at or above the threshold (default 1%) has no hazard data and no active determination	Blocks publishing	All publish paths: finished-good, resale, SDS Updates, private-label, alias (including on-demand for sends), Bulk Publish and scheduled publish	Add an active determination on CAS Determinations or refresh federal data, or raise the threshold or turn off the block in Admin > Settings
Section 14 transport 'Not determined' (unsupported hazard class or Gas state) or incomplete Section 14 override	Blocks publishing	All publish paths, checked in every language; reason also shown in every preview	Correct raw-material hazard data or physical state, or enter UN number, shipping name, class and packing group in every language (SDS editor)

Condition	Effect	Applies to	How to clear it
Trade-secret ingredient whose CAS is on the Prop 65 list	Blocks publishing	All publish paths; also shown in previews and the readiness page	Have the vendor disclose the chemical and clear its trade-secret box (Raw Materials > Edit), or correct the CAS number
Company emergency phone blank	Blocks publishing	Finished-good, resale and alias publish (including send queue and auto-send), SDS Updates; Bulk Publish then runs private-label items only	Enter and save Emergency Phone in Admin > Settings
Private-label manufacturer emergency phone or name blank	Blocks publishing	Every private-label publish: manual, republish, 'Republish Private Labels Only', cascade and bulk; previews show a warning	Enter the Emergency Phone and Name on the Manufacturers screen
Inactive finished good	Blocks publishing	Finished-good publish, SDS Updates, alias publish, private-label republish; Bulk Publish skips it	Set the product's Status to Active (Finished Goods > Edit)
Private-label base checks: no published base SDS, base data missing for a language, alias no longer belongs to the product	Blocks publishing	Private-label publish, republish, cascade and bulk private-label items	Republish the base product in every language, or re-point the item to a valid alias or custom code
Routine controls: edit permission, bulk eligibility (unreviewed raws, freshness, unpublished intermediates), scheduled blackout, auto-publish off, runner lock	Blocks publishing	Permissions: manual actions. Eligibility and scheduling: Bulk Publish and scheduled publish (ineligible products are left out, not failed)	Grant the role edit permission, review raws or publish intermediates, or wait for the window or other run
TSCA inventory not verified, not listed or inactive	Warning only	Previews, readiness page, and the message after manual finished-good and resale publish	Import the CAS into the TSCA inventory, or set a TSCA Review on CAS Determinations
Other data-quality warnings: flash/boiling points, physical state defaults, STOT organs, combustible dust, incomplete ingredients, SG/VOC, technical names, supplier name/phone	Warning only	Preview Warnings box; some also on the readiness page or after publish	Enter the named data (Raw Materials > Edit, CAS Determinations, Product Families, Admin > Settings) or reset the SDS editor edit

WHAT CAN BE CHANGED PER PRODUCT

Text typed in the SDS editor for one product and one language replaces the automatic value for that product and language only.

Field	What a per-product edit does	Limits
Product Identifier, supplier details, emergency phone and logo (Section 1)	No per-product override. No text edit changes these lines.	Change the code and description on Finished Goods or Item Names. Change company details, the emergency phone and the logo on Admin > Settings or Manufacturers.
Recommended Use; Restrictions on Use	A text edit replaces the automatic sentence for that product and language. It takes priority over the product's own field and the family default.	The product's own field holds one value for all languages. Family text on Product Families is set per language and never falls back to English.
Signal Word, Pictograms, Hazard Classes, Hazard and Precautionary Statements (including Combustible Dust and STOT organs)	No text edit. The Hazard Classification Override on Finished Goods adds to or replaces the calculated classes for all languages.	STOT organ wording comes only from CAS Determinations. Replace mode also drops Combustible Dust and the Section 12 component aquatic table.
Other Hazards	A text edit replaces the UV polymerization sentence or 'None known.' for that product and language.	—
Type (Substance / Mixture); Composition table; trade-secret names	No text edit. Type follows the Substance/Mixture field on Finished Goods.	Ingredients, percentages and trade-secret masking come from raw-material constituents. Masking applies in every section and cannot be lifted for one sheet.

Field	What a per-product edit does	Limits
First Aid: Inhalation, Skin, Eye, Ingestion, Most Important Symptoms, Notes to Physician	Each edit replaces its whole line for that language. The Symptoms edit also prints in Section 11.	A Skin edit also removes the UV uncured-product sentence. Section 11 symptoms have no separate edit.
Fire-fighting: Suitable and Unsuitable Media, Specific Hazards, Advice for Firefighters	Each edit replaces its line, including the combustion-products, UV and combustible-dust sentences.	The 'Flash point' sentence in a Specific Hazards edit is refreshed from Section 9 on every generation. A Section 9 flash point edit never changes Suitable Media.
Accidental Release: Personal Precautions, Environmental Precautions, Containment and Cleanup	Each edit replaces the automatic text for that product and language, including the UV waste sentence in Containment.	—
Handling; Storage	Each edit replaces the whole line. A Section 10 Incompatible Materials edit also changes the automatic Storage line.	A Storage edit drops the temperature, freezing and incompatible-materials wording, and a preview warning shows. Without an edit, that wording follows the family SDS profile.
Exposure Limits table; components with a skin notation	No text edit.	Comes from federal data imports and CPDs. The Settings inhalation-only list removes rows from non-powder products.
Engineering Controls; Respiratory, Hand, Eye and Skin Protection (feeds Section 2 Recommended PPE)	Each edit replaces its sentence, UV sentences included. Protection edits also change the Section 2 Recommended PPE line.	Section 2 prints nothing for a baseline tier, even with an edit. UV gloves are always nitrile, because butyl can be worn through. Edits should keep nitrile.
Physical State, Color, Specific Gravity, VOC and Solids	No text edit. These print from product and formula data.	Set Physical State and Color on Finished Goods (or the family default). SG and VOC come from every raw material.
Appearance, Odor, Solubility, Flammability (Solid, Gas)	A text edit replaces the derived value for that product and language.	—
Tested-only properties: Odor Threshold, pH, Melting Point, Evaporation Rate, Flammability Limits, Vapor Pressure and Density, Partition Coefficient, Auto-Ignition, Decomposition Temperature, Viscosity	A text edit enters a tested value for each language. Without one, the sheet prints 'Not determined' (pH prints the family's typical range).	pH is for display only. It is never used for classification or for RCRA D002.
Flash Point; Initial Boiling Point	An edit prints as a measured value for that language. The flash point edit also prints in Section 5.	Must be a temperature that agrees with the Flammable Liquids category (and D001 for flash point). Classification, Section 14 and Suitable Media always use the formula values.
Stability and Reactivity: Reactivity, Chemical Stability, Conditions to Avoid, Incompatible Materials, Hazardous Decomposition Products	Each edit replaces its line. An Incompatible Materials edit also prints in Section 7 Storage while Storage is automatic.	The Section 5 combustion sentence still follows the CAS element flags, not a decomposition edit.
Toxicology: Acute Toxicity, Chronic Effects, Carcinogenicity, UV Acrylate Information	Each edit replaces the automatic text for that product and language.	Likely Routes of Exposure and Component Toxicological Data have no edit. Symptoms change only through the Section 4 edit.
Ecology: Ecotoxicity, Persistence and Degradability, Bioaccumulative Potential, Mobility in Soil	Each edit replaces its line. With a PBT component present, a Persistence edit makes Bioaccumulative print the PBT sentence.	The Component Aquatic Hazard Data table has no edit.
Disposal: Disposal Methods, Contaminated Packaging, RCRA Classification	A Disposal Methods edit replaces the standard sentence and the UV cure-residue sentence.	Contaminated Packaging is fixed text. The RCRA classification is calculated and has no edit.

Field	What a per-product edit does	Limits
Transport: UN Number, Proper Shipping Name, Hazard Class, Packing Group	Edits replace the derived values. The printed Hazard Class also selects the Special Precautions fragments.	Publishing needs all four fields typed. The exceptions are Shipping Name or Packing Group alone when the classifier already regulates the product.
Transport: Environmental Hazards, Special Precautions for User, Note, Transport in Bulk	Environmental Hazards and Special Precautions edits replace their lines. The Environmental Hazards value also selects the marine fragment.	Special Precautions edits never affect classification or publishing. Note and Transport in Bulk have no edit.
Regulatory: OSHA Status, TSCA, SARA 313, HAP, SNUR, California Prop 65	No text edit. Old stored OSHA edits are ignored.	These come from regulatory lists, TSCA determinations and manual raw-material entries. A trade-secret Prop 65 match blocks publishing.
State Regulations	Printed only from a text edit for that product and language, after the Prop 65 block.	No automatic text exists.
Section 16 and footer: Version, Effective Date, Abbreviations, Disclaimer, GHS sections note	No per-product edit. Abbreviations include terms that appear in edited text.	The disclaimer and GHS note are set on Admin > Settings or Manufacturers. Version and date are stamped when the sheet is published.

PRIVATE LABEL SHEETS

- A private-label sheet reuses the base product's content in every language. Only identity and supplier details are replaced.
- Product code and description come from the private-label registry in this order: custom, then shared alias (cut at the first hyphen), then base. They are frozen at publish. Custom codes keep their hyphens on the sheet and in file names.
- The supplier block, logo, emergency phone and disclaimer come only from the manufacturer record. The company logo and phone are never used as a fallback, and a blank manufacturer disclaimer uses the standard disclaimer for that language.
- Publishing is refused when the manufacturer name or emergency phone is blank. All other base-product gates apply too.
- Private-label items have their own version counter: the first publish is 1 even when the base product is at a later version. All languages publish together under one version, or none do.
- File names carry a 'PL_<manufacturer>' tag and the version. The PDF Author is the manufacturer name.
- Sections 2-16 and the use text come from the base product, including its per-product edits. There is no manufacturer-level or item-level text edit.
- Trade-secret identities stay masked in every section. Abbreviations are filtered again against the private-label code, supplier details and disclaimer.

LANGUAGES

- Every sheet is produced in English, Spanish, French and German from per-language translation files. Labels, banners, footer words, the PDF title and subject, and all generated sentences are translated.
- GHS hazard and precautionary statements use the official phrase set for each language. French H410-H412 now read 'entraîne des effets néfastes à long terme' in Sections 2, 6 and 12.
- New ES/FR/DE sentences that do not come from the GHS phrase set (combustion products, skin notation, contaminated packaging, special precautions, Prop 65 clauses, UV polymerization, family profile, STOT organs, combustible dust) are house translations awaiting native-speaker review.
- Chemical names, trade-secret descriptions, regulatory list names and citations print in English on every language. Transport shipping names stay in English with a local gloss.
- '.' decimals and m/d/Y dates are used in every language. Accented section titles are capitalised correctly.
- Family use text and disclaimers are entered per language. A blank language falls back to the translated standard sentence, never to English.
- Per-product text edits are stored per language and do not carry over to other languages. Section 14 edits must be repeated in each language, and every language is checked before publishing.
- Section 16 abbreviations come from each language's own table (for example HDS in Spanish, FDS in French), filtered to the terms printed on that sheet.

TEXT THAT STAYS FIXED

Content that is the same on every sheet, and why.

Item	Why it is fixed
Section titles, field labels, table headings, 'SAFETY DATA SHEET', 'Page X of Y', 'Rev.', the Draft text and the page layout (US Letter, Helvetica)	These are format constants for the 16-section HazCom layout, translated per language. A4 for ES/FR/DE sheets could become a setting but is fixed today.
GHS hazard and precautionary statement wording, including the P308+P313 and P308+P311 first-aid phrases and the official 'Call a POISON CENTER/doctor.'	This is the regulatory phrase set (GHS / HazCom). The owner accepted the official poison-center wording as printed. The data only decides which statements print, and where target organs are entered, which organs and route they name.
UV glove sentence: always nitrile (0.4 mm minimum) on UV inks, corrosive ones included. UV eyewear keeps goggles plus a face shield for H314/H318.	Owner decision 2026-10-12: acrylates can wear chemically through butyl, so the nitrile sentence replaces every glove tier on UV products. A per-product edit can still change it.
Prescribed concentration bands (0.1-1% to 80-100%), the never-understate band rule and the 0.1% disclosure cut-off	HazCom trade-secret and disclosure rules (1910.1200(i), Appendix A). These are deliberate policy, not settings.
Flammable Liquids thresholds (23/35/60/93 °C), the weighted-average flash point rule, the whole-degree display marked '(calculated)', D001 below 60 °C, and Class 3 / packing group cut-offs	GHS, HazCom Appendix B.6, 49 CFR and 40 CFR 261 criteria, plus owner decisions Q1 and A10. The display rounding never moves a product across a category limit.
Base sentences in Sections 4-10 (first aid, extinguishing media, spill, handling, storage, stability)	This is standard SDS wording. Hazard, physical-state, family, UV and dust data add to it or replace it, and each field can be edited per product and language.
Section 9 Appendix D properties with no data field (odor threshold, melting point, evaporation rate, flammability limits, vapor pressure and density, partition coefficient, auto-ignition, decomposition, viscosity) print 'Not determined'	No raw-material or family field exists for them (owner decision Q8: print every property). Only a tested value typed per product changes one. pH now comes from the family's typical range.
Section 12 'Mobility in Soil: No data available.'	No soil-mobility data is stored. It could become data-driven if such a field were added.
Section 13 Contaminated Packaging, the RCRA generator-responsibility sentence, Section 14 Transport in Bulk and the carrier-verification note	Standard HazCom Appendix D and EPA/DOT wording printed on every sheet. The company ships packaged goods only.
Sections 12-15 footnote and the built-in disclaimer paragraph	The footnote is a regulatory statement with an on/off setting. The built-in disclaimer is only a fallback: the live text is set per language in Admin Settings and per manufacturer.
Chemical names, trade-secret descriptions, list names and citations (29 CFR, ANSI, NFPA, 49 CFR shipping names) in English; '.' decimals and m/d/Y dates in every language	The data is stored once and citations keep their official form (owner decision 2026-10-09). ES/FR/DE shipping names add a local gloss.
Last-resort fallbacks: the standard Recommended Use / Restrictions sentences, 'None known.' for Other Hazards on non-UV products, and 'Liquid' when no physical state is known anywhere	They print only when the product, its family and its raw materials provide nothing. The 'Liquid' default raises a preview warning.

CHANGES SINCE THE FIRST AUDIT

Before this update roughly a third of the sheet came from product data. These are the main fields that moved from fixed text to data, or are new.

Field	Change	Note
Section 1 Recommended Use / Restrictions on Use	Now from data	Order: the product's own text, then the active family's text in the sheet language, then the translated standard sentence. A family's English text never prints on ES/FR/DE sheets. Resale sheets use their own sentence or a per-raw-material edit. The old 'Default Product Use' settings are gone, and the Product Family line no longer prints.
Section 1 supplier block, logo and emergency phone	Changed	Taken from Admin Settings only (the config file is used only if a setting was never saved). The address now includes the country, and Email and Website print. Private-label sheets use only the manufacturer's name, logo and phone. A blank emergency phone blocks publishing.

Field	Change	Note
Section 2 hazard classification	Changed	Each triggered class contributes only its own signal word, pictograms and P-codes. Flammable Liquids now comes from the product flash point and boiling point, not from ingredient codes. Bound carbon black and TiO2 are excluded. Acute toxicity Category 5 is dropped. Codes that no class line carries print as their own list. New: the OSHA Combustible Dust class for Powder/Solid products that contain a raw marked Yes.
Target-organ statements H370-H373 (Sections 2, 4, 11)	New	When every CAS behind the code has target organs entered, the statement names those organs, plus the route when all sources agree on one. Otherwise the standard statement prints, with a preview warning if organs were entered for only some sources.
Section 2 Other Hazards	Now from data	Always printed. UV/LED families print the uncured-polymerization sentence, other products print 'None known.', and a per-product edit replaces either.
PPE sentences (Section 2 and Section 8)	Now from data	These were hard-coded English sentences. They are now translated tier sentences chosen from the final hazard codes. On UV products the nitrile glove sentence replaces the glove tier, even on corrosive inks, and the UV eyewear sentence replaces the eyewear tier except for H314/H318, which keep goggles plus a face shield. Respirator and clothing sentences get the UV text added.
Section 3 composition	Changed	'Type' now comes from the Substance/Mixture setting. Bands use each constituent's summed min/max. A Substance sheet always lists its main constituent. A classified product with no listed rows gets a note pointing to Section 2. The raw-material non-hazardous checkbox was removed.
Section 4 First Aid	Now from data	Each route paragraph is built from the hazard codes. A new Symptoms/Effects line (acute and delayed) is also repeated in Section 11. Notes to Physician adds aspiration, corrosive and eye-damage sentences and, for CMR and STOT codes, the official 'If exposed or concerned' phrases. UV products end Skin Contact with an uncured-product sentence.
Section 5 Fire-Fighting	Now from data	Extinguishing media and advice now follow the water-reactive, oxidizer and flammable tiers. The flash point sentence matches Section 9. The combustion sentence names nitrogen oxides, sulfur oxides or hydrogen halides from the CAS element flags, matching Section 10. Combustible-dust and UV sentences were added.
Section 6 Accidental Release	Now from data	Containment text follows the resolved physical state (liquid, paste or gel, solid or powder, gas). The acute-toxicity, corrosive, flammable and ignition add-ons are driven by hazard codes. The aquatic sentences are now word for word the Section 2 statements in each language. UV products get a containment sentence.
Section 7 Handling and Storage	Changed	This used to be a first-match canned paragraph. It is now a base sentence plus every matching hazard sentence. Storage adds the family storage-temperature range, 'Protect from freezing.' for water-based products, 'Store locked up' when P405 prints, and the Section 10 incompatibles list.
Section 8 Exposure Controls	Changed	The limits table prints bands and masks trade secrets. When no limits remain, a 'no exposure limits' sentence prints. A new line names the components with a Skin notation. Engineering Controls is built from state and hazards, and a product with no limits gets a general-ventilation sentence.
Section 9 Physical State and Flash Point	Changed	Physical state: the product's own, then the family default, then the largest raw that has one, then 'Liquid' with a warning. Flash point is the weighted average of the raws that have one (it was the lowest), printed to the whole degree with '(calculated)'. A measured edit prints exactly as typed. No flash point means not flammable.
Section 9 other properties	Now from data	Boiling point (lowest raw), odor (dominant raw), solubility (weighted band) and pH (family typical range) now come from data. Twelve Appendix D lines were added. VOC and solids are divided by the formula total. Two VOC/solids lines were dropped from print.
Section 10 Stability and Reactivity	Now from data	Reactivity, stability, conditions and incompatibles are now additive lists built from hazard classes. Decomposition products come from the element flags. UV/LED families add polymerization, inhibitor and peroxide/initiator lines. The family profile adds a storage maximum and freezing.
Section 11 Toxicology	Now from data	Three acute route lines replace the old constant sentence, with the ATEmix printed only when it set the category. Likely routes, symptoms, chronic fragments from the hazard codes, and banded carcinogen listings (IARC 1/2A/2B, NTP, OSHA only) were added. Trade secrets are masked.

Field	Change	Note
Section 12 Ecology	Now from data	This section prints the product's real aquatic statements with their source, a component table with M-factors, and SARA 313 PBT components at any concentration. Mobility in Soil was added and stays 'No data available.'
Section 13 Disposal	New	A computed RCRA line was added (D001-D003, toxicity-characteristic metals including compounds, and F/K/P/U as reference). Disposal Methods is now one standard sentence, plus a cure-residue sentence on clean UV sheets. Contaminated Packaging was added as a fixed line.
Section 14 Transport	Now from data	This used to copy one ingredient's DOT row. It is now a mixture classification (Class 3 from the product flash point; Classes 8, 6.1 and 9 from hazards) with marine pollutant, technical names, and combustible and viscous notes. Transport in Bulk was added. Special Precautions now adds Class 3, Class 8, marine and UV fragments and can be edited per product.
Section 15 Regulatory	Now from data	OSHA and TSCA status are now driven by data and can no longer be edited per product. SARA 313 prints (it never did before), with PBT and PFAS chemicals at any concentration. HAP has a Clean Air Act 112(b) heading and a banded total. Prop 65 uses correct 'is/are' grammar with the cancer clause first. Trade secrets are masked.
Section 16 and the page footer	Changed	The published version and effective date (in the admin time zone) replace the generation date, and drafts say 'Draft (not yet published)'. Abbreviations are filtered to the terms printed on the sheet. The disclaimer is set per language and per manufacturer. The revision note was removed.
Separate UV acrylate notes in Sections 4-8	Removed	Folded into the Skin Contact, Specific Hazards, Containment, Handling and PPE fields. Only Section 11 keeps an editable UV note. Products count as UV through the active family's UV/LED flag, and the CAS Acrylate flag supplies the acrylate names.
Per-product text edits	Changed	Only text the operator typed is stored. The automatic text shows as a hint, and 'Reset to automatic' clears an edit. Blank and retired rows are ignored, and two cleanup passes removed frozen canned text. Resale raw materials have their own editor.
Publish gates and preview warnings	Changed	The same blocking gates now run on manual, SDS Update, alias, private-label and bulk publishing, in every language. New operator warnings (flash point, physical state, SG/VOC, composition, supplier, STOT organs, combustible dust) appear only in the preview and on the readiness page, and never block.
Summary label on every map row	New	Each row now says 'Generated from real data' or 'Uses fixed text'. Fixed wording that the data selects, or that depends on the data, counts as 'Generated from real data'.